

Architecture

This chapter consists of five sections. The first section describes specific aspects of the device architecture. The second section describes the clocking structure of the microcontrollers. The third section gives an overview of the device memory organization. The fourth section details exceptions on the device, and the last section describes the system and peripheral control registers of the microcontroller.

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2.1 Introduction

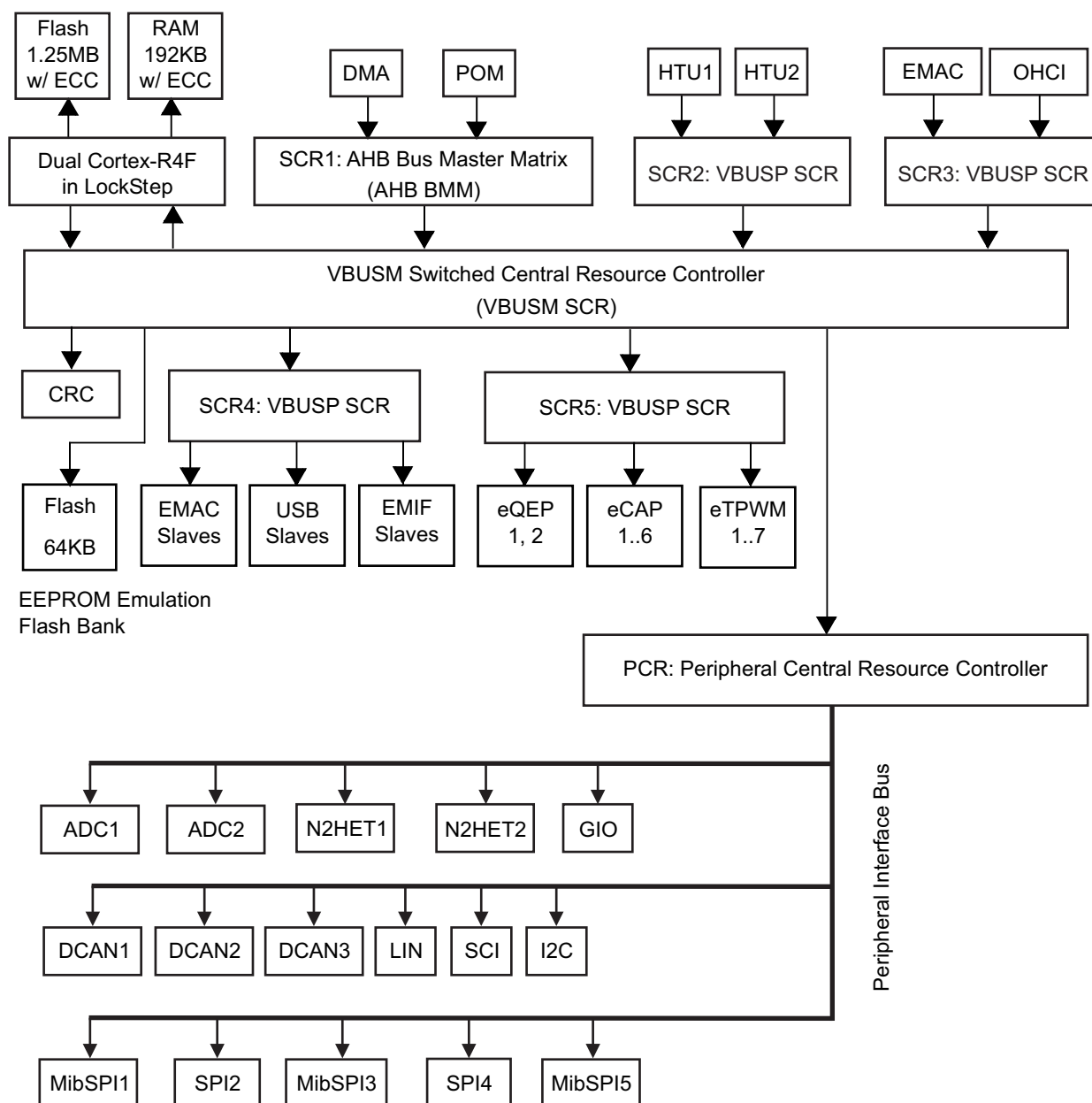
The RM46x family of microcontrollers is based on the Texas Instruments TMS570 Architecture. This chapter describes specific aspects of the architecture as applicable to the RM46x family of microcontrollers.

2.1.1 Architecture Block Diagram

The RM46x microcontrollers are based on the TMS570 Platform architecture, which defines the interconnect between the bus masters and the bus slaves.

Figure 2-1 shows a high-level architectural block diagram for the superset microcontroller.

Figure 2-1. Architectural Block Diagram



2.1.2 Definitions of Terms

Table 2-1 provides a definition of terms used in the architectural block diagram.

Table 2-1. Definition of Terms

Acronym / Term	Full Form	Description
ADCx	Analog-to-Digital Converter	The ADC uses the Successive Approximation Register architecture. It features a selectable 10-bit or 12-bit resolution. The ADC module also includes a RAM to hold the conversion results. A digital logic wrapper manages accesses to the control and status registers. There are two ADC modules on this device.
CRC	Cyclic Redundancy Checker	The CRC module provides two channels to perform background signature verification on any memory region. It also supports maximum-length Parallel Signature Analysis (PSA) based on a 64-bit primitive polynomial. The CRC module is a bus slave in this device.
DAP	Debug Access Port	The DAP allows a tool such as a debugger to read from or write to any region in the device memory-map. The DAP is a bus master in this device.
DCANx	Controller Area Network controller	The DCAN supports the CAN 2.0B protocol standard and uses a serial, multi-master communication protocol that efficiently supports distributed real-time control with robust communication rates of up to 1 megabit per second (Mbps). The DCAN is ideal for applications operating in noisy and harsh environments (for example, automotive and industrial fields) that require reliable serial communication or multiplexed wiring.
DMA	Direct Memory Access	The DMA module is used for transferring 8-, 16-, 32- or 64-bit data across the entire device memory-map. The DMA module is one of the bus masters on the device. That is, it can initiate a read or a write transaction.
eCAP	Enhanced Capture Module	The enhanced Capture (eCAP) module is essential in systems where accurate timing of external events is important.
eFuse	Electronically Programmable Fuse controller	Electrically programmable fuses (eFuses) are used to configure the device after deassertion of PORST. The eFuse values are read and loaded into internal registers as part of the power-on-reset sequence. The eFuse values are protected with Single-Bit Error Correction Double-Bit Error Detection (SECDED) codes. These fuses are programmed during the initial factory test of the device. The eFuse controller is designed so that the state of the eFuses cannot be changed once the device is packaged.
ePWM	Enhanced Pulse Width Modulator	The enhanced pulse width modulator (ePWM) peripheral is a key element in controlling many of the power electronic systems found in both commercial and industrial equipments. These systems include digital motor control, switch mode power supply control, uninterruptible power supplies (UPS), and other forms of power conversion. The ePWM peripheral performs a digital to analog (DAC) function, where the duty cycle is equivalent to a DAC analog value; it is sometimes referred to as a Power DAC.
eQEP	Enhanced Quadrature Encoder Pulse Module	The enhanced quadrature encoder pulse (eQEP) module is used for direct interface with a linear or rotary incremental encoder to get position, direction, and speed information from a rotating machine for use in a high-performance motion and position-control system.
ECC	Error Correction Code	This is a code that is used by the Single-Bit Error Correction Double-Bit Error Detection (SECDED) logic inside the two Cortex-R4F processors (CPUs). There are 8 bits of ECC for every 64 bits of data accessed from the CPU tightly-coupled memories (flash and RAM).
EMAC	Ethernet Media Access Controller	The EMAC has a dedicated DMA-type component that is used to transfer data to / from the EMAC descriptor memory from / to another memory in the device memory-map. This DMA-type component of the EMAC is a bus master in this device.
EMAC slaves	Ethernet Media Access Controller slave ports	There are four EMAC slaves: 1. EMAC Control Module: this provides an interface between the EMAC and MDIO modules and the bus masters. It also includes 8KB of RAM to hold EMAC packet buffer descriptors. 2. EMAC: The EMAC module interfaces to the other devices on the Ethernet Network using the Media Independent Interface (MII) or Reduced Media Independent Interface (RMII). 3. Management Data Input / Output (MDIO): The MDIO module is used to manage the physical layer (PHY) device connected to the EMAC module. 4. Communications Port Programming Interface (CPPI): This is the 8KB of RAM used to hold the EMAC packet buffer descriptors.

Table 2-1. Definition of Terms (continued)

Acronym / Term	Full Form	Description
EEPROM Emulation Flash Bank	Emulated Electrically Erasable Programmable Read-Only Memory	This is a flash bank that is dedicated for use as an emulated EEPROM. This device supports 64KB of flash for emulated EEPROM.
EMIF slaves	External Memory Interface slave ports	There are five EMIF slaves:- External SDRAM memory: EMIF chip select 0- External asynchronous memories: EMIF chip selects 2, 3 and 4- EMIF module control and status registers
GIO	General-purpose Input/Output	The GIO module allows up to 16 terminals to be used as general-purpose Input or Output. Each of these are also capable of generating an interrupt to the CPU.
HTUx	High-end timer Transfer Unit	The HTU is a dedicated transfer unit for the New Enhanced High-End Timer module. The HTU has a native interface to the N2HET RAM, and is used to transfer data to and from the N2HET RAM from and to another region in the device memory-map. There is one HTU per N2HET module, so that there are 2 HTU modules on the device. The HTUx are bus masters in this device.
I2C	Inter-Integrated Circuit controller	The I2C module is a multi-master communication module providing an interface between the device and an I2C-compatible device via the I2C serial bus. The I2C supports both 100 Kbps and 400 Kbps speeds.
LIN	Local Interconnect Network controller	The LIN module supports the Local Interconnect standard revision 2.1 and can be used as a UART in full-duplex mode using the standard Non-Return-to-Zero (NRZ) format.
Lockstep	–	This is the mode of operation of the dual ARM Cortex-R4F CPUs. The outputs of the two CPUs are compared on each CPU clock cycle. Any miscompare is flagged as an error of the highest severity level.
MibSPiX	Multi-Buffered Serial Peripheral Interface	The MibSPiX modules also support the standard SPI communication protocol. The transfers are all grouped into transfer chunks called “transfer groups”. These transfer groups are made up of one or more buffers in the MibSPiX RAM. The RAM is used to hold the control information and data to be transmitted, as well as the status information and data that is received. There are three MibSPI modules in this device.
N2HETx	New Enhanced High-End Timer	The N2HET is an advanced intelligent timer that provides sophisticated timing functions for real-time applications. The timer is software-controlled, using a reduced instruction set, with a specialized timer micromachine and an attached I/O port. The N2HET can be used for pulse width modulated outputs, capture or compare inputs, or general-purpose I/O.
PCR	Peripheral Central Resource controller	The PCR manages the accesses to the peripheral registers and peripheral memories. It provides a global reset for all the peripherals. It also supports the capability to selectively enable or disable the clock for each peripheral individually. The PCR also manages the accesses to the system module registers required to configure the device's clocks, interrupts, and so on. The system module registers also include status flags for indicating exception conditions – resets, aborts, errors, interrupts.
POM	Parameter Overlay Module	The parameter overlay module redirects accesses to a programmable region in flash memory (read-only) to a RAM memory, either on-chip or via the external memory interface (EMIF). This allows a user to evaluate the impact of changing values of constants stored in the flash memory without actually having to erase and reprogram the flash. The POM is also a bus master in this device.
SCI	Serial Communication Interface	The SCI module supports the standard UART in full-duplex mode using the standard Non-Return-to-Zero (NRZ) format.
SCR1: AHB BMM	AMBA High-performance Bus Matrix Module	The DMA, POM and DAP all act as masters on the AMBA High-performance Bus (AHB). The BMM arbitrates between concurrent accesses by these masters using a fixed priority scheme. The modules in descending order of priority are POM → DMA → DAP.
SCR2: VBUSP SCR	VBUSP Switched Central Resource Controller	The SCR2 arbitrates between concurrent accesses by the HTU1 and HTU2. A round-robin priority scheme is used between the HTU1 and HTU2.
SCR3: VBUSP SCR	VBUSP Switched Central Resource Controller	The SCR3 arbitrates between concurrent accesses by the EMAC and another module that is not available in this configuration of the device.
SCR4: VBUSP SCR	VBUSP Switched Central Resource Controller	The SCR4 is used to decode the accesses to the bus slaves for the EMAC and EMIF modules. SCR4 is a bus slave in this device.
SPIx	Serial Peripheral Interface	The SPiX module provides a clocked serial communication interface for reliable communication between the device and other serial devices with the standard SPI interface. There are two SPI modules on this device.

Table 2-1. Definition of Terms (continued)

Acronym / Term	Full Form	Description
USB	Universal Serial Bus	This device provides several varieties of USB functionality, including: - One full-speed USB device port compatible with the USB Specification Revision 2.0 and USB Specification Revision 1.1 - Two USB host ports compatible with USB Specification Revision 2.0, which is based on the OHCI Specification For USB Release 1.0
VBUSM SCR	VBUSM Switched Central Resource Controller	This is the main device SCR. It arbitrates between the accesses from multiple bus masters to the bus slaves using a round robin priority scheme.

2.1.3 Bus Master / Slave Access Privileges

This device implements some restrictions on the bus slave access privileges, see [Table 2-2](#), in order to improve the overall throughput of the interconnect shown in [Figure 2-1](#).

Table 2-2. Bus Master / Slave Access Privileges

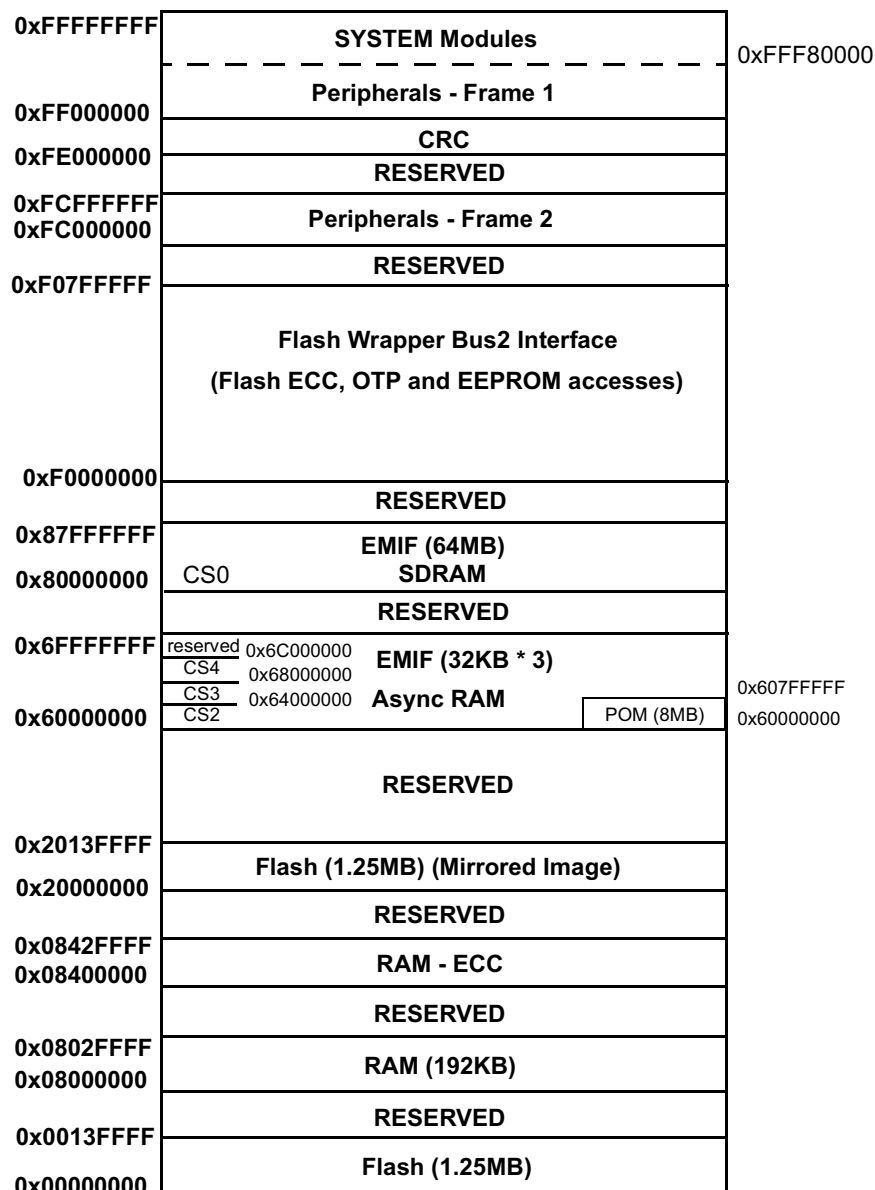
Masters	Master ID	Access Mode	Bus Slaves Being Accessed				
			EEPROM Bank, ECC Bits, OTP Regions	Non-CPU Accesses to CPU Flash and RAM	CRC Module	EMAC, EMIF, and USB Slaves	PCR Modules
CPU Read	0	User/Privilege	Allowed	Allowed	Allowed	Allowed	Allowed
CPU Write	1	User/Privilege	Not allowed	Allowed	Allowed	Allowed	Allowed
POM	2	User	Allowed	Allowed	Allowed	Allowed	Allowed
DMA	3	User	Allowed	Allowed	Allowed	Allowed	Allowed
DAP	5	Privilege	Allowed	Allowed	Allowed	Allowed	Allowed
HTU1	6	Privilege	Not allowed	Allowed	Allowed	Allowed	Allowed
HTU2	8	Privilege	Not allowed	Allowed	Allowed	Allowed	Allowed
EMAC	9	User	Not allowed	Allowed	Not allowed	Allowed	Not allowed
OHCI	10	User	Not allowed	Allowed	Not allowed	Allowed	Not allowed

2.2 Memory Organization

2.2.1 Memory-Map Overview

The Cortex-R4F uses a 32-bit address bus, giving it access to a memory space of 4GB. This space is divided into several regions, each addressed by different memory selects. [Figure 2-2](#) shows the memory-map of the microcontroller.

Figure 2-2. Memory-Map



The main flash instruction memory is addressed starting at 0x00000000 by default. This is also the reset vector location – the ARM Cortex-R4F processor core starts execution from the reset vector address of 0x00000000 whenever the core gets reset.

The CPU data RAM is addressed starting at 0x08000000 by default.

The device also supports the swapping of the CPU instruction memory (flash) and data memory (RAM). This can be done by configuring the MEM SWAP field of the Bus Matrix Module Control Register 1 (BMMCR1).

After swapping, the data RAM is accessed starting from 0x00000000 and the RAM ECC locations are accessed starting from 0x00400000. The flash memory is now accessed starting from 0x08000000.

2.2.2 Memory-Map Table

The control and status registers for each module are mapped within the CPU's 4GB memory space. Some modules also have associated memories, which are also mapped within this space.

Table 2-3 shows the starting and ending addresses of each module's register frame and any associated memory. The table also shows the response generated by the module or the interconnect whenever an access is made to an unimplemented location inside the register or memory frame.

Table 2-3. Module Registers / Memories Memory-Map

Name	Memory Select	Frame Address		Frame Size	Actual Size	Response for Access to Unimplemented Location in Frame
		Start	End			
CPU Tightly-Coupled Memories						
TCM Flash	CS0	0x0000_0000	0x00FF_FFFF	16MB	1.25MB	Abort
TCM RAM + RAM ECC	CSRAM0	0x0800_0000	0x0BFF_FFFF	64MB	192KB	
Mirrored Flash	Flash mirror frame	0x2000_0000	0x20FF_FFFF	16MB	1.25MB	
External Memory Accesses						
EMIF Chip Select 2 (asynchronous)	EMIF select 2	0x6000_0000	0x63FF_FFFF	64MB	16MB	Access to “Reserved” space will generate Abort
EMIF Chip Select 3 (asynchronous)	EMIF select 3	0x6400_0000	0x67FF_FFFF	64MB	16MB	
EMIF Chip Select 4 (asynchronous)	EMIF select 4	0x6800_0000	0x6BFF_FFFF	64MB	16MB	
EMIF Chip Select 0 (synchronous)	EMIF select 0	0x8000_0000	0x87FF_FFFF	128MB	128MB	
Flash Bus2 Interface: OTP, ECC, EEPROM Bank						
Customer OTP, TCM Flash Bank		0xF000_0000	0xF000_1FFF	8KB	4KB	Abort
Customer OTP, EEPROM Bank		0xF000_E000	0xF000_FFFF	8KB	4KB	
Customer OTP–ECC, TCM Flash Bank		0xF004_0000	0xF004_03FF	1KB	512B	
Customer OTP–ECC, EEPROM Bank		0xF004_1C00	0xF004_1FFF	1KB	512B	
TI OTP, TCM Flash Bank		0xF008_0000	0xF008_1FFF	8KB	4KB	
TI OTP, EEPROM Bank		0xF008_E000	0xF008_FFFF	8KB	4KB	
TI OTP–ECC, TCM Flash Bank		0xF00C_0000	0xF00C_03FF	1KB	512B	
TI OTP–ECC, EEPROM Bank		0xF00C_1C00	0xF00C_1FFF	1KB	512B	
EEPROM Bank–ECC		0xF010_0000	0xF013_FFFF	256KB	8KB	
EEPROM Bank		0xF020_0000	0xF03F_FFFF	2MB	64KB	
Flash Data Space ECC		0xF040_0000	0xF04F_FFFF	1MB	160KB	
EMAC and EMIF Slaves						
CPPI Memory Slave (Ethernet RAM)		0xFC52_0000	0xFC52_1FFF	8KB	8KB	Abort
CPGMAC Slave (Ethernet Slave)		0xFCF7_8000	0xFCF7_87FF	2KB	2KB	No error
CPGMACSS Wrapper (Ethernet Wrapper)		0xFCF7_8800	0xFCF7_88FF	256B	256B	No error

Table 2-3. Module Registers / Memories Memory-Map (continued)

Name	Memory Select	Frame Address		Frame Size	Actual Size	Response for Access to Unimplemented Location in Frame
		Start	End			
Ethernet MDIO Interface		0xFCF7_8900	0xFCF7_89FF	256B	256B	No error
W2FC (USB device controller registers)		0xFCF7_8A00	0xFCF7_8A7F	128B	128B	Abort
OHCI (USB Host controller registers)		0xFCF7_8B00	0xFCF7_8BFF	256B	256B	Abort
ePWM1		0xFCF7_8C00	0xFCF7_8CFF	256B	256B	Abort
ePWM2		0xFCF7_8D00	0xFCF7_8DFF	256B	256B	Abort
ePWM3		0xFCF7_8E00	0xFCF7_8EFF	256B	256B	Abort
ePWM4		0xFCF7_8F00	0xFCF7_8FFF	256B	256B	Abort
ePWM5		0xFCF7_9000	0xFCF7_90FF	256B	256B	Abort
ePWM6		0xFCF7_9100	0xFCF7_91FF	256B	256B	Abort
ePWM7		0xFCF7_9200	0xFCF7_92FF	256B	256B	Abort
eCAP1		0xFCF7_9300	0xFCF7_94FF	256B	256B	Abort
eCAP2		0xFCF7_9400	0xFCF7_95FF	256B	256B	Abort
eCAP3		0xFCF7_9500	0xFCF7_96FF	256B	256B	Abort
eCAP4		0xFCF7_9600	0xFCF7_97FF	256B	256B	Abort
eCAP5		0xFCF7_9700	0xFCF7_98FF	256B	256B	Abort
eCAP6		0xFCF7_9800	0xFCF7_99FF	256B	256B	Abort
eQEP1		0xFCF7_9900	0xFCF7_9AFF	256B	256B	Abort
eQEP2		0xFCF7_9A00	0xFCF7_9BFF	256B	256B	Abort
EMIF Registers		0xFCFF_E800	0xFCFF_E8FF	256B	256B	Abort
Cyclic Redundancy Checker (CRC) Module Register Frame						
CRC	CRC frame	0xFE00_0000	0xFEFF_FFFF	16MB	512B	Accesses above 0x200 generate abort.
Peripheral Memories						
MIBSPI5 RAM	PCS[5]	0xFF0A_0000	0xFF0B_FFFF	128KB	2KB	Abort for accesses above 2KB
MIBSPI3 RAM	PCS[6]	0xFF0C_0000	0xFF0D_FFFF	128KB	2KB	Abort for accesses above 2KB
MIBSPI1 RAM	PCS[7]	0xFF0E_0000	0xFF0F_FFFF	128KB	2KB	Abort for accesses above 2KB
DCAN3 RAM	PCS[13]	0xFF1A_0000	0xFF1B_FFFF	128KB	2KB	Wrap around for accesses to unimplemented address offsets lower than 0x7FF. Abort generated for accesses beyond offset 0x800.
DCAN2 RAM	PCS[14]	0xFF1C_0000	0xFF1D_FFFF	128KB	2KB	Wrap around for accesses to unimplemented address offsets lower than 0x7FF. Abort generated for accesses beyond offset 0x800.
DCAN1 RAM	PCS[15]	0xFF1E_0000	0xFF1F_FFFF	128KB	2KB	Wrap around for accesses to unimplemented address offsets lower than 0x7FF. Abort generated for accesses beyond offset 0x800.

Table 2-3. Module Registers / Memories Memory-Map (continued)

Name	Memory Select	Frame Address		Frame Size	Actual Size	Response for Access to Unimplemented Location in Frame
		Start	End			
MIBADC2 RAM	PCS[29]	0xFF3A_0000	0xFF3B_FFFF	128KB	8KB	Wrap around for accesses to unimplemented address offsets lower than 0x1FFF. Abort generated for accesses beyond 0x1FFF.
MIBADC1 RAM	PCS[31]	0xFF3E_0000	0xFF3F_FFFF	128KB	8KB	Wrap around for accesses to unimplemented address offsets lower than 0x1FFF. Abort generated for accesses beyond 0x1FFF.
N2HET2 RAM	PCS[34]	0xFF44_0000	0xFF45_FFFF	128KB	16KB	Wrap around for accesses to unimplemented address offsets lower than 0x3FFF. Abort generated for accesses beyond 0x3FFF.
N2HET1 RAM	PCS[35]	0xFF46_0000	0xFF47_FFFF	128KB	16KB	Wrap around for accesses to unimplemented address offsets lower than 0x3FFF. Abort generated for accesses beyond 0x3FFF.
HET TU2 RAM	PCS[38]	0xFF4C_0000	0xFF4D_FFFF	128KB	1KB	Abort
HET TU1 RAM	PCS[39]	0xFF4E_0000	0xFF4F_FFFF	128KB	1KB	Abort
Debug Components						
CoreSight Debug ROM	CSCS0	0xFFA0_0000	0xFFA0_0FFF	4KB	4KB	Reads return zeros, writes have no effect
Cortex-R4F Debug	CSCS1	0xFFA0_1000	0xFFA0_1FFF	4KB	4KB	Reads return zeros, writes have no effect
POM	CSCS4	0xFFA0_4000	0xFFA0_4FFF	4KB	4KB	Abort
Peripheral Control Registers						
HTU1	PS[22]	0xFFF7_A400	0xFFF7_A4FF	256B	256B	Reads return zeros, writes have no effect
HTU2	PS[22]	0xFFF7_A500	0xFFF7_A5FF	256B	256B	Reads return zeros, writes have no effect
N2HET1	PS[17]	0xFFF7_B800	0xFFF7_B8FF	256B	256B	Reads return zeros, writes have no effect
N2HET2	PS[17]	0xFFF7_B900	0xFFF7_B9FF	256B	256B	Reads return zeros, writes have no effect
GIO	PS[16]	0xFFF7_BC00	0xFFF7_BCFF	256B	256B	Reads return zeros, writes have no effect
MIBADC1	PS[15]	0xFFF7_C000	0xFFF7_C1FF	512B	512B	Reads return zeros, writes have no effect
MIBADC2	PS[15]	0xFFF7_C200	0xFFF7_C3FF	512B	512B	Reads return zeros, writes have no effect
I2C	PS[10]	0xFFF7_D400	0xFFF7_D4FF	256B	256B	Reads return zeros, writes have no effect
DCAN1	PS[8]	0xFFF7_DC00	0xFFF7_DDFD	512B	512B	Reads return zeros, writes have no effect
DCAN2	PS[8]	0xFFF7_DE00	0xFFF7_DFFF	512B	512B	Reads return zeros, writes have no effect

Table 2-3. Module Registers / Memories Memory-Map (continued)

Name	Memory Select	Frame Address		Frame Size	Actual Size	Response for Access to Unimplemented Location in Frame
		Start	End			
DCAN3	PS[7]	0xFFFF7_E000	0xFFFF7_E1FF	512B	512B	Reads return zeros, writes have no effect
LIN	PS[6]	0xFFFF7_E400	0xFFFF7_E4FF	256B	256B	Reads return zeros, writes have no effect
SCI	PS[6]	0xFFFF7_E500	0xFFFF7_E5FF	256B	256B	Reads return zeros, writes have no effect
MibSPI1	PS[2]	0xFFFF7_F400	0xFFFF7_F5FF	512B	512B	Reads return zeros, writes have no effect
SPI2	PS[2]	0xFFFF7_F600	0xFFFF7_F7FF	512B	512B	Reads return zeros, writes have no effect
MibSPI3	PS[1]	0xFFFF7_F800	0xFFFF7_F9FF	512B	512B	Reads return zeros, writes have no effect
SPI4	PS[1]	0xFFFF7_FA00	0xFFFF7_FBFF	512B	512B	Reads return zeros, writes have no effect
MibSPI5	PS[0]	0xFFFF7_FC00	0xFFFF7_FDFF	512B	512B	Reads return zeros, writes have no effect
System Modules Control Registers and Memories						
DMA RAM	PPCS0	0xFFFF8_0000	0xFFFF8_0FFF	4KB	4KB	Abort
VIM RAM	PPCS2	0xFFFF8_2000	0xFFFF8_2FFF	4KB	1KB	Wrap around for accesses to unimplemented address offsets lower than 0x3FF. Abort generated for accesses beyond 0x3FF.
Flash Wrapper	PPCS7	0xFFFF8_7000	0xFFFF8_7FFF	4KB	4KB	Abort
eFuse Farm Controller	PPCS12	0xFFFF8_C000	0xFFFF8_CFFF	4KB	4KB	Abort
Power Management Module (PMM)	PPSE0	0xFFFFF_0000	0xFFFFF_01FF	512B	512B	Abort
Test Controller (FMTM)	PPSE1	0xFFFFF_0400	0xFFFFF_07FF	1KB	1KB	Reads return zeros, writes have no effect
PCR registers	PPS0	0xFFFFF_E000	0xFFFFF_E0FF	256B	256B	Reads return zeros, writes have no effect
System Module - Frame 2 (see platform architecture specification)	PPS0	0xFFFFF_E100	0xFFFFF_E1FF	256B	256B	Reads return zeros, writes have no effect
PBIST	PPS1	0xFFFFF_E400	0xFFFFF_E5FF	512B	512B	Reads return zeros, writes have no effect
STC	PPS1	0xFFFFF_E600	0xFFFFF_E6FF	256B	256B	Reads return zeros, writes have no effect
IOMM Multiplexing Control Module	PPS2	0xFFFFF_EA00	0xFFFFF_EBFF	512B	512B	Generates address error interrupt, if enabled
DCC1	PPS3	0xFFFFF_EC00	0xFFFFF_ECFE	256B	256B	Reads return zeros, writes have no effect
DMA	PPS4	0xFFFFF_F000	0xFFFFF_F3FF	1KB	1KB	Reads return zeros, writes have no effect
DCC2	PPS5	0xFFFFF_F400	0xFFFFF_F4FF	256B	256B	Reads return zeros, writes have no effect
ESM	PPS5	0xFFFFF_F500	0xFFFFF_F5FF	256B	256B	Reads return zeros, writes have no effect
CCMR4	PPS5	0xFFFFF_F600	0xFFFFF_F6FF	256B	256B	Reads return zeros, writes have no effect
RAM ECC even	PPS6	0xFFFFF_F800	0xFFFFF_F8FF	256B	256B	Reads return zeros, writes have no effect

Table 2-3. Module Registers / Memories Memory-Map (continued)

Name	Memory Select	Frame Address		Frame Size	Actual Size	Response for Access to Unimplemented Location in Frame
		Start	End			
RAM ECC odd	PPS6	0xFFFF_F900	0xFFFF_F9FF	256B	256B	Reads return zeros, writes have no effect
RTI + DWWD	PPS7	0xFFFF_FC00	0xFFFF_FCFF	256B	256B	Reads return zeros, writes have no effect
VIM Parity	PPS7	0xFFFF_FD00	0xFFFF_FDFF	256B	256B	Reads return zeros, writes have no effect
VIM	PPS7	0xFFFF_FE00	0xFFFF_FEFF	256B	256B	Reads return zeros, writes have no effect
System Module - Frame 1	PPS7	0xFFFF_FF00	0xFFFF_FFFF	256B	256B	Reads return zeros, writes have no effect

2.2.3 Flash on Microcontrollers

The RM46x microcontrollers support up to 1.25MB of flash for use as program memory. The microcontrollers also support a separate 64KB flash bank for use as emulated EEPROM.

2.2.3.1 Flash Bank Sectoring Configuration

The bank is divided into multiple sectors. A flash sector is the smallest region in the flash bank that must be erased. The sectoring configuration of each flash bank is shown in [Table 2-4](#).

Table 2-4. Flash Memory Banks and Sectors ⁽¹⁾

Memory Arrays (or Banks)	Block Number	Sector Number	Segment	Low Address	High Address
BANK0 (1.25MBytes)	0	0	16K Bytes	0x0000_0000	0x0000_3FFF
		1	16K Bytes	0x0000_4000	0x0000_7FFF
		2	16K Bytes	0x0000_8000	0x0000_BFFF
		3	16K Bytes	0x0000_C000	0x0000_FFFF
		4	16K Bytes	0x0001_0000	0x0001_3FFF
		5	16K Bytes	0x0001_4000	0x0001_7FFF
		6	32K Bytes	0x0001_8000	0x0001_FFFF
		7	128K Bytes	0x0002_0000	0x0003_FFFF
	1	8	128K Bytes	0x0004_0000	0x0005_FFFF
		9	128K Bytes	0x0006_0000	0x0007_FFFF
	2	10	128K Bytes	0x0008_0000	0x0009_FFFF
		11	128K Bytes	0x000A_0000	0x000B_FFFF
	3	12	128K Bytes	0x000C_0000	0x000D_FFFF
		13	128K Bytes	0x000E_0000	0x000F_FFFF
	4	14	128K Bytes	0x0010_0000	0x0011_FFFF
		15	128K Bytes	0x0012_0000	0x0013_FFFF
BANK7 (64KBytes) for EEPROM emulation	0	0	16K Bytes	0xF020_0000	0xF020_3FFF
	1	1	16K Bytes	0xF020_4000	0xF020_7FFF
	2	2	16K Bytes	0xF020_8000	0xF020_BFFF
	3	3	16K Bytes	0xF020_C000	0xF020_FFFF

⁽¹⁾

- The Flash banks are 144-bit wide bank with ECC support.
- The flash bank7 can be programmed while executing code from flash bank0.
- Code execution is not allowed from flash bank7.

Refer to the device datasheet for electrical and timing specifications related to the flash module.

2.2.3.2 ECC Protection for Flash Accesses

The RM46x microcontrollers protect all accesses to the on-chip flash memory by dedicated Single-Bit Error Correction Double-Bit Error Detection (SECDED) logic.

The access to the program memory – flash bank 0, is protected by SECDED logic implemented inside the ARM Cortex-R4F CPU. Accesses to the EEPROM emulation flash bank (bank 7) are protected by dedicated SECDED logic inside the digital interface to the flash banks.

Both the SECDED logic implementations use Error Correction Codes (ECC) for correcting single-bit errors and for detecting double-bit errors in the values read from the flash arrays. There is an 8-bit ECC for every 64 bits of data. The ECC for the flash memory contents needs to be calculated by an external tool such as nowECC. The ECC can then be programmed into the flash array along with the actual application code.

The ECC for the flash array is stored in the flash itself, and is mapped to a region starting at 0xF0400000 for the main flash bank 0, and to a region starting at 0xF0100000 for the EEPROM emulation flash bank 7.

NOTE: ECC Protection Not Enabled By Default

The SECDED logic inside the CPU is not enabled by default and must be enabled by the application.

Code Example for Enabling ECC Protection for Main Flash Accesses:

The following code example can be used to enable the ECC protection for accesses to the main flash array:

```
MRC p15, #0, r1, c1, c0, #1
ORR r1, r1, #0x02000000      ;Enable ECC checking for ATCM
DMB
MCR p15, #0, r1, c1, c0, #1
```

The ECC protection for accesses to the EEPROM emulation flash bank can be enabled by writing 0xA to the EDACEN field of the flash module's Error Correction Control Register 1 (FEDACCTRL1). See [Chapter 5](#) for more details.

When the CPU detects an ECC single-, or double-bit error on a read from the flash memory, it signals this on a dedicated “Event” bus. This event bus signaling is also not enabled by default and must be enabled by the application. The following code example can be used to enable the CPU event signaling:

```
MRC p15,#0,r1,c9,c12,#0      ;Enabling Event monitor states
ORR r1, r1, #0x00000010
MCR p15,#0,r1,c9,c12,#0      ;Set 4th bit ('X') of PMNC register
MRC p15,#0,r1,c9,c12,#0
```

The digital logic that interfaces the ARM Cortex-R4F CPU to the flash banks captures the ECC error events signaled by the CPU, and in turn generates error signals that are input to the central Error Signaling Module (ESM).

2.2.4 On-Chip SRAM

Several SRAM modules are implemented on the device to support the functionality of the modules included.

Reads from the CPU data RAM are protected by ECC calculated inside the CPU. Reads from all other memories are protected by configurable odd or even parity that is evaluated in parallel with the actual read.

The RM46x microcontrollers are targeted towards safety-critical applications, and it is critical for any failures in the on-chip SRAM modules to be identified before these modules are used for safety-critical functions. These microcontrollers support a Programmable Built-In Self-Test (PBIST) mechanism that is used to test each on-chip SRAM module for faults. The PBIST is usually run on device start-up as it is a destructive test and all contents of the tested SRAM module are overwritten during the test.

The microcontrollers also support a hardware-based auto-initialization of on-chip SRAM modules. This process also takes into account the read protection scheme implemented for each SRAM module – ECC or parity.

TI recommends that the PBIST routines be executed on the SRAM modules prior to the auto-initialization. The following section describe these two processes.

2.2.4.1 PBIST RAM Grouping and Algorithm Mapping For On-Chip SRAM Modules

[Table 2-5](#) shows the groupings of the various on-chip memories for PBIST. It also lists the memory types and their assigned RAM Group Select (RGS) and Return Data Select (RDS). See [Chapter 7](#) for more details on the usage of the RGS and RDS information.

Table 2-5. PBIST Memory Grouping

Memory	RAM Group #	Memory Type	RGS	RDS
PBIST_ROM	1	ROM	1	0
STC_ROM	2	ROM	2	0
DCAN1	3	Dual-port	3	0 .. 5
DCAN2	4	Dual-port	4	0 .. 5
DCAN3	5	Dual-port	5	0 .. 5
ESRAM1	6	Single-port	6	0/1 .. 4
MIBSPI1	7	Dual-port	7	0 .. 3
MIBSPI3	8	Dual-port	7	4 .. 7
MIBSPI5	9	Dual-port	7	8 .. 11
VIM	10	Dual-port	8	0 .. 1
MIBADC1	11	Dual-port	9	0
DMA	12	Dual-port	10	0 .. 5
N2HET1	13	Dual-port	11	0 .. 11
HET TU1	14	Dual-port	12	0 .. 5
MIBADC2	18	Dual-port	15	0
N2HET2	19	Dual-port	16	0 .. 11
HET TU2	20	Dual-port	17	0 .. 5
ESRAM5	21	Single-port	18	0/1 .. 4
ESRAM6	22	Single-port	19	0/1 .. 4
ETHERNET	23	Dual-port	20	0 .. 6
	24	Dual-port		
	25	Single-port		
USB	26	Dual-port	21	0 .. 2
	27	Single-port		

Table 2-6 maps the different algorithms supported in application mode for the RAM groups. The table also lists the background pattern options available for each algorithm.

Table 2-6. PBIST Algorithm Mapping

No.	ALGO Register Value	Algorithm	Memories Tested	Available Background Patterns	Valid RAM Groups	Valid RINFO Register Value
1	0x00000001	triple_read_slow_read	ROM		1,2	0x00000003
2	0x00000002	triple_read_fast_read	ROM		1,2	0x00000003
3	0x00000004	march13n	Dual-port	0x00000000, 0x96699669, 0x0F0F0F0F, 0xAA55AA55, 0xC3C3C3C3	3,4,5,7,8,9,10,11,12,13,14, 18,19,20,23,24,26	0x02CE3FDC
4	0x00000008	march13n	Single-port	0x00000000, 0x96699669, 0x0F0F0F0F, 0xAA55AA55, 0xC3C3C3C3	6,21,22,25,27	0x05300020
5	0x00000010	down1A_red	Dual-port	0xFFFFFFFF, 0xAAAAAAAA	3,4,5,7,8,9,10,11,12,13,14, 18,19,20,23,24,26	0x02CE3FDC
6	0x00000020	down1A_red	Single-port	0xFFFFFFFF, 0xAAAAAAAA	6,21,22,25,27	0x05300020
7	0x00000040	mapcolumn	Dual-port	0xFFFFFFFF, 0x00000000	3,4,5,7,8,9,10,11,12,13,14, 18,19,20,23,24,26	0x02CE3FDC
8	0x00000080	mapcolumn	Single-port	0xFFFFFFFF, 0x00000000	6,21,22,25,27	0x05300020
9	0x00000100	precharge	Dual-port	0xFFFFFFFF, 0x00000000	3,4,5,7,8,9,10,11,12,13,14, 18,19,20,23,24,26	0x02CE3FDC
10	0x00000200	precharge	Single-port	0xFFFFFFFF, 0x00000000	6,21,22,25,27	0x05300020
11	0x00000400	dtxn2	Dual-port	0xFFFFFFFF, 0x00000000	3,4,5,7,8,9,10,11,12,13,14, 18,19,20,23,24,26	0x02CE3FDC
12	0x00000800	dtxn2	Single-port	0xFFFFFFFF, 0x00000000	6,21,22,25,27	0x05300020
13	0x00001000	pmos_open	Dual-port	0xFFFFFFFF, 0x00000000	3,4,5,7,8,9,11,14,18,20,24, 26	0x028A25DC
14	0x00002000	pmos_open	Single-port	0xFFFFFFFF, 0x00000000	6,21,22,25,27	0x05300020
15	0x00004000	pmos_open_slice1	Dual-port	0xFFFFFFFF, 0x00000000	10,12,13,19,23,26	0x00441A00
16	0x00008000	pmos_open_slice2	Dual-port	0xFFFFFFFF, 0x00000000	10,12,13,19,23,26	0x00441A00
17	0x00010000	flip10	Dual-port	0xFFFFFFFF	3,4,5,7,8,9,10,11,12,13,14, 18,19,20,23,24,26	0x02CE3FDC
18	0x00020000	flip10	Single-port	0xFFFFFFFF	6,21,22,25,27	0x05300020
19	0x00040000	iddq	Dual-port	0x00000000	3,4,5,7,8,9,10,11,12,13,14, 18,19,20,23,24,26	0x02CE3FDC
20	0x00080000	iddq	Single-port	0x00000000	6,21,22,25,27	0x05300020
21	0x00100000	retention	Dual-port	0x00000000	3,4,5,7,8,9,10,11,12,13,14, 18,19,20,23,24,26	0x02CE3FDC
22	0x00200000	retention	Single-port	0x00000000	6,21,22,25,27	0x05300020
23	0x00400000	iddq	Dual-port	0xFFFFFFFF	3,4,5,7,8,9,10,11,12,13,14, 18,19,20,23,24,26	0x02CE3FDC
24	0x00800000	iddq	Single-port	0xFFFFFFFF	6,21,22,25,27	0x05300020
25	0x01000000	retention	Dual-port	0xFFFFFFFF	3,4,5,7,8,9,10,11,12,13,14, 18,19,20,23,24,26	0x02CE3FDC
26	0x02000000	retention	Single-port	0xFFFFFFFF	6,21,22,25,27	0x05300020

Table 2-6. PBIST Algorithm Mapping (continued)

No.	ALGO Register Value	Algorithm	Memories Tested	Available Background Patterns	Valid RAM Groups	Valid RINFO Register Value
27	0x04000000	iddqrowstripe	Dual-port	0x00000000	3,4,5,7,8,9,10,11,12,13,14,18,19,20,23,24,26	0x02CE3FDC
28	0x08000000	iddqrowstripe	Single-port	0x00000000	6,21,22,25,27	0x05300020
29	0x10000000	iddqrowstripe	Dual-port	0xFFFFFFFF	3,4,5,7,8,9,10,11,12,13,14,18,19,20,23,24,26	0x02CE3FDC
30	0x20000000	iddqrowstripe	Single-port	0xFFFFFFFF	6,21,22,25,27	0x05300020
31	0x40000000	powerup_invpowerup	Dual-port	0xAAAAAAAA	33,34,35,36,37,38,39,40,41,42,43,45,46,47,48,49,50,51	0x0007F7FF
32	0x80000000	powerup_invpowerup	Single-port	0xAAAAAAAA	52,53,54,56,57,58	0x03B80000

NOTE: Recommended Memory Test Algorithm

March13 is the most recommended algorithm for the memory self-test.

For HCLK = 180 MHz, VCLK = 90 MHz, PBIST ROM_CLK = 90 MHz, the March13 algorithm takes 14.02 ms to run on all on-chip SRAMs.

2.2.4.2 Auto-Initialization of On-Chip SRAM Modules

The device system provides the capability to perform a hardware initialization on most memories on the system bus and on the peripheral bus.

The intent of having the hardware initialization is to program the memory arrays with error detection capability to a known state based on their error detection scheme – odd/even parity or ECC. For example, the contents of the CPU data RAM after power-on reset is unknown. A hardware auto-initialization can be started so that there is no ECC error.

NOTE: Effect of ECC or Parity on Memory Auto-Initialization

The ECC or parity should be enabled on the RAMs before hardware auto-initialization starts if parity or ECC is being used.

Auto-Initialization Sequence:

1. Enable the global hardware memory initialization key by programming 0xA into MINITGCR[3:0], the Memory Initialization Key field (MINITGENA) of the Memory Hardware Initialization Global Control Register (MINITGCR) register.
2. Select the module on which the memory hardware initialization has to be performed by programming the appropriate value into the MSINENA(31:0) bits in the MSINENA register. See [Table 2-7](#).
3. If the global auto-initialization scheme is enabled, the corresponding module will initialize its memories based on its memory error checking scheme (even parity or odd parity or ECC).
4. When the memory initialization is complete, the module will signal “memory initialization done”, which sets the corresponding bit in the system module MIDONE field of the MINISTAT register to indicate the completion of its memory initialization.
5. When the memory hardware initialization completes for all modules, (indicated by each module’s MIDONE bit being set), the memory hardware initialization done bit (MINIDONE) is set in the MSTCGSTAT register.

Figure 2-3. Hardware Memory Initialization Protocol

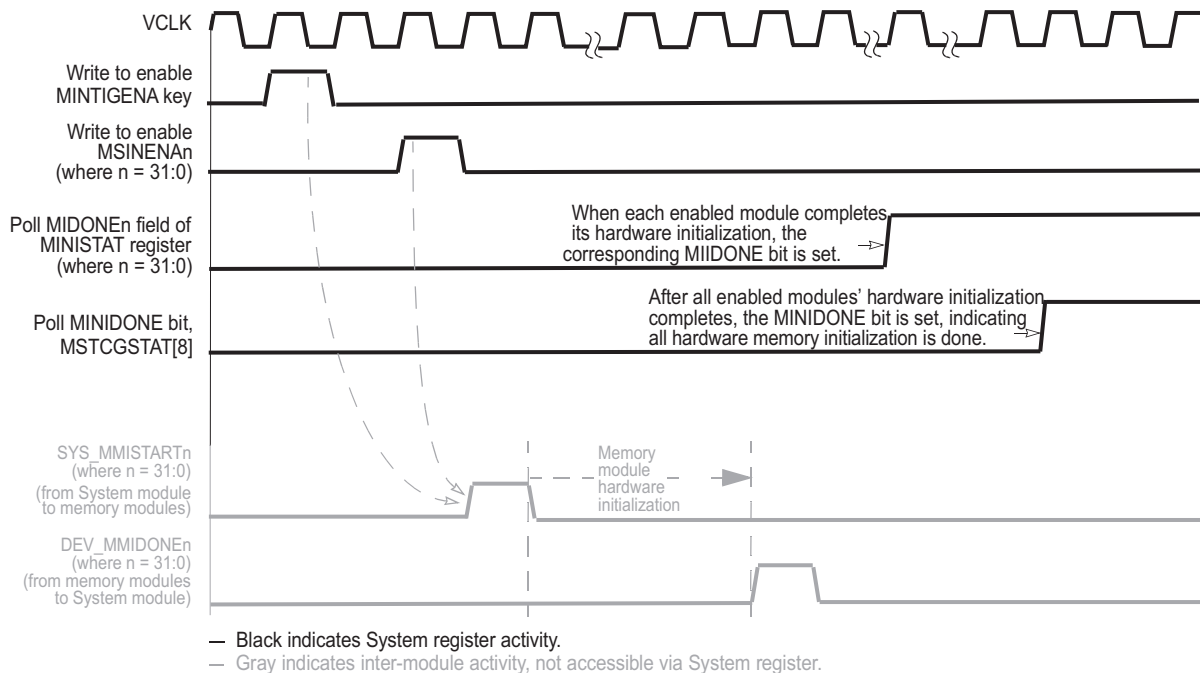


Table 2-7. Memory Initialization Select Mapping ⁽¹⁾⁽²⁾⁽³⁾

Memory	Address Range		MSINENA Register Bit #
	Start	End	
RAM (PD#1)	0x08000000	0x08013FFF	0
RAM (RAM_PD#1)	0x08010000	0x0801FFFF	0
RAM (RAM_PD#2)	0x08020000	0x0802FFFF	0
MIBSPI5 RAM	0xFF0A0000	0xFF0BFFFF	12 ⁽⁴⁾
MIBSPI3 RAM	0xFF0C0000	0xFF0DFFFF	11 ⁽⁴⁾
MIBSPI1 RAM	0xFF0E0000	0xFF0FFFFFFF	7 ⁽⁴⁾
DCAN3 RAM	0xFF1A0000	0xFF1BFFFF	10
DCAN2 RAM	0xFF1C0000	0xFF1DFFFF	6
DCAN1 RAM	0xFF1E0000	0xFF1FFFFFFF	5
MIBADC2 RAM	0xFF3A0000	0xFF3BFFFF	14
MIBADC1 RAM	0xFF3E0000	0xFF3FFFFFFF	8
N2HET2 RAM	0xFF440000	0xFF45FFFF	15
N2HET1 RAM	0xFF460000	0xFF47FFFF	3
HET TU2 RAM	0xFF4C0000	0xFF4DFFFF	16
HET TU1 RAM	0xFF4E0000	0xFF4FFFFFFF	4
DMA RAM	0xFFF80000	0xFFF80FFF	1
VIM RAM	0xFFF82000	0xFFF82FFF	2
USB Device RAM	RAM is not CPU-Addressable		N/A
Ethernet RAM (CPPI Memory Slave)	0xFC520000	0xFC521FFF	N/A

- ⁽¹⁾ If ECC protection is enabled for the CPU data RAM, then the auto-initialization process also initializes the corresponding ECC space.
- ⁽²⁾ If parity protection is enabled for the peripheral SRAM modules, then the parity bits will also be initialized along with the SRAM modules.
- ⁽³⁾ The CPU RAM interface module has separate control bit-fields to select the memory power domain that is to be initialized.
- ⁽⁴⁾ The MibSPIx modules perform an initialization of the transmit and receive RAMs as soon as the multi-buffered mode is enabled. This is independent of whether the application has already initialized these RAMs using the auto-initialization method or not. The MibSPIx modules need to be released from reset by writing 1 to their SPIGCR0 registers before starting auto-initialization on their respective RAMs.

2.3 Exceptions

An “Exception” is an event that makes the processor temporarily halt the normal flow of program execution, for example, to service an interrupt from a peripheral. Before attempting to handle an exception, the processor preserves the critical parts of the current processor state so that the original program can resume when the handler routine has finished.

The following sections describe three exceptions – Reset, Abort and the System Software Interrupts.

For complete details on all exceptions, refer to the [ARM® Cortex®-R4F Technical Reference Manual](#).

2.3.1 Resets

The RM46x microcontroller can be reset by either of the conditions described in [Table 2-8](#). Each reset condition is indicated in the System Exception Status Register (SYSESR).

The device nRST terminal is an I/O. It can be driven low by an external circuit to force a warm reset on the microcontroller. This terminal will be driven low as an output for a minimum of 32 peripheral clock (VCLK) cycles for any device system reset condition. As a result the EXTRST bit in the SYSESR register, SYSESR[3], gets set for all reset conditions listed in [Table 2-8](#). The nRST is driven low as an output for a longer duration during device power-up or whenever the power-on reset (nPORRST) is driven low externally. Refer the device datasheet for the electrical and timing specifications for the nRST.

Table 2-8. Causes of Resets

Condition	Description
Driving nPORRST pin low externally	Cold reset, or power-on reset. This reset signal is typically driven by an external voltage supervisor. This reset is flagged by the PORST bit in the SYSESR register, SYSESR[15].
Voltage Monitor reset	The microcontroller has an embedded voltage monitor that generates a power-on reset when the core voltage gets out of a valid range, or when the I/O voltage falls below a threshold. This reset is also flagged by the PORST bit in the SYSESR register, SYSESR[15]. Note: The voltage monitor is not an alternative for an external voltage supervisor.
Driving nRST pin low externally	Warm reset. This reset input is typically used in a system with multiple ICs and which requires that the microcontroller also gets reset whenever the other IC detects a fault condition. This reset is flagged by the EXTRST bit in the SYSESR, register SYSESR[3].
Oscillator failure	This reset is generated by the system module when the clock monitor detects an oscillator fail condition. Whether or not a reset is generated is also dictated by a register in the system module. This reset is flagged by the OSCRST bit in the SYSESR register, SYSESR[14].
Software reset	This reset is generated by the application software writing a 1 to bit 15 of System Exception Control Register (SYSECR) or a 0 to bit 14 of SYSECR. It is typically used by a bootloader type of code that uses a software reset to allow the code execution to branch to the application code once it is programmed into the program memory. This reset is flagged by the SWRST bit in the SYSESR register, SYSESR[4].
CPU reset	This reset is generated by the CPU self-test controller (LBIST) or by changing the memory protection (MMU/MPU) configuration in the CPURSTCR register. This reset is flagged by the CPURST bit in the SYSESR register, SYSESR[5].
Debug reset	The debug logic implemented on the microcontroller allows a system reset to be generated. This reset is flagged by the WDRST bit in the SYSESR register, SYSESR[13].
Watchdog reset	This reset is generated by the digital windowed watchdog (DWWD) module on the microcontroller. The DWWD can generate a reset whenever the watchdog service window is violated. This reset is flagged by the WDRST bit in the SYSESR register, SYSESR[13].

2.3.2 Aborts

When the ARM Cortex-R4F processor's memory system cannot complete a memory access successfully, an abort is generated. An error occurring on an instruction fetch generates a **prefetch abort**. Errors occurring on data accesses generate **data aborts**. Aborts are also categorized as being either **precise** or **imprecise**.

2.3.2.1 Prefetch Aborts

When a Prefetch Abort (PABT) occurs, the processor marks the prefetched instruction as invalid, but does not take the exception until the instruction is to be executed. If the instruction is not executed, for example because a branch occurs while it is in the pipeline, the abort does not take place.

All prefetch aborts are precise aborts.

2.3.2.2 Data Aborts

An error occurring on a data memory access can generate a data abort. If the instruction generating the memory access is not executed, for example, because it fails its condition codes, or is interrupted, the data abort does not take place.

A Data Abort (DABT) can be either precise or imprecise, depending on the type of fault that caused it.

2.3.2.3 Precise Aborts

A precise abort, also known as a synchronous abort, is one for which the exception is guaranteed to be taken on the instruction that generated the aborting memory access. The abort handler can use the value in the Link Register (r14_abt) to determine which instruction generated the abort, and the value in the Saved Program Status Register (SPSR_abt) to determine the state of the processor when the abort occurred.

2.3.2.4 Imprecise Aborts

An imprecise abort, also known as an asynchronous abort, is one for which the exception is taken on a later instruction to the instruction that generated the aborting memory access. The abort handler cannot determine which instruction generated the abort, or the state of the processor when the abort occurred. Therefore, imprecise aborts are normally fatal.

Imprecise aborts can be generated by store instructions to normal-type or device-type memory. When the store instruction is committed, the data is normally written into a buffer that holds the data until the memory system has sufficient bandwidth to perform the write access. This gives read accesses higher priority. The write data can be held in the buffer for a long period, during which many other instructions can complete. If an error occurs when the write is finally performed, this generates an imprecise abort.

The RM46x microcontroller architecture applies techniques at the system level to mitigate the impact of imprecise aborts. System level adoption of write status sidebands to the data path allow bus masters to comprehend imprecise aborts, turning them into precise aborts. In cases where this approach is not feasible, buffering bridges or other sources of imprecision may build a FIFO of current transactions such that an imprecise abort may be registered at the point of imprecision for later analysis.

Masking Of Imprecise Aborts:

The nature of imprecise aborts means that they can occur while the processor is handling a different abort. If an imprecise abort generates a new exception in such a situation, the banked link register (R14_abt) and the Saved Processor Status Register (SPSR_abt) values are overwritten. If this occurs before the data is pushed to the stack in memory, the state information about the first abort is lost. To prevent this from happening, the Current Processor Status Register (CPSR) contains a mask bit to indicate that an imprecise abort cannot be accepted, the A-bit. When the A-bit is set, any imprecise abort that occurs is held pending by the processor until the A-bit is cleared, when the exception is actually taken. The A-bit is automatically set when abort, IRQ or FIQ exceptions are taken, and on reset. The application must only clear the A-bit in an abort handler after the state information has either been stacked to memory, or is no longer required.

NOTE: Default Behavior for Imprecise Aborts

The A-bit in the CPSR is set by default. This means that no imprecise abort exception will occur. The application must enable imprecise abort exception generation by clearing the A-bit of the CPSR.

2.3.2.5 Conditions That Generate Aborts

An Abort is generated under the following conditions on the RM46x microcontrollers.

- Access to an illegal address (a nonimplemented address)
- Access to a protected address (protection violation)
- Parity / ECC / Time-out Error on a valid access

Illegal Addresses:

The illegal addresses and the responses to an access to these addresses are defined in [Table 2-3](#).

Addresses Protected By MPU:

Memory access permissions can be configured via the ARM Cortex-R4F processor's Memory Protection Unit (MPU). For more details on the MPU configuration, refer to the [ARM® Cortex®-R4F Technical Reference Manual](#).

A memory access violation is logged as a permission fault in the CPU's fault status register and the virtual address of the access is logged into the CPU's fault address register.

Protection of Peripheral Register and Memory Frames:

Accesses to the peripheral register and memory frames can be protected either by configuring the MPU or by configuring the Peripheral Central Resource (PCR) controller registers.

The PCR module PPROTSETx registers contain one bit per peripheral select quadrant. These bits define the access permissions to the peripheral register frames. If the CPU attempts to write to a peripheral register for which it does not have the correct permissions, a protection violation is detected and an Abort occurs.

Some modules also enforce register updates to only be allowed when the CPU is in a privileged mode of operation. If the CPU writes to these registers in user mode, the writes are ignored.

The PCR module PMPROTSETx registers contain one bit per peripheral memory frame. These bits define the access permissions to the peripheral memory frames. If the CPU attempts to write to a peripheral memory for which it does not have the correct permissions, a protection violation is detected and an Abort occurs.

NOTE: No Access Protection for Reads

The PCR PPROTSETx and PMPROTSETx registers protect the peripheral registers and memories against illegal writes by the CPU. The CPU can read from the peripheral registers and memories in both user and privileged modes.

2.3.3 System Software Interrupts

The system module provides the capability of generating up to four software interrupts. A software interrupt is generated by writing the correct key value to either of the four System Software Interrupt Registers (SSIRx). The SSI registers also allow the application to provide a label for that software interrupt. This label is an 8-bit value that can then be used by the interrupt service routine to perform the required task based on the value provided. The source of the system software interrupt is reflected in the system software interrupt vector (SSIVEC) register.

2.4 Clocks

This section describes the clocking structure of the RM46x microcontrollers.

2.4.1 Clock Sources

The devices support up to 7 clock sources. These are shown in [Table 2-9](#). The electrical specifications as well as timing requirements for each of the clock sources are specified in the device datasheet.

Table 2-9. Clock Sources

Clock Source #	Clock Source Name	Description
0	OSCIN	Main oscillator. This is the primary clock for the microcontroller and is the only clock that is input to the phase-locked loops. The oscillator frequency must be between 5 and 20 MHz.
1	PLL1	This is the output of the main PLL. The PLL is capable of modulating its output frequency in a controlled manner to reduce the radiated emissions.
2	Reserved	This clock source is not available and must not be enabled or used as source for any clock domain.
3	EXTCLKIN1	External clock input 1. A square wave input can be applied to this device input and used as a clock source inside the device.
4	LF LPO (Low-Frequency LPO)	This is the low-frequency output of the internal reference oscillator. This is typically an 80 KHz signal which is used by the real-time interrupt module for generating periodic interrupts to wake up from a low power mode.
5	HF LPO (High-Frequency LPO)	This is the high-frequency output of the internal reference oscillator. This is typically a 10 MHz signal which is used by the clock monitor module as a reference clock to monitor the main oscillator frequency.
6	PLL2	This is the output of the second PLL. There is no option of modulating this PLL's output signal. This separate non-modulating PLL allows the generation of an asynchronous clock source that is independent of the CPU clock frequency.
7	EXTCLKIN2	External clock input 2. A square wave input can be applied to this device input and used as a clock source inside the device.

2.4.1.1 Enabling / Disabling Clock Sources

Each clock source can be independently enabled or disabled using the set of Clock Source Disable registers – CSDIS, CSDISSET and CSDISCLR.

Each bit in these registers corresponds to the clock source number indicated in [Table 2-9](#). For example, setting bit 1 in the CSDIS or CSDISSET registers disables the PLL#1.

NOTE: Disabling the Main Oscillator or HF LPO

By default, the clock monitoring circuit is enabled and checks for the main oscillator frequency to be within a certain range using the HF LPO as a reference. If the main oscillator and/or the HF LPO are disabled with the clock monitoring still enabled, the clock monitor will indicate an oscillator fault. The clock monitoring must be disabled before disabling the main oscillator or the HF LPO clock source(s).

The clock source is only disabled once there is no active clock domain that is using that clock source. Also, see [Chapter 10](#) for more information on enabling/disabling the oscillator and PLL.

On the RM46x microcontrollers, the clock sources 0, 4 and 5 are enabled by default.

2.4.1.2 Checking for Valid Clock Sources

The application can check whether a clock source is valid or not by checking the corresponding bit to be set in the Clock Source Valid Status (CSVSTAT) register. For example, the application can check if bit 1 in CSVSTAT is set before using the output of PLL#1 as the source for any clock domain.

2.4.2 Clock Domains

The clocking on this device is divided into multiple clock domains for flexibility in control as well as clock source selection. There are 10 clock domains on this device. Each of these are described in [Table 2-10](#).

Each of the control registers listed in [Table 2-10](#) are defined in [Section 2.5](#). The AC timing characteristics for each clock domain are specified in the device datasheet.

Table 2-10. Clock Domains

Clock Domain #	Clock Domain	Default Source	Source Selection Register	Special Considerations
0	GCLK	OSCIN	GHVSR	• Clock domain used by one of the two Cortex-R4F CPUs operating in lock-step • Always the same frequency as HCLK • In phase with HCLK • Is disabled separately from HCLK via the CDDISx registers bit 0 • Can be divided by 1 up to 8 when running CPU selftest (LBIST) using the CLKDIV field of the STCLKDIV register at address 0xFFFFE108
0	GCLK2	OSCIN	GHVSR	• Clock domain used by the second Cortex-R4F CPU operating in lock-step • Always the same frequency as GCLK • 2 cycles delayed from GCLK • Is disabled along with GCLK • Gets divided by the same divider setting as that for GCLK when running CPU selftest (LBIST)
1	HCLK	OSCIN	GHVSR	• Clock domain used by the high-speed system modules: Flash memory interfaces, TCRAM interface, Error Signaling Module (ESM), DMA • Is disabled via the CDDISx registers bit 1
2	VCLK	OSCIN	GHVSR	• Clock domain used by some system modules (VIM), peripheral modules accessed via the Peripheral Central Resource (PCR) controller, and all other register interfaces also accessed via the PCR • Divided down from HCLK • Can be HCLK/1, HCLK/2,... or HCLK/16 • Is disabled separately from HCLK via the CDDISx registers bit 2
3	VCLK2	OSCIN	GHVSR	• Clock domain used by the timer modules: N2HET1, N2HET2 and the dedicated transfer units: HTU1, HTU2 • Divided down from HCLK • Can be HCLK/1, HCLK/2,... or HCLK/16 • Frequency must be an integer multiple of VCLK frequency • Is disabled separately from HCLK via the CDDISx registers bit 3
8	VCLK3	OSCIN	GHVSR	• Clock domain used for EMAC, USB and EMIF slave interfaces • Divided down from HCLK • Can be HCLK/1, HCLK/2,... or HCLK/16 • Is disabled separately from HCLK via the CDDISx registers bit 8
	VCLK4	OSCIN	GHVSR	• Clock domain used for ePWM, eCAP, and eQEP modules • Divided down from HCLK • Can be HCLK/1, HCLK/2,... or HCLK/16 • Is disabled separately from HCLK via the CDDISx registers bit 9
4	VCLKA1	VCLK	VCLKASR	• Clock domain dedicated for the CAN controllers: DCAN1, DCAN2, DCAN3, used for baud-rate generation • Defaults to VCLK as the source • Frequency can be as fast as HCLK frequency • Is disabled via the CDDISx registers bit 4

Table 2-10. Clock Domains (continued)

Clock Domain #	Clock Domain	Default Source	Source Selection Register	Special Considerations
10	VCLKA3	VCLK	VCLKACON1	• Clock domain dedicated to provide the 48MHz clock to the USB host controller and device controller • Defaults to VCLK as the source • Frequency can be as fast as HCLK frequency • Is disabled via the CDDISx registers bit 10
10	VCLKA3_DIVR	VCLK	VCLKACON1	• Clock domain dedicated to provide the 12MHz clock to the USB host controller • Divided down from VCLKA3 using the VCLKA3R field of the VCLKACON1 register • Frequency can be VCLKA3/1, VCLKA3/2, ..., or VCLKA3/8 • Default frequency is VCLKA3/2 • Is disabled separately via the VCLKACON1 register's VCLKA3_DIV_CDDIS bit, if the VCLKA3 is not already disabled
11	VCLKA4	VCLK	VCLKACON1	• Clock domain dedicated for the Ethernet controller (EMAC) when the TX and RX clocks are internally generated • Defaults to VCLK as the source • Frequency can be as fast as HCLK frequency • Is disabled via the CDDISx registers bit 11
11	VCLKA4_DIVR	VCLK	VCLKACON1	• Divided down from VCLKA4 using the VCLKA4R field of the VCLKACON1 register • Frequency can be VCLKA4/1, VCLKA4/2, ..., or VCLKA4/8 • Default frequency is VCLKA4/2 • Is disabled separately via the VCLKACON1 register's VCLKA4_DIV_CDDIS bit, if the VCLKA4 is not already disabled
6	RTICKL	VCLK	RCLKSRC	• Clock domain dedicated for timebase generation in the Real-Time Interrupt (RTI) generation module • Defaults to VCLK as the source • If a clock source other than VCLK is selected for RTICKL, then the RTICKL frequency must be less than or equal to VCLK/3 • Application can ensure this by programming the RTI1DIV field of the RCLKSRC register, if necessary • Is disabled via the CDDISx registers bit 6

2.4.2.1 Enabling / Disabling Clock Domains

Each clock domain can be independently enabled or disabled using the set of Clock Domain Disable registers – CDDIS, CDDISSET, and CDDISCLR.

Each bit in these registers corresponds to the clock domain number indicated in [Table 2-10](#). For example, setting bit 1 in the CDDIS or CDDISSET registers disables the HCLK clock domain. The clock domain will be turned off only when every module that uses the HCLK domain gives the “permission” for HCLK to be turned off.

All clock domains are enabled by default, or upon a system reset, or whenever a wake up condition is detected.

2.4.2.2 Mapping Clock Sources to Clock Domains

Each clock domain needs to be mapped to a valid clock source. There are control registers that allow an application to choose the clock sources for each clock domain.

- **Selecting clock source for GCLK, HCLK and VCLKx domains**

The CPU clock (GCLK), the system module clock (HCLK), and the peripheral bus clocks (VCLKx) all use the same clock source. This clock source is selected via the GHVSRC register. The default source for the GCLK, HCLK and VCLKx is the main oscillator. That is, after power up, the GCLK and HCLK are running at the OSCIN frequency, while the VCLKx frequency is the OSCIN frequency divided by 2.

- **Selecting clock source for VCLKA1 domains**

The clock source for VCLKA1 domains is selected via the VCLKASRC register. The default source for the VCLKA1 domains is the VCLK.

- **Selecting clock source for VCLKA4 domain**

The clock source for VCLKA4 domain is selected via the VCLKACON1 register. The default source for the VCLKA4 domain is the VCLK.

- **Selecting clock source for RTICLK domain**

The clock source for RTICLK domain is selected via the RCLKSRC register. The default source for the RTICLK domain is the VCLK.

NOTE: **Selecting a clock source for RTICLK that is not VCLK**

When the application chooses a clock source for RTICLK domain that is not VCLK, then the application must ensure that the resulting RTICLK frequency must be less than or equal to VCLK frequency divided by 3. The application can configure the RTI1DIV field of the RCLKSRC register for dividing the selected clock source frequency by 1, 2, 4 or 8 to meet this requirement.

2.4.3 Low Power Modes

All clock domains are active in the normal operating mode. This is the default mode of operation. As described in [Section 2.4.1.1](#) and [Section 2.4.2.1](#), the application can choose to disable any particular clock source and domain that it does not plan to use. Also, the peripheral central resource controller (PCR) has control registers to enable / disable the peripheral clock (VCLK) for each peripheral select. This offers the application a large number of choices for enabling / disabling clock sources, or clock domains, or clocks to specific peripherals.

This section describes three particular low-power modes and their typical characteristics. They are not the only low-power modes configurable by the application, as just described.

Table 2-11. Typical Low-Power Modes

Mode Name	Active Clock Source(s)	Active Clock Domain(s)	Wake Up Options	Suggested Wake Up Clock Source(s)	Wake Up Time(wake up detected -to- CPU code execution start)
Doze	Main oscillator	RTICK	RTI interrupt, GIO interrupt, CAN message, SCI message	Main oscillator	Flash pump sleep -> standby transition time + Flash bank sleep -> standby transition time + Flash bank standby -> active transition time
Snooze	LF LPO	RTICK	RTI interrupt, GIO interrupt, CAN message, SCI message	HF LPO	HF LPO warm start-up time + Flash pump sleep -> standby transition time + Flash bank sleep -> standby transition time + Flash bank standby -> active transition time
Sleep	None	None	GIO interrupt, CAN message, SCI message	HF LPO	HF LPO warm start-up time + Flash pump sleep -> standby transition time + Flash bank sleep -> standby transition time + Flash bank standby -> active transition time

2.4.3.1 Typical Software Sequence to Enter a Low-Power Mode

1. Program the flash banks and flash pump fall-back modes to be “sleep”.
The flash pump transitions from active to sleep mode only after all the flash banks have switched from active to sleep mode. The flash banks start switching from active to sleep mode only after the banks are not accessed for at least a duration defined by the Active Grace Period (AGP) parameter configured for the banks. See [Chapter 5](#) for more details.
2. Disable the clock sources that are not required to be kept active.
A clock source does not get disabled until all clock domains using that clock source are disabled first, or are configured to use an alternate clock source.
3. Disable the clock domains that are not required to be kept active.
A clock domain does not get disabled until all modules using that clock domain “give their permission” for that clock domain to be turned off.
4. Idle the Cortex-R4F core.
The ARM Cortex-R4F CPU has internal power management logic, and requires a dedicated instruction to be used in order to enter a low power mode. This is the Wait For Interrupt (WFI) instruction.
When a WFI instruction is executed, the Cortex-R4F core flushes its pipeline, flushes all write buffers, and completes all pending bus transactions. At this time the core indicates to the system that the clock to the core can be stopped. This indication is used by the Global Clock Module (GCM) to turn off the CPU clock domain (GCLK) if the CDDIS register bit 0 is set.

2.4.3.2 Special Considerations for Entry to Low Power Modes

Some bus master modules – High-End Timer Transfer Units (HTUx), and Parameter Overlay Module (POM), can have ongoing transactions when the application wants to enter a low power mode to turn off the clocks to those modules. This is not recommended as it could leave the device in an unpredictable state. Refer to the individual module user guides for more information about the sequence to be followed to safely enter a low-power mode.

2.4.3.3 Selecting Clock Source Upon Wake Up

The domains for CPU clock (GCLK), the system clock (HCLK) and the peripheral clock (VCLKx) use the same clock source selected via the GHVSRC field of the GHVSRC register. The GHVSRC register also allows the application to choose the clock source after wake up via the GHVWAKE field.

When a wake up condition is detected, if the selected wake up clock source is not already active, the global clock module (GCM) will enable this selected clock source, wait for it to become valid, and then use it for the GCLK, HCLK and VCLKx domains. The other clock domains VCLKAx and RTICLK retain the configuration for their clock source selection registers – VCLKASRC, VCLKACON1 and RCLKSRC.

2.4.4 Clock Test Mode

The RM46x microcontrollers support a test mode which allows a user to bring out several different clock sources and clock domains on to the ECLK terminal. This is very useful information for debug purposes. Each clock source also has a corresponding clock source valid status flag in the Clock Source Valid Status (CSVSTAT) register. The clock source valid status flags can also be brought out on to the N2HET1[12] terminal in this clock test mode.

The clock test mode is controlled by the CLKTEST register ([Section 2.5.1.32](#)) in the system module register frame.

The clock test mode is enabled by writing 5h to the CLK_TEST_EN field.

The signal to be brought out on to the ECLK terminal is defined by the SEL_ECP_PIN field and the signal to be brought out on to the N2HET1[12] terminal is defined by the SEL_GPIO_PIN field.

The choices for these selections are defined in [Table 2-12](#).

Table 2-12. Clock Test Mode Options

SEL_ECP_PIN	Signal on ECLK	SEL_GPIO_PIN	Signal on N2HET1[12]
0000	Oscillator	0000	Oscillator Valid status
0001	PLL1 free-running clock output	0001	PLL1 Valid status
0010	Reserved	0010	Reserved
0011	EXTCLKIN1	0011	Reserved
0100	Low-frequency LPO (Low-Power Oscillator) clock	0100	Reserved
0101	High-frequency LPO (Low-Power Oscillator) clock	0101	HF LPO Valid status
0110	PLL2 free-running clock output	0110	PLL2 Valid Status
0111	EXTCLKIN2	0111	Reserved
1000	GCLK	1000	LF LPO Valid status
1001	RTI Base	1001	Oscillator Valid status
1010	Reserved	1010	Oscillator Valid status
1011	VCLKA1	1011	Oscillator Valid status
1100	Reserved	1100	Oscillator Valid status
1101	VCLKA3_DIVR	1101	VCLKA3
1110	VCLKA4_DIVR	1110	VCLKA4
1111	Reserved	1111	Oscillator Valid status

2.4.5 Safety Considerations for Clocks

The RM46x microcontrollers are targeted for use in several safety-critical applications. The following sections describe the internal or external monitoring mechanisms that detect and signal clock source failures.

2.4.5.1 Oscillator Monitor

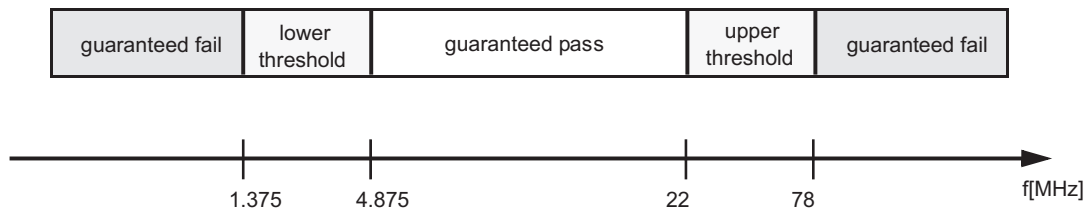
The oscillator clock frequency is monitored by a dedicated circuitry called CLKDET using the HF LPO as the reference clock. The CLKDET flags an oscillator fail condition whenever the OSCIN frequency falls outside of a range which is defined by the HF LPO frequency.

The valid OSCIN range is defined as a minimum of $f_{(\text{HF LPO})} / 4$ to a maximum of $f_{(\text{HF LPO})} \times 4$.

The untrimmed HF LPO frequency on this device can range from 5.5MHz to 19.5MHz. This results in a valid OSCIN frequency range depicted in [Figure 2-4](#).

The application can select the device response to an oscillator fail indication. See [Chapter 10](#) for more details on the oscillator monitoring and the system response choices.

Figure 2-4. LPO and Clock Detection, Untrimmed HF LPO



2.4.5.2 PLL Slip Detector

Both the PLL macros implemented on the microcontrollers have an embedded slip detection circuit. A PLL slip is detected by the slip detector under the following conditions:

1. Reference cycle slip, RFSLIP — the output clock is running *too fast* relative to the reference clock
2. Feedback cycle slip, FBSLIP — the output clock is running *too slow* relative to the reference clock

The device also includes optional filters that can be enabled before a slip indication from the PLL is actually logged in the system module Global Status Register (GLBSTAT). Also, once a PLL slip condition is logged in the system module global status register, the application can choose the device's response to the slip indication. Refer to the Oscillator and PLL chapter for more details on PLL slip and the system response choices.

2.4.5.3 External Clock Monitor

The microcontrollers support a terminal called ECLK – External Clock, which is used to output a slow frequency which is divided down from the device system clock frequency. An external circuit can monitor the ECLK frequency in order to check that the device is operating at the correct frequency.

The frequency of the signal output on the ECLK pin can be divided down by 1 to 65536 from the peripheral clock (VCLK) frequency using the External Clock Prescaler Control Register (ECPCNTL). The actual clock output on ECLK is enabled by setting the ECP CLK FUN bit of the SYSPC1 control register.

2.4.5.4 Dual-Clock Comparators

The microcontrollers include two instances of the dual-clock comparator module. This module includes two down counters which independently count from two separate seed values at the rate of two independent clock frequencies. One of the clock inputs is a reference clock input, selectable between the main oscillator or the HF LPO in functional mode. The second clock input is selectable from among a set of defined signals as described in [Section 2.4.5.4.1](#) and [Section 2.4.5.4.2](#). This mechanism can be used to use a known-good clock to measure the frequency of another clock.

2.4.5.4.1 DCC1

Table 2-13. DCC1 Counter 0 Clock Inputs

Clock Source [3-0]	Clock / Signal Name
others	oscillator (OSCIN)
0x5	HF LPO
0xA	test clock (TCK)

Table 2-14. DCC1 Counter 1 Clock / Signal Inputs

Key [3-0]	Clock Source [3-0]	Clock / Signal Name
0xA	0x0	PLL1 free-running clock output
	0x1	PLL2 free-running clock output
	0x2	LF LPO
	0x3	HF LPO
	0x4	Reserved
	0x5	EXTCLKIN1
	0x6	EXTCLKIN2
	0x7	Reserved
	0x8-0xF	VCLK
others	any value	N2HET1[31]

As can be seen, the main oscillator (OSCIN) can be used for counter 0 as a “known-good” reference clock. The clock for counter 1 can be selected from among 8 options. See [Chapter 11](#) for more details on the DCC usage.

2.4.5.4.2 DCC2

Table 2-15. DCC2 Counter 0 Clock Inputs

Clock Source [3-0]	Clock / Signal Name
others	oscillator (OSCIN)
0xA	test clock (TCK)

Table 2-16. DCC2 Counter 1 Clock / Signal Inputs

Key [3-0]	Clock Source [3-0]	Clock / Signal Name
0xA	0x0-0x7	Reserved
	0x8-0xF	VCLK
others	any value	N2HET2[0]

As can be seen, the main oscillator (OSCIN) can be used for counter 0 as a “known-good” reference clock. The clock for counter 1 can be selected from among 2 options. See [Chapter 11](#) for more details on the DCC usage.

2.5 System and Peripheral Control Registers

The following sections describe the system and peripheral control registers of the RM46x microcontroller.

2.5.1 Primary System Control Registers (SYS)

This section describes the SYSTEM registers. These registers are divided into two separate frames. The start address of the primary system module frame is FFFF FF00h. The start address of the secondary system module frame is FFFF E100h. The registers support 32-, 16-, and 8-bit writes. The offset is relative to the system module frame start address.

Table 2-17 contains a list of the primary system control registers.

Table 2-17. Primary System Control Registers

Offset	Acronym	Register Description	Section
00h	SYSPC1	SYS Pin Control Register 1	Section 2.5.1.1
04h	SYSPC2	SYS Pin Control Register 2	Section 2.5.1.2
08h	SYSPC3	SYS Pin Control Register 3	Section 2.5.1.3
0Ch	SYSPC4	SYS Pin Control Register 4	Section 2.5.1.4
10h	SYSPC5	SYS Pin Control Register 5	Section 2.5.1.5
14h	SYSPC6	SYS Pin Control Register 6	Section 2.5.1.6
18h	SYSPC7	SYS Pin Control Register 7	Section 2.5.1.7
1Ch	SYSPC8	SYS Pin Control Register 8	Section 2.5.1.8
20h	SYSPC9	SYS Pin Control Register 9	Section 2.5.1.9
30h	CSDIS	Clock Source Disable Register	Section 2.5.1.10
34h	CSDISSET	Clock Source Disable Set Register	Section 2.5.1.11
38h	CSDISCLR	Clock Source Disable Clear Register	Section 2.5.1.12
3Ch	CDDIS	Clock Domain Disable Register	Section 2.5.1.13
40h	CDDISSET	Clock Domain Disable Set Register	Section 2.5.1.14
44h	CDDISCLR	Clock Domain Disable Clear Register	Section 2.5.1.15
48h	GHVSR	GCLK, HCLK, VCLK, and VCLK2 Source Register	Section 2.5.1.16
4Ch	VCLKASRC	Peripheral Asynchronous Clock Source Register	Section 2.5.1.17
50h	RCLKSRC	RTI Clock Source Register	Section 2.5.1.18
54h	CSVSTAT	Clock Source Valid Status Register	Section 2.5.1.19
58h	MSTGCR	Memory Self-Test Global Control Register	Section 2.5.1.20
5Ch	MINITGCR	Memory Hardware Initialization Global Control Register	Section 2.5.1.21
60h	MSINENA	Memory Self-Test/Initialization Enable Register	Section 2.5.1.22
64h	MSTFAIL	Memory Self-Test Fail Status Register	Section 2.5.1.23
68h	MSTCGSTAT	MSTC Global Status Register	Section 2.5.1.24
6Ch	MINISTAT	Memory Hardware Initialization Status Register	Section 2.5.1.25
70h	PLLCTL1	PLL Control Register 1	Section 2.5.1.26
74h	PLLCTL2	PLL Control Register 2	Section 2.5.1.27
78h	SYSPC10	SYS Pin Control Register 10	Section 2.5.1.28
7Ch	DIEIDL	Die Identification Register, Lower Word	Section 2.5.1.29
80h	DIEIDH	Die Identification Register, Upper Word	Section 2.5.1.30
88h	LPOMONCTL	LPO/Clock Monitor Control Register	Section 2.5.1.31
8Ch	CLKTEST	Clock Test Register	Section 2.5.1.32
90h	DFTCTRLREG	DFT Control Register	Section 2.5.1.33
94h	DFTCTRLREG2	DFT Control Register 2	Section 2.5.1.34
98h-9Ch	Reserved	Reserved	
A0h	GPREG1	General Purpose Register	Section 2.5.1.35
A8h	IMPFSTS	Imprecise Fault Status Register	Section 2.5.1.36

Table 2-17. Primary System Control Registers (continued)

Offset	Acronym	Register Description	Section
ACh	IMPFTADD	Imprecise Fault Write Address Register	Section 2.5.1.37
B0h	SSIR1	System Software Interrupt Request 1 Register	Section 2.5.1.38
B4h	SSIR2	System Software Interrupt Request 2 Register	Section 2.5.1.39
B8h	SSIR3	System Software Interrupt Request 3 Register	Section 2.5.1.40
BCh	SSIR4	System Software Interrupt Request 4 Register	Section 2.5.1.41
C0h	RAMGCR	RAM Control Register	Section 2.5.1.42
C4h	BMMCR1	Bus Matrix Module Control Register 1	Section 2.5.1.43
C8h	Reserved	Reserved	
CCh	CPURSTCR	CPU Reset Control Register	Section 2.5.1.44
D0h	CLKCNTL	Clock Control Register	Section 2.5.1.45
D4h	ECPCNTL	ECP Control Register	Section 2.5.1.46
DCh	DEVCR1	DEV Parity Control Register 1	Section 2.5.1.47
E0h	SYSECR	System Exception Control Register	Section 2.5.1.48
E4h	SYSESR	System Exception Status Register	Section 2.5.1.49
E8h	SYSTASR	System Test Abort Status Register	Section 2.5.1.50
ECh	GLBSTAT	Global Status Register	Section 2.5.1.51
F0h	DEVID	Device Identification Register	Section 2.5.1.52
F4h	SSIVEC	Software Interrupt Vector Register	Section 2.5.1.53
F8h	SSIF	System Software Interrupt Flag Register	Section 2.5.1.54

2.5.1.1 SYS Pin Control Register 1 (SYSPC1)

The SYSPC1 register, shown in [Figure 2-5](#) and described in [Table 2-18](#), controls the function of the ECLK pin.

Figure 2-5. SYS Pin Control Register 1 (SYSPC1) [offset = 00]

31	Reserved																16	
R-0																		
15	Reserved																1	0
R-0																	ECPCCLKFUN	R/W-0

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 2-18. SYS Pin Control Register 1 (SYSPC1) Field Descriptions

Bit	Field	Value	Description
31-1	Reserved	0	Reads return 0. Writes have no effect.
0	ECPCCLKFUN	0	ECLK function. This bit changes the function of the ECLK pin.
		0	ECLK is in GIO mode.
		1	ECLK is in functional mode as a clock output.
			Note: Proper ECLK duty cycle is not assured until 1 ECLK cycle has elapsed after switching into functional mode.

2.5.1.2 SYS Pin Control Register 2 (SYSPC2)

The SYSPC2 register, shown in [Figure 2-6](#) and described in [Table 2-19](#), controls whether the pin is an input or an output when in GIO mode.

Figure 2-6. SYS Pin Control Register 2 (SYSPC2) [offset = 04h]

31	Reserved															16
R-0																
15	Reserved													1	0	
R-0														ECPLKDIR		
R-0														R/W-0		

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 2-19. SYS Pin Control Register 2 (SYSPC2) Field Descriptions

Bit	Field	Value	Description
31-1	Reserved	0	Reads return 0. Writes have no effect.
0	ECPCCLKDIR	0	ECLK data direction. This bit controls the direction of the ECLK pin when it is configured to be in GIO mode only.
		0	The ECLK pin is an input.
			Note: If the pin direction is set as an input, the output buffer is tristated.
		1	The ECLK pin is an output.
			Note: The ECLK pin is placed into GIO mode by clearing the ECPCCLKFUN bit to 0 in the SYSPC1 register.

2.5.1.3 SYS Pin Control Register 3 (SYSPC3)

The SYSPC3 register, shown in [Figure 2-7](#) and described in [Table 2-20](#), displays the logic state of the ECLK pin when it is in GIO mode.

Figure 2-7. SYS Pin Control Register 3 (SYSPC3) [offset = 08h]

31	Reserved														16
R-0															
15	Reserved										1	0			
R-0											ECPCLKDIN				
R-0											R-U				

LEGEND: R = Read only; -n = value after reset; -U = Undefined

Table 2-20. SYS Pin Control Register 3 (SYSPC3) Field Descriptions

Bit	Field	Value	Description
31-1	Reserved	0	Reads return 0. Writes have no effect.
0	ECPCLKDIN	0	ECLK data in. This bit displays the logic state of the ECLK pin when it is configured to be in GIO mode.
		0	The ECLK pin is at logic low (0).
		1	The ECLK pin is at logic high (1).

2.5.1.4 SYS Pin Control Register 4 (SYSPC4)

The SYSPC4 register, shown in [Figure 2-8](#) and described in [Table 2-21](#), controls the logic level output function of the ECLK pin when it is configured as an output in GIO mode.

Figure 2-8. SYS Pin Control Register 4 (SYSPC4) [offset = 0Ch]

31	Reserved														16
R-0															
15	Reserved										1	0			
R-0											ECPCCLKDOUT				
R-0											R/W-0				

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 2-21. SYS Pin Control Register 4 (SYSPC4) Field Descriptions

Bit	Field	Value	Description
31-1	Reserved	0	Reads return 0. Writes have no effect.
0	ECPCLKDOUT	0	ECLK data out write. This bit is only active when the ECLK pin is configured to be in GIO mode. Writes to this bit will only take effect when the ECLK pin is configured as an output in GIO mode. The current logic state of the ECLK pin will be displayed by this bit in both input and output GIO mode.
		0	The ECLK pin is driven to logic low (0).
		1	The ECLK pin is driven to logic high (1).
			Note: The ECLK pin is placed into GIO mode by setting the ECPCLKFUN bit to 0 in the SYSPC1 register. The ECLK pin is placed in output mode by setting the ECPCLKDIR bit to 1 in the SYSPC2 register.

2.5.1.5 SYS Pin Control Register 5 (SYSPC5)

The SYSPC5 register, shown in [Figure 2-9](#) and described in [Table 2-22](#), controls the set function of the ECLK pin when it is configured as an output in GIO mode.

Figure 2-9. SYS Pin Control Register 5 (SYSPC5) [offset = 10h]

31	Reserved															16
R-0																
15	Reserved										1	0				
R-0											ECPLKSET		R/W-0			

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 2-22. SYS Pin Control Register 5 (SYSPC5) Field Descriptions

Bit	Field	Value	Description
31-1	Reserved	0	Reads return 0. Writes have no effect.
0	ECPLKSET	0	ECLK data out set. This bit drives the output of the ECLK pin high when set in GIO output mode. <i>Write:</i> Writing a 0 has no effect.
		1	<i>Write:</i> The ECLK pin is driven to logic high (1). Note: The current logic state of the ECPLKDOUT bit will also be displayed by this bit when the pin is configured in GIO output mode. Note: The ECLK pin is placed into GIO mode by setting the ECPLKFUN bit to 0 in the SYSPC1 register. The ECLK pin is placed in output mode by setting the ECPLKDIR bit to 1 in the SYSPC2 register.

2.5.1.6 SYS Pin Control Register 6 (SYSPC6)

The SYSPC6 register, shown in [Figure 2-10](#) and described in [Table 2-23](#), controls the clear function of the ECLK pin when it is configured as an output in GIO mode..

Figure 2-10. SYS Pin Control Register 6 (SYSPC6) [offset = 14h]

31	Reserved															16
R-0																
15	Reserved										1	0				
R-0											ECPLKCLR		R/W-0			

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 2-23. SYS Pin Control Register 6 (SYSPC6) Field Descriptions

Bit	Field	Value	Description
31-1	Reserved	0	Reads return 0. Writes have no effect.
0	ECPLKCLR	0	ECLK data out clear. This bit drives the output of the ECLK pin low when set in GIO output mode. <i>Write:</i> The ECLK pin value is unchanged.
		1	<i>Write:</i> The ECLK pin is driven to logic low (0). Note: The current logic state of the ECPLKDOUT bit will also be displayed by this bit when the pin is configured in GIO output mode. Note: The ECLK pin is placed into GIO mode by setting the ECPLKFUN bit to 0 in the SYSPC1 register. The ECLK pin is placed in output mode by setting the ECPLKDIR bit to 1 in the SYSPC2 register.

2.5.1.7 SYS Pin Control Register 7 (SYSPC7)

The SYSPC7 register, shown in [Figure 2-11](#) and described in [Table 2-24](#), controls the open drain function of the ECLK pin.

Figure 2-11. SYS Pin Control Register 7 (SYSPC7) [offset = 18h]

31	Reserved															16
R-0																
15	Reserved										1	0				
R-0											ECPCLKODE		R/W-0			

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 2-24. SYS Pin Control Register 7 (SYSPC7) Field Descriptions

Bit	Field	Value	Description
31-1	Reserved	0	Reads return 0. Writes have no effect.
0	ECPCLKODE	0	ECLK open drain enable. This bit is only active when ECLK is configured to be in GIO mode.
		0	The ECLK pin is configured in push/pull (normal GIO) mode.
		1	The ECLK pin is configured in open drain mode. The ECPCLKDOUT bit in the SYSPC4 register controls the state of the ECLK output buffer: ECPCLKDOUT = 0: The ECLK output buffer is driven low. ECPCLKDOUT = 1: The ECLK output buffer is tristated.
			Note: The ECLK pin is placed into GIO mode by setting the ECPCLKFUN bit to 0 in the SYSPC1 register.

2.5.1.8 SYS Pin Control Register 8 (SYSPC8)

The SYSPC8 register, shown in [Figure 2-12](#) and described in [Table 2-25](#), controls the pull enable function of the ECLK pin when it is configured as an input in GIO mode.

Figure 2-12. SYS Pin Control Register 8 (SYSPC8) [offset = 1Ch]

31	Reserved															16
R-0																
15	Reserved										1	0				
R-0											ECPCLKPUE		R/W-D			

LEGEND: R/W = Read/Write; R = Read only; D = Device Specific; -n = value after reset

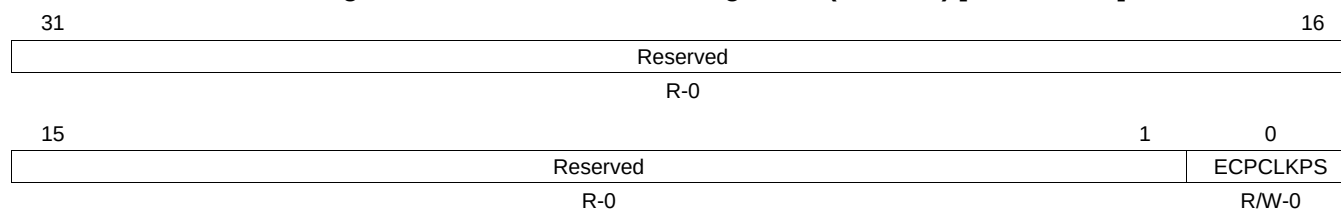
Table 2-25. SYS Pin Control Register 8 (SYSPC8) Field Descriptions

Bit	Field	Value	Description
31-1	Reserved	0	Reads return 0. Writes have no effect.
0	ECPCLKPUE	0	ECLK pull enable. Writes to this bit will only take effect when the ECLK pin is configured as an input in GIO mode.
		0	ECLK pull enable is active.
		1	ECLK pull enable is inactive.
			Note: The pull direction (up/down) is selected by the ECPCLKPS bit in the SYSPC9 register.
			Note: The ECLK pin is placed into GIO mode by setting the ECPCLKFUN bit to 0 in the SYSPC1 register. The ECLK pin is placed in input mode by setting the ECPCLKDIR bit to 0 in the SYSPC2 register.

2.5.1.9 SYS Pin Control Register 9 (SYSPC9)

The SYSPC9 register, shown in [Figure 2-13](#) and described in [Table 2-26](#), controls the pull up/pull down configuration of the ECLK pin when it is configured as an input in GIO mode.

Figure 2-13. SYS Pin Control Register 9 (SYSPC9) [offset = 20h]



LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 2-26. SYS Pin Control Register 9 (SYSPC9) Field Descriptions

Bit	Field	Value	Description
31-1	Reserved	0	Reads return 0. Writes have no effect.
0	ECPCLKPS	0 1	ECLK pull up/pull down select. This bit is only active when ECLK is configured as an input in GIO mode and the pull up/pull down logic is enabled. 0 ECLK pull down is selected, when pull up/pull down logic is enabled. 1 ECLK pull up is selected, when pull up/pull down logic is enabled. Note: The ECLK pin pull up/pull down logic is enabled by setting the ECPCLKPUE bit to 0 in the SYSPC8 register. Note: The ECLK pin is placed into GIO mode by setting the ECPCLKFUN bit to 0 in the SYSPC1 register. The ECLK pin is placed in input mode by setting the ECPCLKDIR bit to 0 in the SYSPC2 register.

2.5.1.10 Clock Source Disable Register (CSDIS)

The CSDIS register, shown in [Figure 2-14](#) and described in [Table 2-27](#), controls and displays the state of the device clock sources.

Figure 2-14. Clock Source Disable Register (CSDIS) [offset = 30h]

31																8															
Reserved																															
R-0																															
7				6				5				4				3				2				1				0			
CLKSR7OFF				CLKSR6OFF				CLKSR5OFF				CLKSR4OFF				CLKSR3OFF				Reserved				CLKSR1OFF				CLKSR0OFF			
R/WP-1				R/WP-1				R/WP-0				R/WP-0				R/WP-1				R-1				R/WP-1				R/WP-0			

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-27. Clock Source Disable Register (CSDIS) Field Descriptions

Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-3	CLKSR[7-3]OFF		Clock source[7-3] off.
		0	Clock source[7-3] is enabled.
		1	Clock source[7-3] is disabled. Note: On wakeup, only clock sources 0, 4, and 5 are enabled.
2	Reserved		Reads return 1. Writes have no effect.
1-0	CLKSR[1-0]OFF		Clock source[1-0] off.
		0	Clock source[1-0] is enabled.
		1	Clock source[1-0] is disabled. Note: On wakeup, only clock sources 0, 4, and 5 are enabled.

Table 2-28. Clock Sources Table

Clock Source #	Clock Source Name
Clock Source 0	Oscillator
Clock Source 1	PLL1
Clock Source 2	Not Implemented
Clock Source 3	EXTCLKIN
Clock Source 4	Low Frequency LPO (Low Power Oscillator) clock
Clock Source 5	High Frequency LPO (Low Power Oscillator) clock
Clock Source 6	PLL2
Clock Source 7	EXTCLKIN2

NOTE: Nonimplemented clock sources should not be enabled or used.

2.5.1.11 Clock Source Disable Set Register (CSDISSET)

The CSDISSET register, shown in [Figure 2-15](#) and described in [Table 2-29](#), sets clock sources to the disabled state.

Figure 2-15. Clock Source Disable Set Register (CSDISSET) [offset = 34h]

Reserved							
R-0							
31	7	6	5	4	3	2	1
SETCLKSR7 OFF	SETCLKSR6 OFF	SETCLKSR5 OFF	SETCLKSR4 OFF	SETCLKSR3 OFF	Reserved	SETCLKSR1 OFF	SETCLKSR0 OFF
R/WP-1	R/WP-1	R/WP-0	R/WP-0	R/WP-1	R-1	R/WP-1	R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-29. Clock Source Disable Set Register (CSDISSET) Field Descriptions

Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-3	SETCLKSR[7-3]OFF	0	Set clock source[7-3] to the disabled state. <i>Read:</i> Clock source[7-3] is enabled. <i>Write:</i> Clock source[7-3] is unchanged.
		1	<i>Read:</i> Clock source[7-3] is disabled. <i>Write:</i> Clock source[7-3] is set to the disabled state. Note: After a new clock source disable bit is set via the CSDISSET register, the new status of the bit will be reflected in the CSDIS register (offset 30h), the CSDISSET register (offset 34h) and the CSDISCLR register (offset 38h).
2	Reserved		Reads return 1. Writes have no effect.
1-0	SETCLKSR[1-0]OFF	0	Set clock source[1-0] to the disabled state. <i>Read:</i> Clock source[1-0] is enabled. <i>Write:</i> Clock source[1-0] is unchanged.
		1	<i>Read:</i> Clock source[1-0] is disabled. <i>Write:</i> Clock source[1-0] is set to the disabled state. Note: After a new clock source disable bit is set via the CSDISSET register, the new status of the bit will be reflected in the CSDIS register (offset 30h), the CSDISSET register (offset 34h) and the CSDISCLR register (offset 38h).

NOTE: A list of the available clock sources is shown in [Table 2-28](#).

2.5.1.12 Clock Source Disable Clear Register (CSDISCLR)

The CSDISCLR register, shown in [Figure 2-16](#) and described in [Table 2-30](#), clears clock sources to the enabled state.

Figure 2-16. Clock Source Disable Clear Register (CSDISCLR) [offset = 38h]

Reserved							
R-0							
7	6	5	4	3	2	1	0
CLRCLKSR7 OFF	CLRCLKSR6 OFF	CLRCLKSR5 OFF	CLRCLKSR4 OFF	CLRCLKSR3 OFF	Reserved	CLRCLKSR1 OFF	CLRCLKSR0 OFF
R/WP-1	R/WP-1	R/WP-0	R/WP-0	R/WP-1	R-1	R/WP-1	R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-30. Clock Source Disable Clear Register (CSDISCLR) Field Descriptions

Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-3	CLRCLKSR[7-3]OFF	0	Enables clock source[7-3]. <i>Read:</i> Clock source[7-3] is enabled. <i>Write:</i> Clock source[7-3] is unchanged.
		1	<i>Read:</i> Clock source[7-3] is enabled. <i>Write:</i> Clock source[7-3] is set to the enabled state. Note: After a new clock source disable bit is set via the CSDISSET register, the new status of the bit will be reflected in the CSDIS register (offset 30h), the CSDISSET register (offset 34h) and the CSDISCLR register (offset 38h).
2	Reserved		Reads return 1. Writes have no effect.
1-0	CLRCLKSR[1-0]OFF	0	Enables clock source[1-0]. <i>Read:</i> Clock source[1-0] is enabled. <i>Write:</i> Clock source[1-0] is unchanged.
		1	<i>Read:</i> Clock source[1-0] is enabled. <i>Write:</i> Clock source[1-0] is set to the enabled state. Note: After a new clock source disable bit is set via the CSDISSET register, the new status of the bit will be reflected in the CSDIS register (offset 30h), the CSDISSET register (offset 34h) and the CSDISCLR register (offset 38h).

NOTE: A list of the available clock sources is shown in [Table 2-28](#).

2.5.1.13 Clock Domain Disable Register (CDDIS)

The CDDIS register, shown in [Figure 2-17](#) and described in [Table 2-31](#), controls the state of the clock domains.

NOTE: All the clock domains are enabled on wakeup.

The application should guarantee that when HCLK and VCLK_sys are turned off through the HCLKOFF bit, the GCLK domain is also turned off.

The register bits in CDDIS are designated as high-integrity bits and have been implemented with error-correcting logic such that each bit, although read and written as a single bit, is actually a multi-bit key with error correction capability. As such, single-bit flips within the “key” can be corrected allowing protection of the system as a whole. An error detected is signaled to the ESM module.

Figure 2-17. Clock Domain Disable Register (CDDIS) [offset = 3Ch]

31																16																			
Reserved																																			
R-0																																			
15												12								11				10				9				8			
Reserved												VCLKA4OFF								VCLKA3OFF				VCLK4OFF				VCLK3OFF							
R-0												R/WP-0								R/WP-0				R/WP-0				R/WP-0							
7				6				5				4				3				2				1				0							
Reserved				RTICKL1OFF				Reserved				VCLKA1OFF				VCLK2OFF				VCLKPOFF				HCLKOFF				GCLKOFF							
R/WP-0				R/WP-0				R/WP-0				R/WP-0				R/WP-0				R/WP-0				R/WP-0				R/WP-0							

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-31. Clock Domain Disable Register (CDDIS) Field Descriptions

Bit	Field	Value	Description
31-12	Reserved	0	Reads return 0. Writes have no effect.
11-10	VCLKA[4-3]OFF	0	VCLKA[4-3] domain off. The VCLKA[4-3] domain is enabled.
		1	The VCLKA[4-3] domain is disabled.
9-8	VCLK[4-3]OFF	0	VCLK[4-3] domain off. The VCLK[4-3] domain is enabled.
		1	The VCLK[4-3] domain is disabled.
7	Reserved	0-1	Reads return 0 or 1 and privilege mode writes allowed.
6	RTICK1OFF	0	RTICK1 domain off. The RTICK1 domain is enabled.
		1	The RTICK1 domain is disabled.
5	Reserved	0-1	Reads return 0 or 1 and privilege mode writes allowed.
4	VCLKA1OFF	0	VCLKA1 domain off. The VCLKA1 domain is enabled.
		1	The VCLKA1 domain is disabled.
3	VCLK2OFF	0	VCLK2 domain off. The VCLK2 domain is enabled.
		1	The VCLK2 domain is disabled.
2	VCLKPOFF	0	VCLK_periph domain off. The VCLK_periph domain is enabled.
		1	The VCLK_periph domain is disabled.

Table 2-31. Clock Domain Disable Register (CDDIS) Field Descriptions (continued)

Bit	Field	Value	Description
1	HCLKOFF		HCLK and VCLK_sys domains off.
		0	The HCLK and VCLK_sys domains are enabled.
		1	The HCLK and VCLK_sys domains are disabled.
0	GCLKOFF		GCLK domain off.
		0	The GCLK domain is enabled.
		1	The GCLK domain is disabled.

2.5.1.14 Clock Domain Disable Set Register (CDDISSET)

This CDDISSET register, shown in [Figure 2-18](#) and described in [Table 2-32](#), sets clock domains to the disabled state.

Figure 2-18. Clock Domain Disable Set Register (CDDISSET) [offset = 40h]

31																	16
Reserved																	
R-0																	
15				12				11		10		9		8			
Reserved								SETVCLKA4 OFF		SETVCLKA3 OFF		SETVCLK4 OFF		SETVCLK3 OFF			
R-0								R/WP-0		R/WP-0		R/WP-0		R/WP-0			
7		6		5		4		3		2		1		0			
Reserved		SETRTI1CLK OFF		Reserved		SETVCLKA1 OFF		SETVCLK2 OFF		SETVCLKP OFF		SETHCLK OFF		SETGCLK OFF			
R/WP-0		R/WP-0		R/WP-0		R/WP-0		R/WP-0		R/WP-0		R/WP-0		R/WP-0			

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-32. Clock Domain Disable Set Register (CDDISSET) Field Descriptions

Bit	Field	Value	Description
31-12	Reserved	0	Reads return 0. Writes have no effect.
11-10	SETVCLKA[4-3]OFF	0	Set VCLKA[4-3] domain. <i>Read:</i> The VCLKA[4-3] domain is enabled. <i>Write:</i> The VCLKA[4-3] domain is unchanged.
		1	<i>Read:</i> The VCLKA[4-3] domain is disabled. <i>Write:</i> The VCLKA[4-3] domain is set to the enabled state.
9-8	SETVCLK[4-3]OFF	0	Set VCLK[4-3] domain. <i>Read:</i> The VCLK[4-3] domain is enabled. <i>Write:</i> The VCLK[4-3] domain is unchanged.
		1	<i>Read:</i> The VCLK[4-3] domain is disabled. <i>Write:</i> The VCLK[4-3] domain is set to the enabled state.
7	Reserved	0-1	Reads return 0 or 1 and privilege mode writes allowed.
6	SETRTI1CLKOFF	0	Set RTICKL1 domain. <i>Read:</i> The RTICKL1 domain is enabled. <i>Write:</i> The RTICKL1 domain is unchanged.
		1	<i>Read:</i> The RTICKL1 domain is disabled. <i>Write:</i> The RTICKL1 domain is set to the enabled state.
5	Reserved	0-1	Reads return 0 or 1 and privilege mode writes allowed.
4	SETVCLKA1OFF	0	Set VCLKA1 domain. <i>Read:</i> The VCLKA1 domain is enabled. <i>Write:</i> The VCLKA1 domain is unchanged.
		1	<i>Read:</i> The VCLKA1 domain is disabled. <i>Write:</i> The VCLKA1 domain is set to the enabled state.
3	SETVCLK2OFF	0	Set VCLK2 domain. <i>Read:</i> The VCLK2 domain is enabled. <i>Write:</i> The VCLK2 domain is unchanged.
		1	<i>Read:</i> The VCLK2 domain is disabled. <i>Write:</i> The VCLK2 domain is set to the enabled state.

Table 2-32. Clock Domain Disable Set Register (CDDISSET) Field Descriptions (continued)

Bit	Field	Value	Description
2	SETVCLKPOFF	0	Set VCLK_periph domain. <i>Read:</i> The VCLK_periph domain is enabled. <i>Write:</i> The VCLK_periph domain is unchanged.
		1	<i>Read:</i> The VCLK_periph domain is disabled. <i>Write:</i> The VCLK_periph domain is set to the enabled state.
1	SETHCLKOFF	0	Set HCLK and VCLK_sys domains. <i>Read:</i> The HCLK and VCLK_sys domain is enabled. <i>Write:</i> The HCLK and VCLK_sys domain is unchanged.
		1	<i>Read:</i> The HCLK and VCLK_sys domain is disabled. <i>Write:</i> The HCLK and VCLK_sys domain is set to the enabled state.
0	SETGCLKOFF	0	Set GCLK domain. <i>Read:</i> The GCLK domain is enabled. <i>Write:</i> The GCLK domain is unchanged.
		1	<i>Read:</i> The GCLK domain is disabled. <i>Write:</i> The GCLK domain is set to the enabled state.

2.5.1.15 Clock Domain Disable Clear Register (CDDISCLR)

The CDDISCLR register, shown in [Figure 2-19](#) and described in [Table 2-33](#), clears clock domains to the enabled state.

Figure 2-19. Clock Domain Disable Clear Register (CDDISCLR) [offset = 44h]

31	Reserved																16
R-0																	
15				12				11		10		9		8			
Reserved								CLRVLKA4 OFF		CLRVLKA3 OFF		CLRVLK4 OFF		CLRVLK3 OFF			
R-0								R/WP-0		R/WP-0		R/WP-0		R/WP-0			
7		6		5		4		3		2		1		0			
Reserved		CLRRT1CLK OFF		Reserved		CLRVLKA1 OFF		CLRVLK2 OFF		CLRVLKP OFF		CLRCLK OFF		CLRGCLK OFF			
R/WP-0		R/WP-0		R/WP-0		R/WP-0		R/WP-0		R/WP-0		R/WP-0		R/WP-0			

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-33. Clock Domain Disable Clear Register (CDDISCLR) Field Descriptions

Bit	Field	Value	Description
31-12	Reserved	0	Reads return 0. Writes have no effect.
11-10	CLRVLKA[4-3]OFF	0	Clear VLKA[4-3] domain. <i>Read:</i> The VLKA[4-3] domain is enabled. <i>Write:</i> The VLKA[4-3] domain is unchanged.
		1	<i>Read:</i> The VLKA[4-3] domain is disabled. <i>Write:</i> The VLKA[4-3] domain is cleared to the enabled state.
9-8	CLRVLK[4-3]OFF	0	Clear VLK[4-3] domain. <i>Read:</i> The VLK[4-3] domain is enabled. <i>Write:</i> The VLK[4-3] domain is unchanged.
		1	<i>Read:</i> The VLK[4-3] domain is disabled. <i>Write:</i> The VLK[4-3] domain is cleared to the enabled state.
7	Reserved	0-1	Reads return 0 or 1 and privilege mode writes allowed.
6	CLRRTI1CLKOFF	0	Clear RTICK1 domain. <i>Read:</i> The RTICK1 domain is enabled. <i>Write:</i> The RTICK1 domain is unchanged.
		1	<i>Read:</i> The RTICK1 domain is disabled. <i>Write:</i> The RTICK1 domain is cleared to the enabled state.
5	Reserved	0-1	Reads return 0 or 1 and privilege mode writes allowed.
4	CLRVLKA1OFF	0	Clear VLKA1 domain. <i>Read:</i> The VLKA1 domain is enabled. <i>Write:</i> The VLKA1 domain is unchanged.
		1	<i>Read:</i> The VLKA1 domain is disabled. <i>Write:</i> The VLKA1 domain is cleared to the enabled state.
3	CLRVLK2OFF	0	Clear VLK2 domain. <i>Read:</i> The VLK2 domain is enabled. <i>Write:</i> The VLK2 domain is unchanged.
		1	<i>Read:</i> The VLK2 domain is disabled. <i>Write:</i> The VLK2 domain is cleared to the enabled state.

Table 2-33. Clock Domain Disable Clear Register (CDDISCLR) Field Descriptions (continued)

Bit	Field	Value	Description
2	CLRVCLKPOFF	0	Clear VCLK_periph domain. <i>Read:</i> The VCLK_periph domain is enabled. <i>Write:</i> The VCLK_periph domain is unchanged.
		1	<i>Read:</i> The VCLK_periph domain is disabled. <i>Write:</i> The VCLK_periph domain is cleared to the enabled state.
1	CLRHCLKOFF	0	Clear HCLK and VCLK_sys domains. <i>Read:</i> The HCLK and VCLK_sys domain is enabled. <i>Write:</i> The HCLK and VCLK_sys domain is unchanged.
		1	<i>Read:</i> The HCLK and VCLK_sys domain is disabled. <i>Write:</i> The HCLK and VCLK_sys domain is cleared to the enabled state.
0	CLRGCLKOFF	0	Clear GCLK domain. <i>Read:</i> The GCLK domain is enabled. <i>Write:</i> The GCLK domain is unchanged.
		1	<i>Read:</i> The GCLK domain is disabled. <i>Write:</i> The GCLK domain is cleared to the enabled state.

2.5.1.16 GCLK, HCLK, VCLK, and VCLK2 Source Register (GHVSRC)

The GHVSRC register, shown in [Figure 2-20](#) and described in [Table 2-34](#), controls the clock source configuration for the GCLK, HCLK, VCLK and VCLK2 clock domains.

Figure 2-20. GCLK, HCLK, VCLK, and VCLK2 Source Register (GHVSRC) [offset = 48h]

31	28	27	24	23	20	19	16
Reserved				GHVWAKE			
R-0				R/WP-0			
15	Reserved				4	3	0
R-0				GHVSRC			
R-0				R/WP-0			

LEGEND: R = Read only; R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-34. GCLK, HCLK, VCLK, and VCLK2 Source Register (GHVSRC) Field Descriptions

Bit	Field	Value	Description
31-28	Reserved	0	Reads return 0. Writes have no effect.
27-24	GHVWAKE	0 1h 2h 3h 4h 5h 6h 7h 8h-Fh	GCLK, HCLK, VCLK, VCLK2 source on wakeup. Clock source0 is the source for GCLK, HCLK, VCLK, VCLK2 on wakeup. Clock source1 is the source for GCLK, HCLK, VCLK, VCLK2 on wakeup. Clock source2 is the source for GCLK, HCLK, VCLK, VCLK2 on wakeup. Clock source3 is the source for GCLK, HCLK, VCLK, VCLK2 on wakeup. Clock source4 is the source for GCLK, HCLK, VCLK, VCLK2 on wakeup. Clock source5 is the source for GCLK, HCLK, VCLK, VCLK2 on wakeup. Clock source6 is the source for GCLK, HCLK, VCLK, VCLK2 on wakeup. Clock source7 is the source for GCLK, HCLK, VCLK, VCLK2 on wakeup. Reserved
23-20	Reserved	0	Reads return 0. Writes have no effect.
19-16	HVLPM	0 1h 2h 3h 4h 5h 6h 7h 8h-Fh	HCLK, VCLK, VCLK2 source on wakeup when GCLK is turned off. Clock source0 is the source for HCLK, VCLK, VCLK2 on wakeup. Clock source1 is the source for HCLK, VCLK, VCLK2 on wakeup. Clock source2 is the source for HCLK, VCLK, VCLK2 on wakeup. Clock source3 is the source for HCLK, VCLK, VCLK2 on wakeup. Clock source4 is the source for HCLK, VCLK, VCLK2 on wakeup. Clock source5 is the source for HCLK, VCLK, VCLK2 on wakeup. Clock source6 is the source for HCLK, VCLK, VCLK2 on wakeup. Clock source7 is the source for HCLK, VCLK, VCLK2 on wakeup. Reserved
15-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	GHVSRC	0 1h 2h 3h 4h 5h 6h 7h 8h-Fh	GCLK, HCLK, VCLK, VCLK2 current source. Note: The GHVSRC[3-0] bits are updated with the HVLPM[3-0] setting when GCLK is turned off, and are updated with the GHVWAKE[3-0] setting on system wakeup. Clock source0 is the source for GCLK, HCLK, VCLK, VCLK2. Clock source1 is the source for GCLK, HCLK, VCLK, VCLK2. Clock source2 is the source for GCLK, HCLK, VCLK, VCLK2. Clock source3 is the source for GCLK, HCLK, VCLK, VCLK2. Clock source4 is the source for GCLK, HCLK, VCLK, VCLK2. Clock source5 is the source for GCLK, HCLK, VCLK, VCLK2. Clock source6 is the source for GCLK, HCLK, VCLK, VCLK2. Clock source7 is the source for GCLK, HCLK, VCLK, VCLK2. Reserved

NOTE: Non implemented clock sources should not be enabled or used. A list of the available clock sources is shown in [Table 2-28](#).

2.5.1.17 Peripheral Asynchronous Clock Source Register (VCLKASRC)

The VCLKASRC register, shown in [Figure 2-21](#) and described in [Table 2-35](#), sets the clock source for the asynchronous peripheral clock domains to be configured to run from a specific clock source.

Figure 2-21. Peripheral Asynchronous Clock Source Register (VCLKASRC) [offset = 4Ch]

31																16																																															
Reserved																																																															
R-0																																																															
15								12								11								8								7								4								3								0							
Reserved																Reserved																Reserved																VCLKA1S															
R-0																R/WP-9h																R-0																R/WP-9h															

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-35. Peripheral Asynchronous Clock Source Register (VCLKASRC) Field Descriptions

Bit	Field	Value	Description
31-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	VCLKA1S		Peripheral asynchronous clock1 source.
		0	Clock source0 is the source for peripheral asynchronous clock1.
		1h	Clock source1 is the source for peripheral asynchronous clock1.
		2h	Clock source2 is the source for peripheral asynchronous clock1.
		3h	Clock source3 is the source for peripheral asynchronous clock1.
		4h	Clock source4 is the source for peripheral asynchronous clock1.
		5h	Clock source5 is the source for peripheral asynchronous clock1.
		6h	Clock source6 is the source for peripheral asynchronous clock1.
		7h	Clock source7 is the source for peripheral asynchronous clock1.
		8h-Fh	VCLK is the source for peripheral asynchronous clock1.

NOTE: Non implemented clock sources should not be enabled or used. A list of the available clock sources is shown in [Table 2-28](#).

2.5.1.18 RTI Clock Source Register (RCLKSRC)

The RCLKSRC register, shown in [Figure 2-22](#) and described in [Table 2-36](#), controls the RTI (Real Time Interrupt) clock source selection.

NOTE: Important constraint when the RTI clock source is not VCLK

If the RTIx clock source is chosen to be anything other than the default VCLK, then the RTI clock needs to be at least three times slower than the VCLK. This can be achieved by configuring the RTIxCLK divider in this register. This divider is internally bypassed when the RTIx clock source is VCLK.

Figure 2-22. RTI Clock Source Register (RCLKSRC) [offset = 50h]

31	26	25	24	23	20	19	16
Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
R-0	R/WP-1h	R-0	R/WP-9h				
15	10	9	8	7	4	3	0
Reserved	RTI1DIV	Reserved	RTI1SRC				
R-0	R/WP-1h	R-0	R/WP-9h				

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-36. RTI Clock Source Register (RCLKSRC) Field Descriptions

Bit	Field	Value	Description
31-26	Reserved	0	Reads return 0. Writes have no effect.
25-24	Reserved	0	Reads return value and privilege mode writes allowed.
23-20	Reserved	0	Reads return 0. Writes have no effect.
19-16	Reserved	0	Reads return value and privilege mode writes allowed.
15-10	Reserved	0	Reads return 0. Writes have no effect.
9-8	RTI1DIV	0 1h 2h 3h	RTI clock1 Divider. RTICK1 divider value is 1. RTICK1 divider value is 2. RTICK1 divider value is 4. RTICK1 divider value is 8.
7-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	RTI1SRC	0 1h 2h 3h 4h 5h 6h 7h 8h-Fh	RTI clock1 source. Clock source0 is the source for RTICK1. Clock source1 is the source for RTICK1. Clock source2 is the source for RTICK1. Clock source3 is the source for RTICK1. Clock source4 is the source for RTICK1. Clock source5 is the source for RTICK1. Clock source6 is the source for RTICK1. Clock source7 is the source for RTICK1. VCLK is the source for RTICK1.

NOTE: A list of the available clock sources is shown in [Table 2-28](#).

2.5.1.19 Clock Source Valid Status Register (CSVSTAT)

The CSVSTAT register, shown in [Figure 2-23](#) and described in [Table 2-37](#), indicates the status of usable clock sources.

Figure 2-23. Clock Source Valid Status Register (CSVSTAT) [offset = 54h]

31																8															
Reserved																															
R-0																															
7				6				5				4				3				2				1				0			
CLKSR7V				CLKSR6V				CLKSR5V				CLKSR4V				CLKSR3V				Reserved				CLKSR1V				CLKSR0V			
R-1				R-0				R-0				R-1				R-1				R-0				R-0				R-0			

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 2-37. Clock Source Valid Register (CSVSTAT) Field Descriptions

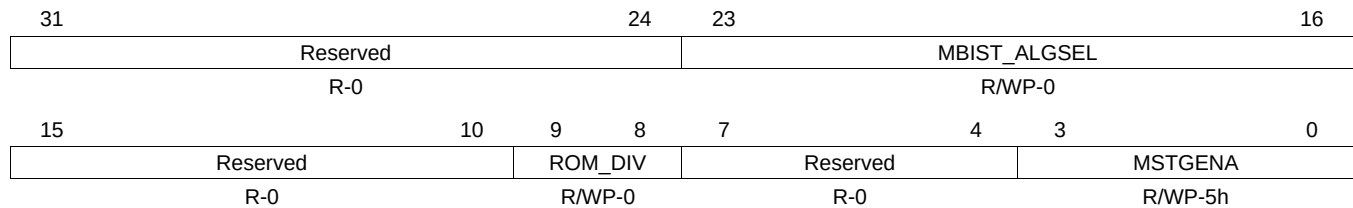
Bit	Field	Value	Description
31-8	Reserved.	0	Reads return 0. Writes have no effect.
7-3	CLKSR[7-3]V	0 1	Clock source[7-3] valid. Clock source[7-3] is not valid. Clock source[7-3] is valid. Note: If the valid bit of the source of a clock domain is not set (that is, the clock source is not fully stable), the respective clock domain is disabled by the Global Clock Module (GCM).
2	Reserved.	0	Reads return 0. Writes have no effect.
1-0	CLKSR[1-0]V	0 1	Clock source[1-0] valid. Clock source[1-0] is not valid. Clock source[1-0] is valid. Note: If the valid bit of the source of a clock domain is not set (that is, the clock source is not fully stable), the respective clock domain is disabled.

NOTE: A list of the available clock sources is shown in [Table 2-28](#).

2.5.1.20 Memory Self-Test Global Control Register (MSTGCR)

The MSTGCR register, shown in [Figure 2-24](#) and described in [Table 2-38](#), controls several aspects of the PBIST (Programmable Built-In Self Test) memory controller.

Figure 2-24. Memory Self-Test Global Control Register (MSTGCR) [offset = 58h]



LEGEND: R = Read only; R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

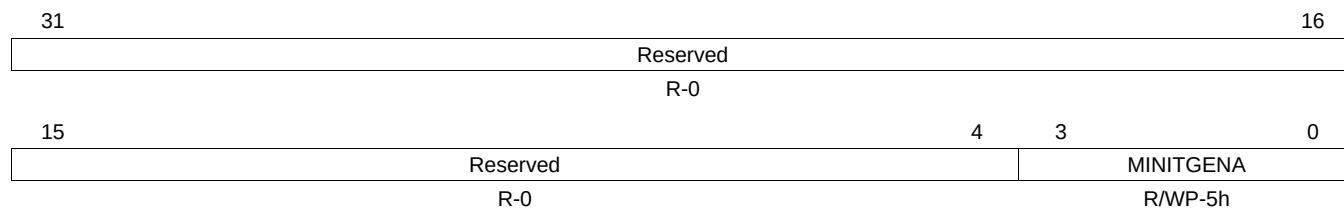
Table 2-38. Memory Self-Test Global Control Register (MSTGCR) Field Descriptions

Bit	Field	Value	Description
31-24	Reserved	0	Reads return 0. Writes have no effect.
23-16	MBIST_ALGSEL	0 No Algorithm is selected. 1h Checkerboard is selected. 2h March 13N with background of all 0s and all 1s is selected. 4h March 11N with background of hex 5 and As is selected. 8h March 13N with backgrounds of hex 3 and Cs, hex 0F and F0s, and 69 and 96s are selected. 10h PMOS Open Address Decode is selected. 20h No Algorithm is selected. 40h No Algorithm is selected. 80h No Algorithm is selected.	
15-10	Reserved	0	Reads return 0. Writes have no effect.
9-8	ROM_DIV	0 ROM clock source is HCLK divided by 1. PBIST will reset for 16 VBUS cycles. 1h ROM clock source is HCLK divided by 2. PBIST will reset for 32 VBUS cycles. 2h ROM clock source is HCLK divided by 4. PBIST will reset for 64 VBUS cycles. 3h ROM clock source is HCLK divided by 8. PBIST will reset for 96 VBUS cycles.	
7-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	MSTGENA	Ah Memory self-test controller is enabled. Others Memory self-test controller is disabled.	Memory self-test controller global enable key Note: Enabling the MSTGENA key will generate a reset to the state machine of the selected PBIST controller. Note: It is recommended that a value of 5h be used to disable the memory self-test controller. This value will give maximum protection from a bit flip inducing event that would inadvertently enable the controller.

2.5.1.21 Memory Hardware Initialization Global Control Register (MINITGCR)

The MINITGCR register, shown in [Figure 2-25](#) and described in [Table 2-39](#), enables automatic hardware memory initialization.

Figure 2-25. Memory Hardware Initialization Global Control Register (MINITGCR) [offset = 5Ch]



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-39. Memory Hardware Initialization Global Control Register (MINITGCR) Field Descriptions

Bit	Field	Value	Description
31-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	MINITGENA	Ah Others	Memory hardware initialization global enable key. Global memory hardware initialization is enabled. Global memory hardware initialization is disabled. Note: It is recommended that a value of 5h be used to disable memory hardware initialization. This value will give maximum protection from an event that would inadvertently enable the controller.

2.5.1.22 MBIST Controller/ Memory Initialization Enable Register (MSINENA)

The MSINENA register, shown in [Figure 2-26](#) and described in [Table 2-40](#), enables PBIST controllers for memory self test and the memory modules initialized during automatic hardware memory initialization.

Figure 2-26. MBIST Controller/Memory Initialization Enable Register (MSINENA) [offset = 60h]

31	MSIENA	16
	R/WP-0	
15	MSIENA	0
	R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-40. MBIST Controller/Memory Initialization Enable Register (MSINENA) Field Descriptions

Bit	Field	Value	Description
31-0	MSIENA	0	PBIST controller and memory initialization enable register. In memory self-test mode, all the corresponding bits of the memories to be tested should be set before enabling the global memory self-test controller key (MSTGENA) in the MSTGCR register (offset 58h). The reason for this is that MSTGENA, in addition to being the global enable for all individual PBIST controllers, is the source for the reset generation to all the PBIST controller state machines. Disabling the MSTGENA or MINITGENA key (by writing from Ah to any other value) will reset all the MSIENA[31-0] bits to their default values. In memory self-test mode (MSTGENA = Ah): PBIST controller [31-0] is disabled. In memory Initialization mode (MINITGENA = Ah): Memory module [31-0] auto hardware initialization is disabled.
		1	In memory self-test mode (MSTGENA = Ah): PBIST controller [31-0] is enabled. In memory Initialization mode (MINITGENA = Ah): Memory module [31-0] auto hardware initialization is enabled. Note: Software should ensure that both the memory self-test global enable key (MSTGENA) and the memory hardware initialization global key (MINITGENA) are not enabled at the same time.

2.5.1.23 Memory Self-Test Fail Status Register (MSTFAIL)

The MSTFAIL register, shown in [Figure 2-27](#) and described in [Table 2-41](#), shows the fail status of the memory self-tests.

Figure 2-27. Memory Self-Test Fail Status Register (MSTFAIL) [offset = 64h]

31		16
MSTF		
R/WP-0		
15		0
MSTF		
R/WP-0		

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

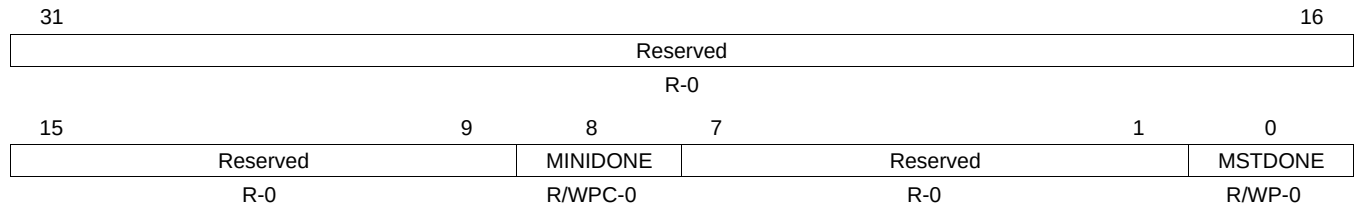
Table 2-41. Memory Self-Test Fail Status Register (MSTFAIL) Field Descriptions

Bit	Field	Value	Description
31-0	MSTF	0	Memory self-test fail status bit. <i>Read:</i> PBIST controller [31-0] run did not fail. <i>Write:</i> A write of 0 has no effect.
		1	<i>Read:</i> PBIST controller [31-0] run failed. <i>Write:</i> The bit is cleared to 0. Note: Disabling the MSTGENA key (by writing from Ah to any other value) will reset all the individual fail status bits to their default values.

2.5.1.24 MSTC Global Status Register (MSTCGSTAT)

The MSTCGSTAT register, shown in [Figure 2-28](#) and described in [Table 2-42](#), shows the status of the memory hardware initialization and the memory self-test.

Figure 2-28. MSTC Global Status Register (MSTCGSTAT) [offset = 68h]



LEGEND: R/W = Read/Write; R = Read only; C = Clear; WP = Write in privileged mode only; -n = value after reset

Table 2-42. MSTC Global Status Register (MSTCGSTAT) Field Descriptions

Bit	Field	Value	Description
31-9	Reserved	0	Reads return 0. Writes have no effect.
8	MINIDONE	0	Memory hardware initialization complete status. Note: Disabling the MINITGENA key (By writing from Ah to any other value) will clear the MINIDONE status bit to 0. Note: Individual memory initialization status is shown in the MINISTAT register. <i>Read:</i> Memory hardware initialization is not complete for all memory. <i>Write:</i> A write of 0 has no effect.
		1	<i>Read:</i> Hardware initialization of all memory is completed. <i>Write:</i> The bit is cleared to 0.
7-1	Reserved	0	Reads return 0. Writes have no effect.
0	MSTDONE	0	Memory self-test run complete status. Note: Disabling the MSTGENA key (by writing from Ah to any other value) will clear the MSTDONE status bit to 0. <i>Read:</i> Memory self-test is not completed. <i>Write:</i> A write of 0 has no effect.
		1	<i>Read:</i> Memory self-test is completed. <i>Write:</i> The bit is cleared to 0.

Table 2-44. PLL Control Register 1 (PLLCTL1) Field Descriptions

Bit	Field	Value	Description
31	ROS	0 1	Reset on PLL Slip Do not reset system when PLL slip is detected. Reset when PLL slip is detected . Note: BPOS (Bits 30-29) must also be enabled for ROS to be enabled.
30-29	MASK_SLIP	2h Others	Mask detection of PLL slip Bypass on PLL Slip is disabled. If a PLL Slip is detected no action is taken. Bypass on PLL Slip is enabled. If a PLL Slip is detected the device will automatically bypass the PLL and use the oscillator to provide the device clock. Note: If ROS (Bit 31) is set to 1, the device will be reset if a PLL Slip and the PLL will be bypassed after the reset occurs.
28-24	PLLDIV	0 1h : 1Fh	PLL Output Clock Divider $R = PLLDIV + 1$ $f_{PLL\ CLK} = f_{post-ODCLK} / R$ $f_{PLL\ CLK} = f_{post-ODCLK} / 1$ $f_{PLL\ CLK} = f_{post-ODCLK} / 2$: $f_{PLL\ CLK} = f_{post-ODCLK} / 32$
23	ROF	0 1	Reset on Oscillator Fail Do not reset system when oscillator is out of range. Reset when oscillator is out of range.
22	Reserved	0	Value has no effect on PLL operation.
21-16	REFCLKDIV	0 1h : 3Fh	Reference Clock Divider $NR = REFCLKDIV + 1$ $f_{INT\ CLK} = f_{OSCIN} / NR$ $f_{INT\ CLK} = f_{OSCIN} / 1$ $f_{INT\ CLK} = f_{OSCIN} / 2$: $f_{INT\ CLK} = f_{OSCIN} / 64$
15-0	PLLMUL	0h 100h : 5B00h 5C00h : FF00h	PLL Multiplication Factor $NF = (PLLMUL / 256) + 1$, valid multiplication factors are from 1 to 256. $f_{VCO\ CLK} = f_{INT\ CLK} \times NF$ $f_{VCO\ CLK} = f_{INT\ CLK} \times 1$ $f_{VCO\ CLK} = f_{INT\ CLK} \times 2$: $f_{VCO\ CLK} = f_{INT\ CLK} \times 92$ $f_{VCO\ CLK} = f_{INT\ CLK} \times 93$: $f_{VCO\ CLK} = f_{INT\ CLK} \times 256$

2.5.1.27 PLL Control Register 2 (PLLCTL2)

The PLLCTL2 register, shown in [Figure 2-31](#) and described in [Table 2-45](#), controls the modulation characteristics and the output divider of the PLL.

Figure 2-31. PLL Control Register 2 (PLLCTL2) [offset = 74h]

31	30																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
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LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-45. PLL Control Register 2 (PLLCTL2) Field Descriptions

Bit	Field	Value	Description
31	FMENA	0 1	Frequency Modulation Enable. Disable frequency modulation. Enable frequency modulation.
30-22	SPREADINGRATE	0 1h : 1FFh	$NS = SPREADINGRATE + 1$ $f_{mod} = f_s = f_{INT_CLK} / (2 \times NS)$ $f_{mod} = f_s = f_{INT_CLK} / (2 \times 1)$ $f_{mod} = f_s = f_{INT_CLK} / (2 \times 2)$: $f_{mod} = f_s = f_{INT_CLK} / (2 \times 512)$
21	Reserved	0	Value has no effect on PLL operation.
20-12	MULMOD	0 8h 9h : 1FFh	Multiplier Correction when Frequency Modulation is enabled When FMENA = 0, MUL_when_MOD = 0; when FMENA = 1, MUL_when_MOD = (MULMOD / 256) No adder to NF MUL_when_MOD = 8/256 MUL_when_MOD = 9/256 : MUL_when_MOD = 511/256
11-9	ODPLL	0 1h : 7h	Internal PLL Output Divider. $OD = ODPLL + 1$ $f_{post-ODCLK} = f_{VCO_CLK} / OD$ Note: PLL output clock is gated off if ODPLL is changed while the PLL is active. $f_{post-ODCLK} = f_{VCO_CLK} / 1$ $f_{post-ODCLK} = f_{VCO_CLK} / 2$: $f_{post-ODCLK} = f_{VCO_CLK} / 8$ Note: PLL output clock is gated off, if ODPLL is changed while the PLL is active.

Table 2-45. PLL Control Register 2 (PLLCTL2) Field Descriptions (continued)

Bit	Field	Value	Description
8-0	SPR_AMOUNT		Spreading Amount. $NV = (SPR_AMOUNT + 1)/2048$ NV ranges from 1/2048 to 512/2048 Note that the PLL output clock is disabled for 1 modulation period if the SPR_AMOUNT field is changed while the frequency modulation is enabled. If frequency modulation is disabled and SPR_AMOUNT is changed, there is no effect on the PLL output clock.
		0	NV = 1/2048
		1h	NV = 2/2048
		:	:
		1FFh	NV = 512/2048

2.5.1.28 SYS Pin Control Register 10 (SYSPC10)

The SYSPC10 register, shown in [Figure 2-32](#) and described in [Table 2-46](#), controls the function of the ECPCLK slew mode.

Figure 2-32. SYS Pin Control Register 10 (SYSPC10) [offset = 78h]

31	Reserved	16
	R-0	
15	Reserved	1
	R-0	0
		ECPCLK_SLEW
		R/W-0

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 2-46. SYS Pin Control Register 10 (SYSPC10) Field Descriptions

Bit	Field	Value	Description
31-1	Reserved	0	Reads return 0. Writes have no effect.
0	ECPCLK_SLEW	0	ECCLK slew control. This bit controls between the fast or slow slew mode. Fast mode is enabled; the normal output buffer is used for this pin.
		1	Slow mode is enabled; slew rate control is used for this pin.

2.5.1.29 Die Identification Register Lower Word (DIEIDL)

The DIEIDL register, shown in [Figure 2-33](#) and described in [Table 2-47](#), contains information about the die lot number, wafer number, and X, Y wafer coordinates.

Figure 2-33. Die Identification Register, Lower Word (DIEIDL) [offset = 7Ch]

31		24	23		16
WAFAER #				Y WAFAER COORDINATE	
R-D				R-D	
15	12	11			0
Y WAFAER COORDINATE		X WAFAER COORDINATE			
R-D		R-D			

LEGEND: R = Read only; -n = value after reset; D = value is device specific

Table 2-47. Die Identification Register, Lower Word (DIEIDL) Field Descriptions

Bit	Field	Description
31-24	WAFAER #	These read-only bits contain the wafer number of the device.
23-12	Y WAFAER COORDINATE	These read-only bits contain the Y wafer coordinate of the device.
11-0	X WAFAER COORDINATE	These read-only bits contain the X wafer coordinate of the device.

NOTE: Die Identification Information

The die identification information will vary from unit to unit. This information is programmed by TI as part of the initial device test procedure.

2.5.1.30 Die Identification Register Upper Word (DIEIDH)

The DIEIDH register, shown in [Figure 2-34](#) and described in [Table 2-48](#), contains information about the die lot number.

Figure 2-34. Die Identification Register, Upper Word (DIEIDH) [offset = 80h]

31		24	23		16
Reserved				LOT #	
R-0				R-D	
15					0
LOT #					
R-D					

LEGEND: R = Read only; -n = value after reset; D = value is device specific

Table 2-48. Die Identification Register, Upper Word (DIEIDH) Field Descriptions

Bit	Field	Description
31-24	Reserved	Reserved for TI use. Writes have no effect.
23-0	LOT #	This read-only register contains the device lot number.

NOTE: Die Identification Information

The die identification information will vary from unit to unit. This information is programmed by TI as part of the initial device test procedure.

2.5.1.31 LPO/Clock Monitor Control Register (LPOMONCTL)

The LPOMONCTL register, shown in [Figure 2-35](#) and described in [Table 2-49](#), controls the Low Frequency (Clock Source 4) and High Frequency (Clock Source 5) Low Power Oscillator's trim values.

Figure 2-35. LPO/Clock Monitor Control Register (LPOMONCTL) [offset = 88h]

31	25	24	23	17	16		
Reserved		BIAS ENABLE	Reserved		OSCFRQCONFIGCNT		
R-0		R/WP-1	R-0		R-0		
15	13	12	8	7	5	4	0
Reserved		HFTRIM		Reserved		LFTRIM	
R-0		R/WP-10h		R-0		R/WP-10h	

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-49. LPO/Clock Monitor Control Register (LPOMONCTL) Field Descriptions

Bit	Field	Value	Description
31-25	Reserved	0	Reads return 0. Writes have no effect.
24	BIAS ENABLE	0 1	Bias enable. The bias circuit inside the low-power oscillator (LPO) is disabled. The bias circuit inside LPO is enabled.
23-17	Reserved	0	Reads return 0. Writes have no effect.
16	OSCFRQCONFIGCNT	0 1	Configures the counter based on OSC frequency. OSC freq is $\leq$ 20MHz OSC freq is $>$ 20MHz and $\leq$ 80MHz
15-13	Reserved	0	Reads return 0. Writes have no effect.

Table 2-49. LPO/Clock Monitor Control Register (LPOMONCTL) Field Descriptions (continued)

Bit	Field	Value	Description																																																																
12-8	HFTRIM		High-frequency oscillator trim value. This four-bit value is used to center the HF oscillator's frequency. Caution: This value should only be changed when the HF oscillator is not the source for a clock domain, otherwise a system failure could result. The following values are the ratio: f / fo in the F021 process. <table><tr><td>0</td><td>29.52</td></tr><tr><td>1h</td><td>34.24%</td></tr><tr><td>2h</td><td>38.85%</td></tr><tr><td>3h</td><td>43.45%</td></tr><tr><td>4h</td><td>47.99%</td></tr><tr><td>5h</td><td>52.55%</td></tr><tr><td>6h</td><td>57.02%</td></tr><tr><td>7h</td><td>61.46%</td></tr><tr><td>8h</td><td>65.92%</td></tr><tr><td>9h</td><td>70.17</td></tr><tr><td>Ah</td><td>74.55%</td></tr><tr><td>Bh</td><td>78.92%</td></tr><tr><td>Ch</td><td>83.17%</td></tr><tr><td>Dh</td><td>87.43%</td></tr><tr><td>Eh</td><td>91.75%</td></tr><tr><td>Fh</td><td>95.89%</td></tr><tr><td>10h</td><td>100.00% (Default at Reset)</td></tr><tr><td>11h</td><td>104.09</td></tr><tr><td>12h</td><td>108.17</td></tr><tr><td>13h</td><td>112.32</td></tr><tr><td>14h</td><td>116.41</td></tr><tr><td>15h</td><td>120.67</td></tr><tr><td>16h</td><td>124.42</td></tr><tr><td>17h</td><td>128.38</td></tr><tr><td>18h</td><td>132.24</td></tr><tr><td>19h</td><td>136.15</td></tr><tr><td>1Ah</td><td>140.15</td></tr><tr><td>1Bh</td><td>143.94</td></tr><tr><td>1Ch</td><td>148.02</td></tr><tr><td>1Dh</td><td>151.80x</td></tr><tr><td>1Eh</td><td>155.50x</td></tr><tr><td>1Fh</td><td>159.35%</td></tr></table>	0	29.52	1h	34.24%	2h	38.85%	3h	43.45%	4h	47.99%	5h	52.55%	6h	57.02%	7h	61.46%	8h	65.92%	9h	70.17	Ah	74.55%	Bh	78.92%	Ch	83.17%	Dh	87.43%	Eh	91.75%	Fh	95.89%	10h	100.00% (Default at Reset)	11h	104.09	12h	108.17	13h	112.32	14h	116.41	15h	120.67	16h	124.42	17h	128.38	18h	132.24	19h	136.15	1Ah	140.15	1Bh	143.94	1Ch	148.02	1Dh	151.80x	1Eh	155.50x	1Fh	159.35%
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1h	34.24%																																																																		
2h	38.85%																																																																		
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15h	120.67																																																																		
16h	124.42																																																																		
17h	128.38																																																																		
18h	132.24																																																																		
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1Bh	143.94																																																																		
1Ch	148.02																																																																		
1Dh	151.80x																																																																		
1Eh	155.50x																																																																		
1Fh	159.35%																																																																		
7-5	Reserved	0	Reads return 0. Writes have no effect.																																																																

Table 2-49. LPO/Clock Monitor Control Register (LPOMONCTL) Field Descriptions (continued)

Bit	Field	Value	Description
4-0	LFTRIM		Low-frequency oscillator trim value. This four-bit value is used to center the LF oscillator's frequency. Caution: This value should only be changed when the LF oscillator is not the source for a clock domain, otherwise a system failure could result. The following values are the ratio: f / f_0 in the F021 process.
		0	20.67
		1h	25.76
		2h	30.84
		3h	35.90
		4h	40.93
		5h	45.95
		6h	50.97
		7h	55.91
		8h	60.86
		9h	65.78
		Ah	70.75
		Bh	75.63
		Ch	80.61
		Dh	85.39
		Eh	90.23
		Fh	95.11
		10h	100.00% (Default at Reset)
		11h	104.84
		12h	109.51
		13h	114.31
		14h	119.01
		15h	123.75
		16h	128.62
		17h	133.31
		18h	138.03
		19h	142.75
		1Ah	147.32
		1Bh	152.02
		1Ch	156.63
		1Dh	161.38
		1Eh	165.90
		1Fh	170.42

2.5.1.32 Clock Test Register (CLKTEST)

The CLKTEST register, shown in [Figure 2-36](#) and described in [Table 2-50](#), controls the clock signal that is supplied to the ECLK pin for test and debug purposes.

NOTE: Clock Test Register Usage

This register should only be used for test and debug purposes.

Figure 2-36. Clock Test Register (CLKTEST) [offset = 8Ch]

31	27	26	25	24
Reserved	ALTLIMPCLOCK ENABLE	RANGEDET CTRL	RANGEDET ENASSEL	
R-0	R/WP-0	R/WP-0	R/WP-0	
23	20	19	16	
Reserved	CLK_TEST_EN			
R-0	R/WP-Ah			
15	12	11	8	7
Reserved	SEL_GPIO_PIN	Reserved	SEL_ECP_PIN	
R-0	R/WP-0	R-0	R/WP-0	

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-50. Clock Test Register (CLKTEST) Field Descriptions

Bit	Field	Value	Description
31-27	Reserved	0	Reads return 0. Writes have no effect.
26	ALTLIMPCLOCKENABLE	0 1	This bit selects a clock driven by the GIOB[0] pin as an alternate limp clock to the clock monitor phase frequency detect (PFD). The 10-MHz LPO fast clock is the compare clock for the clock detect PFD circuit and the source to limp clock on a clock fail. The ALTLIMPCLOCK driven on the GIOB[0] pin is the compare clock for the clock detect PFD circuit and the source to limp clock on a clock fail.
25	RANGEDETCTRL	0 1	Range detection control. This bit's functionality is dependant on the state of the RANGEDETENSSEL bit (Bit 24) of the CLKTEST register. The clock monitor range detection circuitry (RANGEDETECTENABLE) is disabled. The clock monitor range detection circuitry (RANGEDETECTENABLE) is enabled.
24	RANGEDETENASSEL	0 1	Selects range detection enable. This bit resets asynchronously on power on reset. The range detect enable is generated by the hardware in the clock monitor wrapper. The range detect enable is controlled by the RANGEDETCTRL bit (Bit 25) of the CLKTEST register.
23-20	Reserved	0	Reads return 0. Writes have no effect.
19-16	CLK_TEST_EN	5h Others	Clock test enable. This bit enables the clock going to the ECLK pin. Note: The ECLK pin must also be placed into Functional mode by setting the ECPCLKFUN bit to 1 in the SYSPC1 register. Clock going to ECLK pin is enabled. Clock going to ECLK pin is disabled.
15-12	Reserved	0	Reads return 0. Writes have no effect.

Table 2-50. Clock Test Register (CLKTEST) Field Descriptions (continued)

Bit	Field	Value	Description
11-8	SEL_GPIO_PIN		GIOB[0] pin clock source valid, clock source select
		0	Oscillator valid status
		1h	PLL1 valid status
		2h-4h	Reserved
		5h	High-frequency LPO (Low-Power Oscillator) clock valid status
		6h	PLL2 valid status
		7h	Reserved
		8h	Low-frequency LPO (Low-Power Oscillator) clock valid status
		9h-Ch	Oscillator valid status
		Dh	VCLKA3
		Eh	VCLKA4
		Fh	Oscillator valid status
7-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	SEL_ECP_PIN		ECLK pin clock source select
			Note: Only valid clock sources can be selected for the ECLK pin. Valid clock sources are displayed by the CSVSTAT register.
		0	Oscillator clock
		1h	PLL1 clock output
		2h	Reserved
		3h	EXTCLKIN1
		4h	Low-frequency LPO (Low-Power Oscillator) clock
		5h	High-frequency LPO (Low-Power Oscillator) clock
		6h	Secondary PLL free-running clock output
		7h	EXTCLKIN2
		8h	GCLK
		9h	RTI Base
		Ah	Reserved
		Bh	VCLKA1
		Ch	Reserved
		Dh	VCLKA3_DIVR
		Eh	VCLKA4
		Fh	Reserved

NOTE: Nonimplemented clock sources should not be enabled or used.

2.5.1.33 DFT Control Register (DFTCTRLREG)

This register is shown in [Figure 2-37](#) and described in [Table 2-51](#).

Figure 2-37. DFT Control Register (DFTCTRLREG) [offset = 90h]

31															16															
Reserved																														
R-0																														
15		14		13		12		11		10		9		8		7						4		3		0				
Reserved		DFTWRITE		Reserved		DFTREAD		Reserved				TEST_MODE_KEY																		
R-0		R/WP-1h		R-0		R/WP-1h		R-0				R/WP-5h																		

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

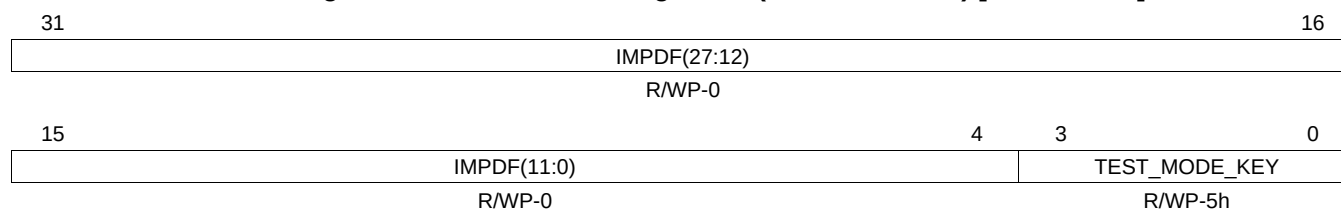
Table 2-51. DFT Control Register (DFTCTRLREG) Field Descriptions

Bit	Field	Value	Description
31-14	Reserved	0	Reads return 0. Writes have no effect.
13-12	DFTWRITE		DFT logic access. For F021: DFTWRITE[0] = 0 and DFTREAD[0] = 0 configured in stress mode DFTWRITE[1] = 0 and DFTREAD[1] = 0 configured in stress mode DFTWRITE[0] = 0 and DFTREAD[0] = 0 configured in fast mode DFTWRITE[1] = 1 and DFTREAD[1] = 1 configured in fast mode DFTWRITE[0] = 1 and DFTREAD[0] = 1 configured in slow mode DFTWRITE[1] = 0 and DFTREAD[1] = 0 configured in slow mode DFTWRITE[0] = 1 and DFTREAD[0] = 1 configured in screen mode DFTWRITE[1] = 1 and DFTREAD[1] = 1 configured in screen mode
11-10	Reserved	0	Reads return 0. Writes have no effect.
9-8	DFTREAD		DFT logic access. For F021: DFTWRITE[0] = 0 and DFTREAD[0] = 0 configured in stress mode DFTWRITE[1] = 0 and DFTREAD[1] = 0 configured in stress mode DFTWRITE[0] = 0 and DFTREAD[0] = 0 configured in fast mode DFTWRITE[1] = 1 and DFTREAD[1] = 1 configured in fast mode DFTWRITE[0] = 1 and DFTREAD[0] = 1 configured in slow mode DFTWRITE[1] = 0 and DFTREAD[1] = 0 configured in slow mode DFTWRITE[0] = 1 and DFTREAD[0] = 1 configured in screen mode DFTWRITE[1] = 1 and DFTREAD[1] = 1 configured in screen mode
7-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	TEST_MODE_KEY	0-Fh (except Ah) Ah	Test mode key. This register is for internal TI use only. Register key disable. All bits in this register will maintain their default value and cannot be written. Register key enable. All the bits can be written to only when the key is enabled. On reset, these bits will be set to 5h.

2.5.1.34 DFT Control Register 2 (DFTCTRLREG2)

This register is shown in [Figure 2-38](#) and described in [Table 2-52](#).

Figure 2-38. DFT Control Register 2 (DFTCTRLREG2) [offset = 94h]



LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-52. DFT Control Register 2 (DFTCTRLREG2) Field Descriptions

Bit	Field	Value	Description
31-4	IMPDPF[27:0]		DFT Implementation defined bits.
		0	IMPDPF[27:0] - Disabled
		1	IMPDPF[27:0] - Enabled
3-0	TEST_MODE_KEY		Test mode key. This register is for internal TI use only.
		0-Fh (except Ah)	Register key disable. All bits in this register will maintain their default value and cannot be written.
		Ah	Register key enable. All the bits can be written to only when the key is enabled.

2.5.1.35 General Purpose Register (GPREG1)

This register is shown in [Figure 2-39](#) and described in [Table 2-53](#).

Figure 2-39. General Purpose Register (GPREG1) [offset = A0h]

31	30	26	25	20	19	16
EMIF_FUNC	Reserved	PLL1_FBSLIP_FILTER_COUNT	PLL1_FBSLIP_FILTER_KEY			
R/WP-0	R-0	R/WP-0	R/WP-5h			
15						0
OUTPUT_BUFFER_LOW_EMI_MODE						
R/WP-FFFFh						

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-53. General Purpose Register (GPREG1) Field Descriptions

Bit	Field	Value	Description
31	EMIF_FUNC	0	Enable EMIF functions to be output. EMIF_ADDR[0], EMIF_ADDR[1], EMIF_ADDR[6], EMIF_ADDR[7], EMIF_ADDR[8], EMIF_BA[1], EMIF_nCS[0], and EMIF_nCS[3] are multiplexed with N2HET2 signals. By default, these terminals are tri-stated and pulled down. Any application that requires the EMIF functionality must set the EMIF_FUNC bit. This allows these 8 EMIF module outputs to be driven on to the assigned balls.
30-26	Reserved	0	Reads return 0. Writes have no effect.
25-20	PLL1_FBSLIP_FILTER_COUNT	0-3Fh	FBSLIP down counter programmed value.
		0	Configures the system response when a FBSLIP is indicated by the PLL macro. When PLL1_FBSLIP_FILTER_KEY is not Ah, the down counter counts from the programmed value on every LPO high frequency clock once PLL macro indicates FBSLIP. When the count reaches 0, if the synchronized FBSLIP signal is still high, an FBSLIP condition is indicated to the system module and is captured in the global status register. When the FBSLIP signal from the PLL macro is de-asserted before the count reaches 0, the counter is reloaded with the programmed value.
		0	On reset, counter value is 0. Counter must be programmed to a non-zero value and enabled for the filtering to be enabled.
		1h	Filtering is disabled.
		2h	Filtering is enabled. Every slip is recognized.
		2h	Filtering is enabled. The slip must be at least 2 HF LPO cycles wide in order to be recognized as a slip.
		:	:
		3Fh	Filtering is enabled. The slip must be at least 63 HF LPO cycles wide in order to be recognized as a slip.
19-16	PLL1_FBSLIP_FILTER_KEY	5h	Enable the FBSLIP filtering.
		5h	On reset, the FBSLIP filter is disabled and the FBSLIP passes through.
		Fh	This is an unsupported value. You should avoid writing this value to this bit field.
		All other values	FBSLIP filtering is enabled. Recommended to program Ah in this bit field. Enabling of the FBSLIP occurs when the KEY is programmed and a non-zero value is present in the COUNT field.

Table 2-53. General Purpose Register (GPREG1) Field Descriptions (continued)

Bit	Field	Value	Description
15-0	OUTPUT_BUFFER_LOW_EMI_MODE		Control field for the low-EMI mode of output buffers for module/signals: bit[0] controls MiBSP1 bit[1] controls SPI2 bit[2] controls MiBSP3 bit[3] controls SPI4 bit[4] controls MiBSP5 bit[5] Reserved bit[6] controls EMIF bit[7] Reserved bit[8] controls signal TMS bit[9] controls signal TDI bit[10] controls signal TDO bit[11] controls signal RTCK bit[12] controls signal TEST bit[13] controls signal nERROR bit[14] controls signal ADEVT bit[15] Reserved
		0	Enable EMI mode for each connected output buffers.
		1h-FFFFh	Enable/Disable EMI mode for connected output buffers.
		FFFFh	Disable EMI mode for each connected output buffer.

2.5.1.36 Imprecise Fault Status Register (IMPFASTS)

The IMPFASTS register, shown in [Figure 2-40](#) and described in [Table 2-54](#), displays information about imprecise aborts that have occurred.

Figure 2-40. Imprecise Fault Status Register (IMPFASTS) [offset = A8h]

31				24				23				16									
Reserved								MASTERID													
R-0								R-0													
15				11				10		9		8		7		1				0	
Reserved						EMIFA		NCBA		VBUSA		Reserved						ATYPE			
R-0						R-0		R-0		R-0		R-0						R/WC-0			

LEGEND: R/W = Read/Write; R = Read only; C = Clear; -n = value after reset

Table 2-54. Imprecise Fault Status Register (IMPFASTS) Field Descriptions

Bit	Field	Value	Description
31-24	Reserved	0	Reads return 0. Writes have no effect.
23-16	MASTERID	0-FFh	Master ID. This register indicates which master is responsible for the imprecise abort. The master ID value depends on device implementation- see Table 2-2 for MASTERID values for each bus master. Notes: - These bits are only updated when an imprecise abort occurs - These bits are cleared to 0 only on power-on reset. The value of these bits remains unchanged after all other resets.
15-11	Reserved	0	Reads return 0. Writes have no effect.
10	EMIFA	0 1	EMIF imprecise abort. This register indicates the imprecise abort was generated writing into the EMIF. Notes: - This bit is only updated when an imprecise abort occurs - This bit is cleared to 0 only on power-on reset. The value of this register remains unchanged after all other resets. 0 EMIF did not generate the last imprecise abort. 1 EMIF was written with an illegal address and generated an imprecise abort.
9	NCBA	0 1	Non-cacheable, bufferable abort (NCBA). This register indicates the imprecise abort was generated by a non-cacheable, bufferable write or shared device write through the write buffer of the CPU. Notes: - This bit is only updated when an imprecise abort generated by a non-cacheable, bufferable write or shared device write occurs. - This bit is cleared to 0 only on power-on reset. The value of this register remains unchanged after all other resets. 0 A NCBA is not responsible for the last imprecise abort. 1 A NCBA was written with an illegal address and generated an imprecise abort.
8	VBUSA	0 1	VBUS abort. This register indicates the imprecise abort was generated when writing into the peripheral frame. Notes: - This bit is only updated when an imprecise abort is generated when writing into the peripheral frame - This bit is cleared to 0 only on power-on reset. The value of this register remains unchanged after all other resets. 0 The peripheral frame did not generate the last imprecise abort. 1 The peripheral frame was written with an illegal address and generated an imprecise abort.
7-1	Reserved	0	Reads return 0. Writes have no effect.

Table 2-54. Imprecise Fault Status Register (IMPFASTS) Field Descriptions (continued)

Bit	Field	Value	Description
0	ATYPE		Abort type. This bit indicates to the CPU whether the last abort was an imprecise abort or a precise abort. Notes: - This bit is updated after each abort is generated to the CPU. - This bit is cleared on CPU read. - This bit is cleared to 0 only on power-on reset. The value of this bit remains unchanged after all other resets 0 The last abort generated was a precise abort. MASTERID, VBUSA, NCBA, EMIFA and IMPFTADD were not updated. 1 The last abort generated was an imprecise abort. MASTERID, VBUSA, NCBA, EMIFA and IMPFTADD were updated. Note: Once ATYPE is set, the IMPFAWADD and IMPFASTS bits are not updated by subsequent ABORT signals.

NOTE: The DMA and the peripheral master port will also generate an imprecise abort to the CPU when writing to the peripheral region or to the EMIF region. This will be indicated in the Master ID field of this register.

2.5.1.37 Imprecise Fault Address Register (IMPFTADD)

This IMPFTADD register, shown in [Figure 2-41](#) and described in [Table 2-55](#), shows the address at which an imprecise abort occurred.

Figure 2-41. Imprecise Fault Write Address Register (IMPFTADD) [offset = ACh]

31	16
IMPFTADD	
R-0	
15	0
IMPFTADD	
R-0	

LEGEND: R = Read only; -n = value after reset

Table 2-55. Imprecise Fault Write Address Register (IMPFTADD) Field Descriptions

Bit	Field	Value	Description
31-0	IMPFTADD	0- FFFF FFFFh	These bits contain the fault address when an imprecise abort occurs. Note: These bits are only updated when an imprecise abort occurs. Note: These bits are cleared to 0 only on power-on reset. The value of this register remains unchanged after all other resets.

2.5.1.38 System Software Interrupt Request 1 Register (SSIR1)

The SSIR1 register, shown in [Figure 2-42](#) and described in [Table 2-56](#), is used for software interrupt generation.

Figure 2-42. System Software Interrupt Request 1 Register (SSIR1) [offset = B0h]

31	Reserved																16
R-0																	
15	SSKEY1								8	7	SSDATA1						0
R/W-0									R/W-0								

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 2-56. System Software Interrupt Request 1 Register (SSIR1) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-8	SSKEY1	0-FFh	System software interrupt request key. A 075h written to these bits initiates IRQ/FIQ interrupts. Data in this field is always read as 0. The SSKEY1 field can be written into only if the write data matches the key (75h). The SSDATA1 field can only be written into if the write data into this field, the SSKEY1 field, matches the key (75h).
7-0	SSDATA1	0-FFh	System software interrupt data. These bits contain user read/write register bits. They may be used by the application software as different entry points for the interrupt routine. The SSDATA1 field cannot be written into unless the write data into the SSKEY1 field matches the key (75h); therefore, byte writes cannot be performed on the SSDATA1 field.

NOTE: This register is mirrored at offset FCh for compatibility reasons.

2.5.1.39 System Software Interrupt Request 2 Register (SSIR2)

The SSIR2 register, shown in [Figure 2-43](#) and described in [Table 2-57](#), is used for software interrupt generation.

Figure 2-43. System Software Interrupt Request 2 Register (SSIR2) [offset = B4h]

31	Reserved																16
R-0																	
15	SSKEY2								8	7	SSDATA2						0
R/W-0									R/W-0								

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 2-57. System Software Interrupt Request 2 Register (SSIR2) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-8	SSKEY2	0-FFh	System software interrupt2 request key. A 84h written to these bits initiates IRQ/FIQ interrupts. Data in this field is always read as 0. The SSKEY2 field can be written into only if the write data matches the key (84h). The SSDATA2 field can only be written into if the write data into this field, the SSKEY2 field, matches the key (84h).
7-0	SSDATA2	0-FFh	System software interrupt data. These bits contain user read/write register bits. They may be used by the application software as different entry points for the interrupt routine. The SSDATA2 field cannot be written into unless the write data into the SSKEY2 field matches the key (84h); therefore, byte writes cannot be performed on the SSDATA2 field.

2.5.1.40 System Software Interrupt Request 3 Register (SSIR3)

The SSIR3 register, shown in [Figure 2-44](#) and described in [Table 2-58](#), is used for software interrupt generation.

Figure 2-44. System Software Interrupt Request 3 Register (SSIR3) [offset = B8h]

31	Reserved																16
R-0																	
15	SSKEY3								8	7	SSDATA3						0
R/W-0									R/W-0								

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 2-58. System Software Interrupt Request 3 Register (SSIR3) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-8	SSKEY3	0-FFh	System software interrupt request key. A 93h written to these bits initiates IRQ/FIQ interrupts. Data in this field is always read as 0. The SSKEY3 field can be written into only if the write data matches the key (93h). The SSDATA3 field can only be written into if the write data into this field, the SSKEY3 field, matches the key (93h).
7-0	SSDATA3	0-FFh	System software interrupt data. These bits contain user read/write register bits. They may be used by the application software as different entry points for the interrupt routine. The SSDATA3 field cannot be written into unless the write data into the SSKEY3 field matches the key (93h); therefore, byte writes cannot be performed on the SSDATA3 field.

2.5.1.41 System Software Interrupt Request 4 Register (SSIR4)

The SSIR4 register, shown in [Figure 2-45](#) and described in [Table 2-59](#), is used for software interrupt generation.

Figure 2-45. System Software Interrupt Request 4 Register (SSIR4) [offset = BCh]

31	Reserved																16
R-0																	
15	SSKEY4								8	7	SSDATA4						0
R/W-0								R/W-0									

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 2-59. System Software Interrupt Request 4 Register (SSIR4) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-8	SSKEY4	0-FFh	System software interrupt2 request key. A A2h written to these bits initiates IRQ/FIQ interrupts. Data in this field is always read as 0. The SSKEY4 field can be written into only if the write data matches the key (A2h). The SSDATA4 field can only be written into if the write data into this field, the SSKEY4 field, matches the key (A2h).
7-0	SSDATA4	0-FFh	System software interrupt data. These bits contain user read/write register bits. They may be used by the application software as different entry points for the interrupt routine. The SSDATA4 field cannot be written into unless the write data into the SSKEY4 field matches the key (A2h); therefore, byte writes cannot be performed on the SSDATA4 field.

2.5.1.42 RAM Control Register (RAMGCR)

The RAMGCR register, shown in [Figure 2-46](#) and described in [Table 2-60](#), is used to configure eSRAM data and address wait states.

NOTE: The RAM_DFT_EN bits are for TI internal use only.

The contents of the RAM_DFT_EN field should not be changed.

Figure 2-46. RAM Control Register (RAMGCR) [offset = C0h]

31		20	19	16
Reserved				RAM_DFT_EN
R-0				R/WP-5h
15				8
Reserved				
R-0				
7		3	2	1
Reserved		WST_AENAO	Reserved	WST_DENAO
R-0		R/WP-0	R-0	R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-60. RAM Control Register (RAMGCR) Field Descriptions

Bit	Field	Value	Description
31-20	Reserved	0	Reads return 0. Writes have no effect.
19-16	RAM_DFT_EN	Ah Others	Functional mode RAM DFT (Design For Test) port enable key. Note: For TI internal use only. RAM DFT port is enabled. RAM DFT port is disabled. Note: It is recommended that a value of 5h be used to disable the RAM DFT port. This value will give maximum protection from a bit flip inducing event that would inadvertently enable the controller.
15-3	Reserved	0	Reads return 0. Writes have no effect.
2	WST_AENAO	0 1	eSRAM data phase wait state enable bit. The default address setup time for eSRAM0 is used. The eSRAM address setup time is increased by one HCLK cycle.
1	Reserved	0	Reads return 0. Writes have no effect.
0	WST_DENAO	0 1	eSRAM data phase wait state enable bit. There are no wait states for eSRAM during the data phase. The eSRAM data phase setup time is increased by one HCLK cycle.

2.5.1.43 Bus Matrix Module Control Register 1 (BMMCR1)

The BMMCR1 register, shown in [Figure 2-47](#) and described in [Table 2-61](#), allows RAM and Program (Flash) memory addresses to be swapped.

Figure 2-47. Bus Matrix Module Control Register 1 (BMMCR) [offset = C4h]

31																16	
Reserved																	
R-0																	
15													4	3			0
Reserved													MEMSW				
R-0													R/WP-Ah				

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-61. Bus Matrix Module Control Register 1 (BMMCR) Field Descriptions

Bit	Field	Value	Description
31-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	MEMSW		Memory swap key. Note: A CPU reset must be issued after the memory swap key has been changed for the memory swap to occur. A CPU reset can be initiated by changing the state of the CPU RESET bit in the CPURSTCR register.
		5h	Swapped memory-map: eSRAM starts at address 0. Program memory (Flash) starts at address 800 0000h.
		Ah	Default memory-map: Program memory (Flash) starts at address 0. eSRAM starts at address 800 0000h.
		Others	The device memory-map is unchanged.

2.5.1.44 CPU Reset Control Register (CPURSTCR)

The CPURSTCR register shown in [Figure 2-48](#) and described in [Table 2-62](#) allows a reset to the Cortex-R4F CPU to be generated.

NOTE: The register bits in CPURSTCR are designated as high-integrity bits and have been implemented with error-correcting logic such that each bit, although read and written as a single bit, is actually a multi-bit key with error correction capability. As such, single-bit flips within the "key" can be corrected allowing protection of the system as a whole. An error detected is signaled to the ESM module.

Figure 2-48. CPU Reset Control Register (CPURSTCR) [offset = CCh]

31	Reserved															16
R-0																
15	Reserved										1	0				
R-0											CPU RESET					
R-0											R/WP-0					

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-62. CPU Reset Control Register (CPURSTGCR) Field Descriptions

Bit	Field	Value	Description
31-1	Reserved	0	Reads return 0. Writes have no effect.
0	CPU RESET		CPU Reset. Only the CPU is reset whenever this bit is toggled. There is no system reset.

2.5.1.45 Clock Control Register (CLKCNTL)

The CLKCNTL register, shown in [Figure 2-49](#) and described in [Table 2-63](#), controls peripheral reset and the peripheral clock divide ratios.

NOTE: VCLK and VCLK2 clock ratio restrictions.

The VCLK2 frequency must always be greater than or equal to the VCLK frequency. The VCLK2 frequency must be an integer multiple of the VCLK frequency.

In addition, the VCLK and VCLK2 clock ratios must not be changed simultaneously. When increasing the frequency (decreasing the divider), first change the VCLK2R field and then change the VCLKR field. When reducing the frequency (increasing the divider), first change the VCLKR field and then change the VCLK2R field.

You should do a read-back between the two writes. This assures that there are enough clock cycles between the two writes.

Figure 2-49. Clock Control Register (CLKCNTL) [offset = D0h]

31	28	27	24	23	20	19	16
Reserved		VCLK2R		Reserved		VCLKR	
R-0		R/WP-1h		R-0		R/WP-1h	
15	9	8	7	0			
Reserved		PENA		Reserved			
R-0		R/WP-0		R-0			

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-63. Clock Control Register (CLKCNTL) Field Descriptions

Bit	Field	Value	Description
31-28	Reserved	0	Reads return 0. Writes have no effect.
27-24	VCLK2R	0 : Fh	VBUS clock2 ratio. Note: The VCLK2 frequency must always be greater than or equal to the VCLK frequency. The VCLK2 frequency must be an integer multiple of the VCLK frequency. In addition, the VCLK and VCLK2 clock ratios must not be changed simultaneously. The VCLK2 speed is HCLK divided by 1. : The VCLK2 speed is HCLK divided by 16.
23-20	Reserved	0	Reads return 0. Writes have no effect.
19-16	VCLKR	0 : Fh	VBUS clock ratio. Note: The VCLK2 frequency must always be greater than or equal to the VCLK frequency. The VCLK2 frequency must be an integer multiple of the VCLK frequency. In addition, the VCLK and VCLK2 clock ratios must not be changed simultaneously. The VCLK speed is HCLK divided by 1. : The VCLK speed is HCLK divided by 16.
15-9	Reserved	0	Reads return 0. Writes have no effect.
8	PENA	0 1	Peripheral enable bit. The application must set this bit before accessing any peripheral The global peripheral/peripheral memory frames are in reset. All peripheral/peripheral memory frames are out of reset.
7-0	Reserved	0	Reads return 0. Writes have no effect.

2.5.1.46 ECP Control Register (ECPCNTL)

The ECP register, shown in [Figure 2-50](#) and described in [Table 2-64](#), configures the ECLK pin in functional mode.

NOTE: ECLK Functional mode configuration.

The ECLK pin must be placed into Functional mode by setting the ECPCLKFUN bit to 1 in the SYSPC1 register before a clock source will be visible on the ECLK pin.

Figure 2-50. ECP Control Register (ECPCNTL) [offset = D4h]

31	25	24	23	22	18	17	16
Reserved	ECPSSEL	ECPCOS	Reserved	ECPINSEL			
R-0	R/W-0	R/W-0	R-0	R/W-0			
15							0
ECPDIV							
R/W-0							

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 2-64. ECP Control Register (ECPCNTL) Field Descriptions

Bit	Field	Value	Description
31-25	Reserved	0	Reads return 0. Writes have no effect.
24	ECPSSEL	0 1	This bit allows the selection between VCLK and OSCIN as the clock source for ECLK. Note: Other ECLK clock sources are available for debug purposes by configuring the CLKTEST register. VCLK is selected as the ECP clock source. OSCIN is selected as the ECP clock source.
23	ECPCOS	0 1	ECP continue on suspend. Note: Suspend mode is entered while performing certain JTAG debugging operations. ECLK output is disabled in suspend mode. ECLK output will be shut off and will not be seen on the I/O pin of the device. ECLK output is not disabled in suspend mode. ECLK output will not be shut off and will be seen on the I/O pin of the device.
22-18	Reserved	0	Reads return 0. Writes have no effect.
17-16	ECPINSEL	0 1h 2h 3h	Select ECP input clock source. Tied Low HCLK External clock Tied Low
15-0	ECPDIV	0-FFFFh	ECP divider value. The value of ECPDIV bits determine the external clock (ECP clock) frequency as a ratio of VBUS clock or OSCIN as shown in the formula: $ECLK = \frac{VCLK \text{ or } OSCIN}{(ECPDIV + 1)}$

2.5.1.47 DEV Parity Control Register 1 (DEVCR1)

This register is shown in [Figure 2-51](#) and described in [Table 2-65](#).

Figure 2-51. DEV Parity Control Register 1 (DEVCR1) [offset = DCh]

31																16	
Reserved																	
R-0																	
15											4	3					0
Reserved											DEVPARSEL						
R-0											R/WP-Ah						

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-65. DEV Parity Control Register 1 (DEVCR1) Field Descriptions

Bit	Field	Value	Description
31-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	DEVPARSEL		Device parity select bit key. Note: After an odd (DEVPARSEL = 5h) or even (DEVPARSEL = Ah) scheme is programmed into the DEVPARSEL register, any one bit change can be detected and will retain its programmed scheme. More than one bit changes in DEVPARSEL will cause a default to odd parity scheme.
		5h	The device parity is even.
		Ah	The device parity is odd.

2.5.1.48 System Exception Control Register (SYSECR)

The SYSECR register, shown in [Figure 2-52](#) and described in [Table 2-66](#), is used to generate a software reset.

NOTE: The register bits in SYSECR are designated as high-integrity bits and have been implemented with error-correcting logic such that each bit, although read and written as a single bit, is actually a multi-bit key with error correction capability. As such, single-bit flips within the “key” can be corrected allowing protection of the system as a whole. An error detected is signaled to the ESM module.

Figure 2-52. System Exception Control Register (SYSECR) [offset = E0h]

31		Reserved														16	
R-0																	
15		14		13												0	
RESET1	RESET0	Reserved															
R/WP-0	R/WP-1	R-0															

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-66. System Exception Control Register (SYSECR) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-14	RESET[1-0]		Software reset bits. Setting RESET1 or clearing RESET0 causes a system software reset.
		1h	No reset will occur.
		1x, x0	A global system reset will occur.
13-0	Reserved	0	Reads return 0. Writes have no effect.

2.5.1.49 System Exception Status Register (SYSESR)

The SYSESR register, shown in [Figure 2-53](#) and described in [Table 2-67](#), shows the source for different resets encountered. Previous reset source status bits are not automatically cleared if new resets occur. After reading this register, the software should clear any flags that are set so that the source of future resets can be determined. Any bit in this register can be cleared by writing a 1 to the bit.

Figure 2-53. System Exception Status Register (SYSESR) [offset = E4h]

Reserved															
R-0															
Reserved				Reserved				Reserved				Reserved			
R/WC-X				R/WC-X*				R/WC-X*				R-0			
Reserved				CPURST				SWRST				EXTRST			
R-0				R/WC-X*				R/WC-X*				R/WC-X*			
R-0				R-0				R-0				R-0			

LEGEND: R/W = Read/Write; R = Read only; C = Clear; X = value unchanged after reset; X* = 0 after PORST but unchanged after other resets; -n = value after reset

Table 2-67. System Exception Status Register (SYSESR) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15	PORST	0 1	Power-on reset. This bit is set when a power-on reset occurs, either internally asserted by the VMON or externally asserted by the nPORRST pin. No power-on reset has occurred since this bit was last cleared. A reset was caused by a power-on reset. (This bit should be cleared after being read so that subsequent resets can be properly identified as not being power-on resets.)
14	OSCRST	0 1	Reset caused by an oscillator failure or PLL cycle slip. This bit is set when a reset is caused by an oscillator failure or PLL slip. Note: The action taken when an oscillator failure or PLL slip is detected must be configured in the PLLCTL1 register. No reset has occurred due to an oscillator failure or a PLL cycle slip. A reset was caused by an oscillator failure or a PLL cycle slip.
13	WDRST	0 1	Watchdog reset flag. This bit is set when the last reset was caused by the digital windowed watchdog. During debugging, the ICEPICK logic implemented on the microcontroller also allows a system reset to be generated via the debug logic (DBGST). This DBGST reset is also indicated on the WDRST bit of the SYSESR. This flag can also be set via a reset driven by ICEPICK. No reset has occurred because of the DWWD. A reset was caused by the DWWD.
12-6	Reserved	0	Reads return 0. Writes have no effect.
5	CPURST	0 1	CPU reset flag. This bit is set when the CPU is reset. Note: A CPU reset can be initiated by the CPU self-test controller (LBIST) or by changing the memory protection (MMU/MPU) configuration in CPURSTCR register. No CPU reset has occurred. A CPU reset occurred.
4	SWRST	0 1	Software reset flag. This bit is set when a software system reset has occurred. Note: A software system reset can be initiated by writing to the RESET bits in the SYSECR register. No software reset has occurred. A software reset occurred.

Table 2-67. System Exception Status Register (SYSESR) Field Descriptions (continued)

Bit	Field	Value	Description
3	EXTRST	0 1	External reset flag. This bit is set when a reset is caused by the external reset pin nRST or by any reset that also asserts the nRST pin (PORST, OSCRST, WDRST and SWRST). The external reset pin has not asserted a reset. A reset has been caused by the external reset pin.
2-0	Reserved	0	Reads return 0. Writes have no effect.

2.5.1.50 System Test Abort Status Register (SYSTASR)

This register is shown in [Figure 2-54](#) and described in [Table 2-68](#).

Figure 2-54. System Test Abort Status Register (SYSTASR) [offset = E8h]

31											16	
Reserved												
R-0												
15						5	4					0
Reserved							EFUSE_Abort					
R-0							R/WPC-X/0					

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; C = Clear; -X = Value unchanged after reset; -n = value after reset

Table 2-68. System Test Abort Status Register (SYSTASR) Field Descriptions

Bit	Field	Value	Description
31-5	Reserved	0	Reads return 0. Writes have no effect.
4-0	EFUSE_Abort	0 1h 2h 3h 4h Others 1Fh	Test Abort status flag. These bits are set when test abort occurred: <i>Read:</i> The last operation (if any) completed successfully. This is also the value that the error/status register is set to after reset. <i>Read:</i> Controller times out because there is no last row sent from the FuseROM. <i>Read:</i> The autoloader machine was started, either through the SYS_INITZ signal from the system or the JTAG data register. In either case, the autoloader machine did not find enough FuseROM data to fill the scan chain. <i>Read:</i> The autoloader machine was started, either through the SYS_INITZ signal from the system or the JTAG data register. In either case, the autoloader machine starts the scan chain with a signature it expects to see after the scan chain is full. The autoloader machine was able to fill the scan chain, but the wrong signature was returned. <i>Read:</i> The autoloader machine was started, either through the SYS_INITZ signal from the system or the JTAG data register. In either case, the autoloader machine was not able or not allowed to complete its operation. <i>Read:</i> Reserved. <i>Write:</i> These bits are cleared to 0.

2.5.1.51 Global Status Register (GLBSTAT)

The GLBSTAT register, shown in [Figure 2-55](#) and described in [Table 2-69](#), indicates the FMzPLL (PLL1) slip status and the oscillator fail status.

NOTE: PLL and OSC fail behavior

The device behavior after a PLL slip or an oscillator failure is configured in the PLLCTL1 register.

Figure 2-55. Global Status Register (GLBSTAT) [offset = ECh]

31																16
Reserved																
R-0																
15	10				9	8	7					1	0			
Reserved					FBSLIP		RFSLIP		Reserved				OSCFAIL			
R-0					R/W1C-n		R/W1C-n		R-0				R/W1C-n			

LEGEND: R/W = Read/Write; R = Read only; W1C = Write 1 to Clear; -n = value after reset

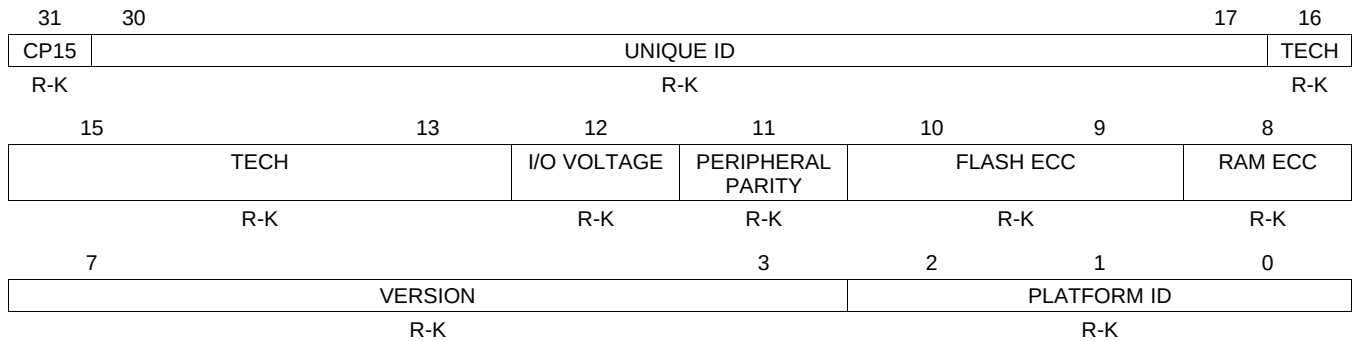
Table 2-69. Global Status Register (GLBSTAT) Field Descriptions

Bit	Field	Value	Description
31-10	Reserved	0	Reads return 0. Writes have no effect.
9	FBSLIP	0	PLL over cycle slip detection. (cleared by nPORRST, maintains its previous value for all other resets) <i>Read:</i> No PLL over cycle slip has been detected. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> A PLL over cycle slip has been detected. <i>Write:</i> The bit is cleared to 0.
8	RFSLIP	0	PLL under cycle slip detection. (cleared by nPORRST, maintains its previous value for all other resets) <i>Read:</i> No PLL under cycle slip has been detected. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> A PLL under cycle slip has been detected. <i>Write:</i> The bit is cleared to 0.
7-1	Reserved	0	Reads return 0. Writes have no effect.
0	OSCFAIL	0	Oscillator fail flag bit. (cleared by nPORRST, maintains its previous value for all other resets) <i>Read:</i> No oscillator failure has been detected. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> An oscillator failure has been detected. <i>Write:</i> The bit is cleared to 0.

2.5.1.52 Device Identification Register (DEVID)

The DEVID is a read-only register. It contains device-specific information that is hard-coded during device manufacture. For the initial silicon version, the device identification code value is 802A AD05h. This register is shown in [Figure 2-56](#) and described in [Table 2-70](#).

Figure 2-56. Device Identification Register (DEVID) [offset = F0h]



LEGEND: R = Read only; -n = value after reset; K = Constant value

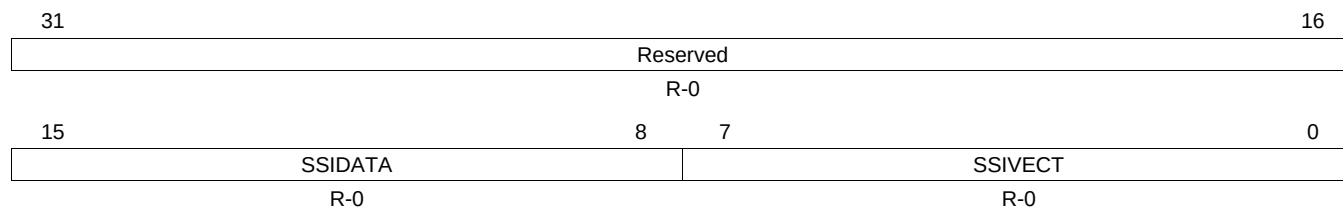
Table 2-70. Device Identification Register (DEVID) Field Descriptions

Bit	Field	Value	Description
31	CP15	0 1	CP15 CPU. This bit indicates whether the CPU has a coprocessor 15 (CP15). The CPU has no CP15 present. The CPU has a CP15 present. The CPU ID can be read using the CP15 C0,C0,0 register.
30-17	UNIQUE ID	0-3FFFh	Device ID. The device ID is unique by device configuration.
16-13	TECH	0 1h 2h 3h 4h 5h 6h-Fh	These bits define the process technology by which the device was manufactured. Device manufactured in the C05 process technology. Device manufactured in the F05 process technology. Device manufactured in the C035 process technology. Device manufactured in the F035 process technology. Device manufactured in the C021 process technology. Device manufactured in the F021 process technology. Reserved
12	I/O VOLTAGE	0 1	Input/output voltage. This bit defines the I/O voltage of the device. The I/O voltage is 3.3 V. The I/O voltage is 5 V.
11	PERIPHERAL PARITY	0 1	Peripheral parity. This bit indicates whether or not peripheral memory parity is present. The peripheral memories have no parity. The peripheral memories have parity.
10-9	FLASH ECC	0 1h 2h 3h	These bits indicate which parity is present for the program memory. No memory protection is present. The program memory (Flash) has single-bit parity. The program memory (Flash) has ECC. This combination is reserved.
8	RAM ECC	0 1	RAM ECC. This bit indicates whether or not RAM memory ECC is present. The RAM memories do not have ECC. The RAM memories have ECC.
7-3	VERSION	0-1Fh	Version. These bits provide the revision of the device.
2-0	PLATFORM ID	5h	The device is part of the TMS570 family. The TMS570 ID is always 5h.

2.5.1.53 Software Interrupt Vector Register (SSIVEC)

The SSIVEC register, shown in [Figure 2-57](#) and described in [Table 2-71](#), contains information about software interrupts.

Figure 2-57. Software Interrupt Vector Register (SSIVEC) [offset = F4h]



LEGEND: R = Read only; -n = value after reset

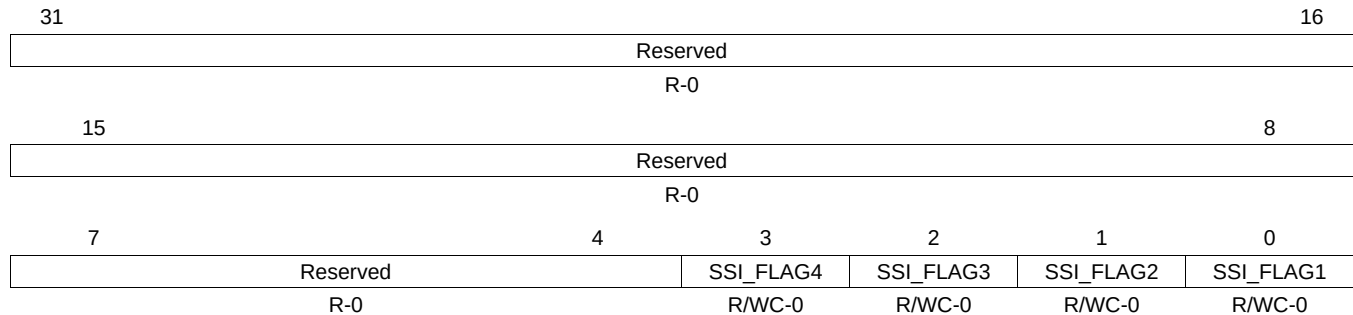
Table 2-71. Software Interrupt Vector Register (SSIVEC) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-8	SSIDATA	0-FFh	System software interrupt data key. These bits contain the data key value of the source for the system software interrupt, which is indicated by the vector in the SSIVEC[7-0] field.
7-0	SSIVECT		These bits contain the source for the system software interrupt. Note: A read from the SSIVECT bits clears the corresponding SSI_FLAG[4-1] bit in the SSIF register, corresponding to the source vector of the system software interrupt. Note: The SSIR[4-1] interrupt has the following priority order: SSIR1 has the highest priority. SSIR4 has the lowest priority.
		0	No software interrupt is pending.
		1h	A software interrupt has been generated by writing the correct key value to The SSIR1 register.
		2h	A software interrupt has been generated by writing the correct key value to The SSIR2 register.
		3h	A software interrupt has been generated by writing the correct key value to The SSIR3 register.
		4h	A software interrupt has been generated by writing the correct key value to The SSIR4 register.
		5h-FFh	Reserved

2.5.1.54 System Software Interrupt Flag Register (SSIF)

The SSIF register, shown in [Figure 2-58](#) and described in [Table 2-72](#), contains software interrupt flag status information.

Figure 2-58. System Software Interrupt Flag Register (SSIF) [offset = F8h]



LEGEND: R/W = Read/Write; R = Read only; C = Clear; -n = value after reset

Table 2-72. System Software Interrupt Flag Register (SSIF) Field Descriptions

Bit	Field	Value	Description
31-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	SSI_FLAG[4-1]		System software interrupt flag[4-1]. This flag is set when the correct SSKEY is written to the SSIR register[4-1]. Note: A read from the SSIVEC register clears the corresponding SSI_FLAG [4-1] bit in the SSIF, corresponding to the source vector of the system software interrupt.
		0	<i>Read:</i> No IRQ/FIQ interrupt was generated since the bit was last cleared. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> An IRQ/FIQ interrupt was generated. <i>Write:</i> The bit is cleared to 0.

2.5.2 Secondary System Control Registers (SYS2)

[Table 2-73](#) lists the secondary frame of system registers. The start address of the second system module frame is FFFF E100h. The registers support 32-, 16-, and 8-bit writes. The offset is relative to the system module frame start address.

Table 2-73. Secondary System Control Registers

Offset	Acronym	Register Description	Section
00h	PLLCTL3	PLL Control Register 3	Section 2.5.2.1
04h	Reserved	Reserved	
08h	STCCLKDIV	CPU Logic BIST Clock Divider	Section 2.5.2.2
0Ch-20h	Reserved	Reserved	
24h	ECPCNTL	ECP Control Register	Section 2.5.1.46
28h-38h	Reserved	Reserved	
3Ch	CLK2CNTRL	Clock 2 Control Register	Section 2.5.2.3
40h	VCLKACON1	Peripheral Asynchronous Clock Configuration 1 Register	Section 2.5.2.4
44h-6Ch	Reserved	Reserved	
70h	CLKSLIP	Clock Slip Register	Section 2.5.2.5
74h-E8h	Reserved	Reserved	
ECh	EFC_CTLREG	EFUSE Controller Control Register	Section 2.5.2.6
F0h	DIEIDL_REG0	Die Identification Register Lower Word	Section 2.5.2.7
F4h	DIEIDH_REG1	Die Identification Register Upper Word	Section 2.5.2.8
F8h	DIEIDL_REG2	Die Identification Register Lower Word	Section 2.5.2.9
FCh	DIEIDH_REG3	Die Identification Register Upper Word	Section 2.5.2.10

NOTE: All additional registers in the secondary system frame are reserved.

2.5.2.1 PLL Control Register 3 (PLLCTL3)

The PLLCTL3 register is shown in [Figure 2-59](#) and described in [Table 2-74](#); controls the settings of PLL2 (Clock Source 6 - FPLL).

Figure 2-59. PLL Control Register 3 (PLLCTL3) [offset = 00]

31	29	28	24	23	22	21	16
ODPLL2		PLLDIV2		Reserved		REFCLKDIV2	
R/WP-1h		R/WP-Fh		R-0		R/WP-2h	
15							0
PLL_MUL2							
R/WP-5F00h							

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-74. PLL Control Register 3 (PLLCTL3) Field Descriptions

Bit	Field	Value	Description
31-29	ODPLL2	0 1h : 7h	Internal PLL Output Divider $OD2 = ODPLL2 + 1$, ranges from 1 to 8. $f_{post_ODCLK2} = f_{output_CLK2} / OD2$ $f_{post_ODCLK2} = f_{output_CLK2} / 1$ $f_{post_ODCLK2} = f_{output_CLK2} / 2$ $f_{post_ODCLK2} = f_{output_CLK2} / 8$ Note: PLL output clock is gated off if ODPLL2 is changed while the PLL2 is active.
28-24	PLLDIV2	0 1h : 1Fh	PLL#2 Output Clock Divider $R2 = PLLDIV2 + 1$, ranges from 1 to 32. $f_{PLL2\ CLK} = f_{post_ODCLK2} / R2$ $f_{PLL2\ CLK} = f_{post_ODCLK2} / 1$ $f_{PLL2\ CLK} = f_{post_ODCLK2} / 2$ $f_{PLL2\ CLK} = f_{post_ODCLK2} / 32$
23-22	Reserved	0	Value has no effect on PLL operation.
21-16	REFCLKDIV2	0 1h : 3Fh	Reference Clock Divider $NR2 = REFCLKDIV2 + 1$, ranges from 1 to 64. $f_{INTCLK2} = f_{OSCIN} / NR2$ $f_{INTCLK2} = f_{OSCIN} / 1$ $f_{INTCLK2} = f_{OSCIN} / 2$ $f_{INTCLK2} = f_{OSCIN} / 64$ Note: This value should not be changed while the PLL2 is active.
15-0	PLLMUL2	100h : 5B00h 5C00h : FF00h	PLL2 Multiplication Factor $NF2 = (PLLMUL2 / 256) + 1$, valid multiplication factors are from 1 to 256. $f_{VCOCLK2} = f_{INTCLK2} \times NF2$ User and privileged mode (read): Privileged mode (write): $f_{VCOCLK2} = f_{INTCLK2} \times 1$ $f_{VCOCLK2} = f_{INTCLK2} \times 92$ $f_{VCOCLK2} = f_{INTCLK2} \times 93$ $f_{VCOCLK2} = f_{INTCLK2} \times 256$

2.5.2.2 CPU Logic Bist Clock Divider (STCLKDIV)

This register is shown in [Figure 2-60](#) and described in [Table 2-75](#).

Figure 2-60. CPU Logic BIST Clock Prescaler (STCLKDIV) [offset = 08h]

31	27	26	24	23	16
Reserved		CLKDIV		Reserved	
R-0		R/WP-0		R-0	
15					0
Reserved					
R-0					

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-75. CPU Logic BIST Clock Prescaler (STCLKDIV) Field Descriptions

Bit	Field	Value	Description
31-27	Reserved	0	Reads return 0. Writes have no effect.
26-24	CLKDIV	0	Clock divider/prescaler for CPU clock during logic BIST
23-0	Reserved	0	Reads return 0. Writes have no effect.

2.5.2.3 Clock 2 Control Register (CLK2CNTRL)

This register is shown in [Figure 2-61](#) and described in [Table 2-76](#).

Figure 2-61. Clock 2 Control Register (CLK2CNTRL) [offset = 3Ch]

31											16				
Reserved															
R-0															
15	12				11	8				7	4		3	0	
Reserved				VCLK4R				Reserved				VCLK3R			
R-0				R/WP-1h				R-0				R/WP-1h			

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-76. Clock 2 Control Register (CLK2CNTRL) Field Descriptions

Bit	Field	Value	Description
31-12	Reserved	0	Reads return 0. Writes have no effect.
11-8	VCLK4R	0	VBUS clock4 ratio.
		:	The ratio is HCLK divide by 1.
		Fh	The ratio is HCLK divided by 16.
7-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	VCLK3R	0	VBUS clock3 ratio.
		:	The ratio is HCLK divide by 1.
		Fh	The ratio is HCLK divided by 16.

2.5.2.4 Peripheral Asynchronous Clock Configuration 1 Register (VCLKACON1)

This register is shown in [Figure 2-62](#) and described in [Table 2-77](#).

Figure 2-62. Peripheral Asynchronous Clock Configuration 1 Register (VCLKACON1) [offset = 40h]

31		27	26	24
Reserved			VCLKA4R	
R-0			R/WP-1h	
23	21	20	19	16
Reserved		VCLKA4_DIV_CDDIS	VCLKA4S	
R-0		R/WP-0	R/WP-9h	
15		11	10	8
Reserved			VCLKA3R	
R-0			R/WP-1h	
7	5	4	3	0
Reserved		VCLKA3_DIV_CDDIS	VCLKA3S	
R-0		R/WP-0	R/WP-9h	

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-77. Peripheral Asynchronous Clock Configuration 1 Register (VCLKACON1) Field Descriptions

Bit	Field	Value	Description
31-27	Reserved	0	Reads return 0. Writes have no effect.
26-24	VCLKA4R	0 : 7h	Clock divider for the VCLKA4 source. Output will be present on VCLKA4_DIVR. VCLKA4 domain will be enabled by writing to the CDDIS register and VCLKA4_DIV_CDDIS bit. It can be inferred that VCLKA4_DIV clock is disabled when VCLKA4 clock is disabled. The ratio is VCLKA4 divided by 1. : The ratio is VCLKA4 divided by 8.
23-21	Reserved	0	Reads return 0. Writes have no effect.
20	VCLKA4_DIV_CDDIS	0 1	Disable the VCLKA4 divider output. VCLKA4 domain will be enabled by writing to the CDDIS register Enable the prescaled VCLKA4 clock on VCLKA4_DIVR. Disable the prescaled VCLKA4 clock on VCLKA4_DIVR.
19-16	VCLKA4S	0 1h 2h 3h 4h 5h 6h 7h 8h-Fh	Peripheral asynchronous clock4 source. Clock source0 is the source for peripheral asynchronous clock4. Clock source1 is the source for peripheral asynchronous clock4. Clock source2 is the source for peripheral asynchronous clock4. Clock source3 is the source for peripheral asynchronous clock4. Clock source4 is the source for peripheral asynchronous clock4. Clock source5 is the source for peripheral asynchronous clock4. Clock source6 is the source for peripheral asynchronous clock4. Clock source7 is the source for peripheral asynchronous clock4. VCLK or a divided VCLK is the source for peripheral asynchronous clock4. See the device-specific data manual for details.
15-11	Reserved	0	Reads return 0. Writes have no effect.

**Table 2-77. Peripheral Asynchronous Clock Configuration 1 Register (VCLKACON1)
Field Descriptions (continued)**

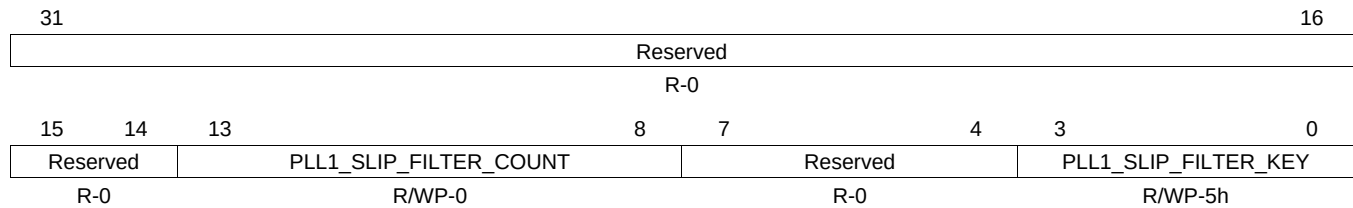
Bit	Field	Value	Description
10-8	VCLKA3R	0 : 7h	Clock divider for the VCLKA3 source. Output will be present on VCLKA3_DIVR. VCLKA3 domain will be enabled by writing to the CDDIS register and VCLKA3_DIV_CDDIS bit. It can inferred that VCLKA3_DIV clock is disabled when VCLKA3 clock is disabled. The ratio is VCLKA3 divided by 1. : The ratio is VCLKA3 divided by 8.
7-5	Reserved	0	Reads return 0. Writes have no effect.
4	VCLKA3_DIV_CDDIS	0 1	Disable the VCLKA3 divider output. VCLKA3 domain will be enabled by writing to the CDDIS register Enable the prescaled VCLKA3 clock on VCLKA3_DIVR. Disable the prescaled VCLKA3 clock on VCLKA3_DIVR.
3-0	VCLKA3S	0 1h 2h 3h 4h 5h 6h 7h 8h-Fh	Peripheral asynchronous clock3 source. Clock source0 is the source for peripheral asynchronous clock3. Clock source1 is the source for peripheral asynchronous clock3. Clock source2 is the source for peripheral asynchronous clock3. Clock source3 is the source for peripheral asynchronous clock3. Clock source4 is the source for peripheral asynchronous clock3. Clock source5 is the source for peripheral asynchronous clock3. Clock source6 is the source for peripheral asynchronous clock3. Clock source7 is the source for peripheral asynchronous clock3. VCLK is the source for peripheral asynchronous clock3.

NOTE: Nonimplemented clock sources should not be enabled or used. A list of the available clock sources is shown in [Table 2-9](#).

2.5.2.5 Clock Slip Register (CLKSLIP)

This register is shown in [Figure 2-63](#) and described in [Table 2-78](#).

Figure 2-63. Clock Slip Register (CLKSLIP) [offset = 70h]



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-78. Clock Slip Register (CLKSLIP) Field Descriptions

Bit	Field	Value	Description
31-14	Reserved	0	Reads return 0. Writes have no effect.
13-8	PLL1_SLIP_FILTER_COUNT	0 1h 2h : 3Fh	PLL SLIP down counter programmed value. Count is on 10M clock. On reset, counter value is 0. Counter must be programmed to a non-zero value and enabled for the filtering to be enabled. Filtering is disabled. Filtering is enabled. Every slip is recognized. Filtering is enabled. The slip must be at least 2 HF LPO cycles wide in order to be recognized as a slip. : Filtering is enabled. The SLIP must be at least 63 HF LPO cycles wide in order to be recognized as a slip.
7-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	PLL1_SLIP_FILTER_KEY	5h Fh Others	Enable the PLL SLIP filtering. On reset, the PLL SLIP filter is disabled and the PLL SLIP passes through. This is an unsupported value. You should avoid writing this value to this bit field. PLL SLIP filtering is enabled. Recommended to program Ah in this bit field. Enabling of the PLL SLIP occurs when the KEY is programmed and a non-zero value is present in the COUNT field.

2.5.2.6 EFUSE Controller Control Register (EFC_CTLREG)

This register is shown in [Figure 2-64](#) and described in [Table 2-79](#).

Figure 2-64. EFUSE Controller Control Register (EFC_CTLREG) [offset = ECh]

31																16	
Reserved																	
R-0																	
15											4	3					0
Reserved											EFC_INSTR_WEN						
R-0											R/WP-5h						

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 2-79. EFUSE Controller Control Register (EFC_CTLREG) Field Descriptions

Bit	Field	Value	Description
31-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	EFC_INSTR_WEN	Ah Others	Enable user write of 4 EFUSE controller instructions. SYS module generates the enable signal that will be tied to OCP_FROM_WRITE_DISABLE on efuse controller port Writing of instructions (Program, ProgramCRA, RunAutoload, and LoadFuseScanchain) to EFC registers is enabled. Writing of instructions (Program, ProgramCRA, RunAutoload, and LoadFuseScanchain) in EFC registers is blocked.

2.5.2.7 Die Identification Register Lower Word (DIEIDL_REG0)

The DIEIDL_REG0 register is a duplicate of the DIEIDL register, see [Section 2.5.1.29](#). The DIEIDL_REG0 register, shown in [Figure 2-65](#) and described in [Table 2-80](#), contains information about the die wafer number, and X, Y wafer coordinates.

Figure 2-65. Die Identification Register, Lower Word (DIEIDL_REG0) [offset = F0h]

31											24	23					16
WAFER #											Y WAFER COORDINATE						
R-D											R-D						
15											12	11					0
Y WAFER COORDINATE					X WAFER COORDINATE												
R-D					R-D												

LEGEND: R = Read only; -n = value after reset; D = value is device specific

Table 2-80. Die Identification Register, Lower Word (DIEIDL_REG0) Field Descriptions

Bit	Field	Description
31-24	WAFER #	These read-only bits contain the wafer number of the device.
23-12	Y WAFER COORDINATE	These read-only bits contain the Y wafer coordinate of the device.
11-0	X WAFER COORDINATE	These read-only bits contain the X wafer coordinate of the device.

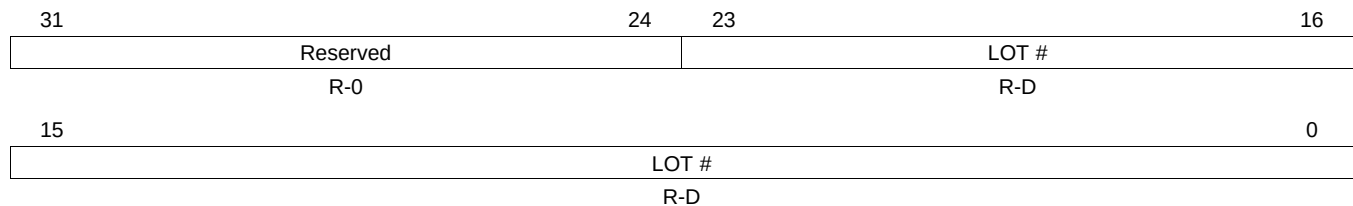
NOTE: Die Identification Information

The die identification information will vary from unit to unit. This information is programmed by TI as part of the initial device test procedure.

2.5.2.8 Die Identification Register Upper Word (DIEIDH_REG1)

The DIEIDH_REG1 register is a duplicate of the DIEIDH register, see [Section 2.5.1.30](#). The DIEIDH_REG1 register, shown in [Figure 2-66](#) and described in [Table 2-81](#), contains information about the die lot number.

Figure 2-66. Die Identification Register, Upper Word (DIEIDH_REG1) [offset = F4h]



LEGEND: R = Read only; -n = value after reset; D = value is device dependent

Table 2-81. Die Identification Register, Upper Word (DIEIDH_REG1) Field Descriptions

Bit	Field	Description
31-24	Reserved	Reserved for TI use. Writes have no effect.
23-0	LOT #	This read-only register contains the device lot number.

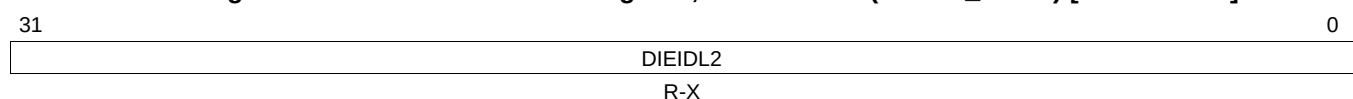
NOTE: Die Identification Information

The die identification information will vary from unit to unit. This information is programmed by TI as part of the initial device test procedure.

2.5.2.9 Die Identification Register Lower Word (DIEIDL_REG2)

This register is shown in [Figure 2-67](#) and described in [Table 2-82](#).

Figure 2-67. Die Identification Register, Lower Word (DIEIDL_REG2) [offset = F8h]



LEGEND: R = Read only; -n = value after reset; X = value is unchanged after reset

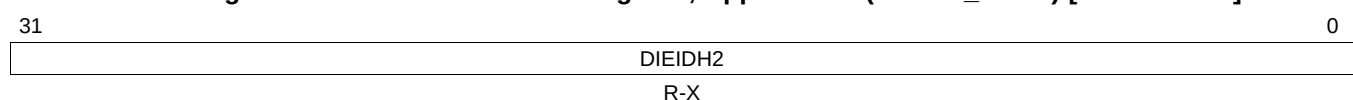
Table 2-82. Die Identification Register, Lower Word (DIEIDL_REG2) Field Descriptions

Bit	Field	Value	Description
31-0	DIEIDL2(95:64)	0-FFFF FFFFh	This read-only register contains the lower word (95:64) of the die ID information. See the device data sheet for the content of the register.

2.5.2.10 Die Identification Register Upper Word (DIEIDH_REG3)

This register is shown in [Figure 2-68](#) and described in [Table 2-83](#).

Figure 2-68. Die Identification Register, Upper Word (DIEIDH_REG3) [offset = FCh]



LEGEND: R = Read only; -n = value after reset; X = value is unchanged after reset

Table 2-83. Die Identification Register, Upper Word (DIEIDH_REG3) Field Descriptions

Bit	Field	Value	Description
31-0	DIEIDH2(127-96)	0-FFFF FFFFh	This read-only register contains the upper word (127:97) of the die ID information. See the device data sheet for the content of the register.

2.5.3 Peripheral Central Resource (PCR) Control Registers

This section describes the Peripheral Central Resource (PCR) control registers. The start address of the PCR register frame is FFFF E000h. [Table 2-84](#) summarizes the registers in the PCR, which are used to configure protection to the peripherals in PCS and PS regions. Not all chip selects exist on this device.

Table 2-84. Peripheral Central Resource Control Registers

Offset	Acronym	Register Description	Section
00h	PMPROTSET0	Peripheral Memory Protection Set Register 0	Section 2.5.3.1
04h	PMPROTSET1	Peripheral Memory Protection Set Register 1	Section 2.5.3.2
08h-0Ch	Reserved	Reserved	
10h	PMPROTCLR0	Peripheral Memory Protection Clear Register 0	Section 2.5.3.3
14h	PMPROTCLR1	Peripheral Memory Protection Clear Register 1	Section 2.5.3.4
18h-1Ch	Reserved	Reserved	
20h	PPROTSET0	Peripheral Protection Set Register 0	Section 2.5.3.5
24h	PPROTSET1	Peripheral Protection Set Register 1	Section 2.5.3.6
28h	PPROTSET2	Peripheral Protection Set Register 2	Section 2.5.3.7
2Ch	PPROTSET3	Peripheral Protection Set Register 3	Section 2.5.3.8
30h-3Ch	Reserved	Reserved	
40h	PPROTCLR0	Peripheral Protection Clear Register 0	Section 2.5.3.9
44h	PPROTCLR1	Peripheral Protection Clear Register 1	Section 2.5.3.10
48h	PPROTCLR2	Peripheral Protection Clear Register 2	Section 2.5.3.11
4Ch	PPROTCLR3	Peripheral Protection Clear Register 3	Section 2.5.3.12
50h-5Ch	Reserved	Reserved	
60h	PCSPWRDWNSET0	Peripheral Memory Power-Down Set Register 0	Section 2.5.3.13
64h	PCSPWRDWNSET1	Peripheral Memory Power-Down Set Register 1	Section 2.5.3.14
68h-6Ch	Reserved	Reserved	
70h	PCSPWRDWNCLR0	Peripheral Memory Power-Down Clear Register 0	Section 2.5.3.15
74h	PCSPWRDWNCLR1	Peripheral Memory Power-Down Clear Register 1	Section 2.5.3.16
78h-7Ch	Reserved	Reserved	
80h	PSPWRDWNSET0	Peripheral Power-Down Set Register 0	Section 2.5.3.17
84h	PSPWRDWNSET1	Peripheral Power-Down Set Register 1	Section 2.5.3.18
88h	PSPWRDWNSET2	Peripheral Power-Down Set Register 2	Section 2.5.3.19
8Ch	PSPWRDWNSET3	Peripheral Power-Down Set Register 3	Section 2.5.3.20
90h-9Ch	Reserved	Reserved	
A0h	PSPWRDWNCLR0	Peripheral Power-Down Clear Register 0	Section 2.5.3.21
A4h	PSPWRDWNCLR1	Peripheral Power-Down Clear Register 1	Section 2.5.3.22
A8h	PSPWRDWNCLR2	Peripheral Power-Down Clear Register 2	Section 2.5.3.23
ACh	PSPWRDWNCLR3	Peripheral Power-Down Clear Register 3	Section 2.5.3.24

2.5.3.1 Peripheral Memory Protection Set Register 0 (PMPROTSET0)

This register is shown in [Figure 2-69](#) and described in [Table 2-85](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-69. Peripheral Memory Protection Set Register 0 (PMPROTSET0) [offset = 00]

31	0
PCS[31-0]PROTSET	
R/WP-0	

LEGEND: R/W = Read/Write: WP = Write in privileged mode only: -n = value after reset

Table 2-85. Peripheral Memory Protection Set Register 0 (PMPROTSET0) Field Descriptions

Bit	Field	Value	Description
31-0	PCS[31-0]PROTSET	0	Peripheral memory frame protection set. <i>Read:</i> The peripheral memory frame can be written to and read from in both user and privileged modes. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The peripheral memory frame can be written to only in privileged mode, but it can be read in both user and privileged modes. <i>Write:</i> The corresponding bit in PMPROTSET0 and PMPROTCLR0 registers is set to 1.

2.5.3.2 Peripheral Memory Protection Set Register 1 (PMPROTSET1)

This register is shown in [Figure 2-70](#) and described in [Table 2-86](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-70. Peripheral Memory Protection Set Register 1 (PMPROTSET1) [offset = 04h]

31	0
PCS[63-32]PROTSET	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-86. Peripheral Memory Protection Set Register 1 (PMPROTSET1) Field Descriptions

Bit	Field	Value	Description
31-0	PCS[63-32]PROTSET	0	Peripheral memory frame protection set. <i>Read:</i> The peripheral memory frame can be written to and read from in both user and privileged modes. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The peripheral memory frame can be written to only in privileged mode, but it can be read in both user and privileged modes. <i>Write:</i> The corresponding bit in PMPROTSET1 and PMPROTCLR1 registers is set to 1.

2.5.3.3 Peripheral Memory Protection Clear Register 0 (PMPROTCLR0)

This register is shown in [Figure 2-71](#) and described in [Table 2-87](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-71. Peripheral Memory Protection Clear Register 0 (PMPROTCLR0) [offset = 10h]

31	0
PCS[31-0]PROTCLR	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-87. Peripheral Memory Protection Clear Register 0 (PMPROTCLR0) Field Descriptions

Bit	Field	Value	Description
31-0	PCS[31-0]PROTCLR	0	Peripheral memory frame protection clear. <i>Read:</i> The peripheral memory frame can be written to and read from in both user and privileged modes. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The peripheral memory frame can be written to only in privileged mode, but it can be read in both user and privileged modes. <i>Write:</i> The corresponding bit in PMPROTSET0 and PMPROTCLR0 registers is cleared to 0.

2.5.3.4 Peripheral Memory Protection Clear Register 1 (PMPROTCLR1)

This register is shown in [Figure 2-72](#) and described in [Table 2-88](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-72. Peripheral Memory Protection Clear Register 1 (PMPROTCLR1) [offset = 14h]

31	0
PCS[63-32]PROTCLR	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-88. Peripheral Memory Protection Clear Register 1 (PMPROTCLR1) Field Descriptions

Bit	Field	Value	Description
31-0	PCS[63-32]PROTCLR	0	Peripheral memory frame protection clear. <i>Read:</i> The peripheral memory frame can be written to and read from in both user and privileged modes. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The peripheral memory frame can be written to only in privileged mode, but it can be read in both user and privileged modes. <i>Write:</i> The corresponding bit in PMPROTSET1 and PMPROTCLR1 registers is cleared to 0.

2.5.3.5 Peripheral Protection Set Register 0 (PPROTSET0)

There is one bit for each quadrant for PS0 to PS7.

The following are the ways in which quadrants are used within a PS frame-

- a. The slave uses all the four quadrants.

Only the bit corresponding to the quadrant 0 of PS_n is implemented. It protects the whole 1K-byte frame. The remaining three bits are not implemented.

- b. The slave uses two quadrants.

Each quadrant has to be in one of these groups: (Quad 0 and Quad 1) or (Quad 2 and Quad 3).

For the group Quad0/Quad1, the bit quadrant 0 protects both quadrants 0 and 1. The bit quadrant 1 is not implemented.

For the group Quad2/Quad3, the bit quadrant 2 protects both quadrants 2 and 3. The bit quadrant 3 is not implemented.

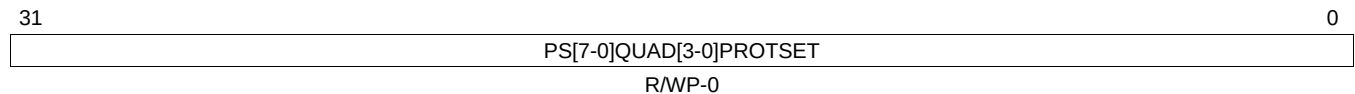
- c. The slave uses only one quadrant.

In this case, the bit, as specified in [Table 2-89](#), protects the slave.

The above arrangement is true for all the peripheral select (PS0 to PS31), presented in [Section 2.5.3.6 - Section 2.5.3.12](#). This register holds bits for PS0 to PS7 and is shown in [Figure 2-73](#) and described in [Table 2-89](#).

NOTE: Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-73. Peripheral Protection Set Register 0 (PPROTSET0) [offset = 20h]



LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-89. Peripheral Protection Set Register 0 (PPROTSET0) Field Descriptions

Bit	Field	Value	Description
31-0	PS[7-0]QUAD[3-0]PROTSET	0	Peripheral select quadrant protection set. <i>Read:</i> The peripheral select quadrant can be written to and read from in both user and privileged modes. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The peripheral select quadrant can be written to only in privileged mode, but it can be read in both user and privileged modes. <i>Write:</i> The corresponding bit in PPROTSET0 and PPROTCLR0 registers is set to 1.

2.5.3.6 Peripheral Protection Set Register 1 (PPROTSET1)

There is one bit for each quadrant for PS8 to PS15. The protection scheme is described in [Section 2.5.3.5](#). This register is shown in [Figure 2-74](#) and described in [Table 2-90](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-74. Peripheral Protection Set Register 1 (PPROTSET1) [offset = 24h]

31	0
PS[15-8]QUAD[3-0]PROTSET	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-90. Peripheral Protection Set Register 1 (PPROTSET1) Field Descriptions

Bit	Field	Value	Description
31-0	PS[15-8]QUAD[3-0]PROTSET	0	Peripheral select quadrant protection set. <i>Read:</i> The peripheral select quadrant can be written to and read from in both user and privileged modes. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The peripheral select quadrant can be written to only in privileged mode, but it can be read in both user and privileged modes. <i>Write:</i> The corresponding bit in PPROTSET1 and PPROTCLR1 registers is set to 1.

2.5.3.7 Peripheral Protection Set Register 2 (PPROTSET2)

There is one bit for each quadrant for PS16 to PS23. The protection scheme is described in [Section 2.5.3.5](#). This register is shown in [Figure 2-75](#) and described in [Table 2-91](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-75. Peripheral Protection Set Register 2 (PPROTSET2) [offset = 28h]

31	0
PS[23-16]QUAD[3-0]PROTSET	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-91. Peripheral Protection Set Register 2 (PPROTSET2) Field Descriptions

Bit	Field	Value	Description
31-0	PS[23-16]QUAD[3-0]PROTSET	0	Peripheral select quadrant protection set. <i>Read:</i> The peripheral select quadrant can be written to and read from in both user and privileged modes. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The peripheral select quadrant can be written to only in privileged mode, but it can be read in both user and privileged modes. <i>Write:</i> The corresponding bit in PPROTSET2 and PPROTCLR2 registers is set to 1.

2.5.3.8 Peripheral Protection Set Register 3 (PPROTSET3)

There is one bit for each quadrant for PS24 to PS31. The protection scheme is described in [Section 2.5.3.5](#). This register is shown in [Figure 2-76](#) and described in [Table 2-92](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-76. Peripheral Protection Set Register 3 (PPROTSET3) [offset = 2Ch]

31	0
PS[31-24]QUAD[3-0]PROTSET	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-92. Peripheral Protection Set Register 3 (PPROTSET3) Field Descriptions

Bit	Field	Value	Description
31-0	PS[31-24]QUAD[3-0]PROTSET	0	Peripheral select quadrant protection set. <i>Read:</i> The peripheral select quadrant can be written to and read from in both user and privileged modes. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The peripheral select quadrant can be written to only in privileged mode, but it can be read in both user and privileged modes. <i>Write:</i> The corresponding bit in PPROTSET3 and PPROTCLR3 registers is set to 1.

2.5.3.9 Peripheral Protection Clear Register 0 (PPROTCLR0)

There is one bit for each quadrant for PS0 to PS7. The protection scheme is described in [Section 2.5.3.5](#). This register is shown in [Figure 2-77](#) and described in [Table 2-93](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-77. Peripheral Protection Clear Register 0 (PPROTCLR0) [offset = 40h]

31	0
PS[7-0]QUAD[3-0]PROTCLR	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-93. Peripheral Protection Clear Register 0 (PPROTCLR0) Field Descriptions

Bit	Field	Value	Description
31-0	PS[7-0]QUAD[3-0]PROTCLR	0	Peripheral select quadrant protection clear. <i>Read:</i> The peripheral select quadrant can be written to and read from in both user and privileged modes. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The peripheral select quadrant can be written to only in privileged mode, but it can be read in both user and privileged modes. <i>Write:</i> The corresponding bit in PPROTSET0 and PPROTCLR0 registers is cleared to 0.

2.5.3.10 Peripheral Protection Clear Register 1 (PPROTCLR1)

There is one bit for each quadrant for PS8 to PS15. The protection scheme is described in [Section 2.5.3.5](#). This register is shown in [Figure 2-78](#) and described in [Table 2-94](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-78. Peripheral Protection Clear Register 1 (PPROTCLR1) [offset = 44h]

31	0
PS[15-8]QUAD[3-0]PROTCLR	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-94. Peripheral Protection Clear Register 1 (PPROTCLR1) Field Descriptions

Bit	Field	Value	Description
31-0	PS[15-8]QUAD[3-0]PROTCLR	0	Peripheral select quadrant protection clear. <i>Read:</i> The peripheral select quadrant can be written to and read from in both user and privileged modes. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The peripheral select quadrant can be written to only in privileged mode, but it can be read in both user and privileged modes. <i>Write:</i> The corresponding bit in PPROTSET1 and PPROTCLR1 registers is cleared to 0.

2.5.3.11 Peripheral Protection Clear Register 2 (PPROTCLR2)

There is one bit for each quadrant for PS16 to PS23. The protection scheme is described in [Section 2.5.3.5](#). This register is shown in [Figure 2-79](#) and described in [Table 2-95](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-79. Peripheral Protection Clear Register 2 (PPROTCLR2) [offset = 48h]

31	0
PS[23-16]QUAD[3-0]PROTCLR	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-95. Peripheral Protection Clear Register 2 (PPROTCLR2) Field Descriptions

Bit	Field	Value	Description
31-0	PS[23-16]QUAD[3-0]PROTCLR	0	Peripheral select quadrant protection clear. <i>Read:</i> The peripheral select quadrant can be written to and read from in both user and privileged modes. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The peripheral select quadrant can be written to only in privileged mode, but it can be read in both user and privileged modes. <i>Write:</i> The corresponding bit in PPROTSET2 and PPROTCLR2 registers is cleared to 0.

2.5.3.12 Peripheral Protection Clear Register 3 (PPROTCLR3)

There is one bit for each quadrant for PS24 to PS31. The protection scheme is described in [Section 2.5.3.5](#). This register is shown in [Figure 2-80](#) and described in [Table 2-96](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-80. Peripheral Protection Clear Register 3 (PPROTCLR3) [offset = 4Ch]

31	0
PS[31-24]QUAD[3-0]PROTCLR	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-96. Peripheral Protection Clear Register 3 (PPROTCLR3) Field Descriptions

Bit	Field	Value	Description
31-0	PS[31-24]QUAD[3-0] PROTCLR	0	Peripheral select quadrant protection clear. <i>Read:</i> The peripheral select quadrant can be written to and read from in both user and privileged modes. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The peripheral select quadrant can be written to only in privileged mode, but it can be read in both user and privileged modes. <i>Write:</i> The corresponding bit in PPROTSET3 and PPROTCLR3 registers is cleared to 0.

2.5.3.13 Peripheral Memory Power-Down Set Register 0 (PCSPWRDWNSET0)

Each bit corresponds to a bit at the same index in the PMPROT register in that they both relate to the same peripheral. This register is shown in [Figure 2-81](#) and described in [Table 2-97](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-81. Peripheral Memory Power-Down Set Register 0 (PCSPWRDWNSET0) [offset = 60h]

31	0
PCS[31-0]PWRDWNSET	
R/WP-1	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-97. Peripheral Memory Power-Down Set Register 0 (PCSPWRDWNSET0) Field Descriptions

Bit	Field	Value	Description
31-0	PCS[31-0]PWRDWNSET	0	Peripheral memory clock power-down set. <i>Read:</i> The peripheral memory clock[31-0] is active. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The peripheral memory clock[31-0] is inactive. <i>Write:</i> The corresponding bit in the PCSPWRDWNSET0 and PCSPWRDWNCLR0 registers is set to 1.

2.5.3.14 Peripheral Memory Power-Down Set Register 1 (PCSPWRDWNSET1)

This register is shown in [Figure 2-82](#) and described in [Table 2-98](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-82. Peripheral Memory Power-Down Set Register 1 (PCSPWRDWNSET1) [offset = 64h]

31	0
PCS[63-32]PWRDWNSET	
R/WP-1	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-98. Peripheral Memory Power-Down Set Register 1 (PCSPWRDWNSET1) Field Descriptions

Bit	Field	Value	Description
31-0	PCS[63-32]PWRDWNSET	0	Peripheral memory clock power-down set. <i>Read:</i> The peripheral memory clock[63-32] is active. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The peripheral memory clock[63-32] is inactive. <i>Write:</i> The corresponding bit in the PCSPWRDWNSET1 and PCSPWRDWNCLR1 registers is set to 1.

2.5.3.15 Peripheral Memory Power-Down Clear Register 0 (PCSPWRDWNCLR0)

This register is shown in [Figure 2-83](#) and described in [Table 2-99](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-83. Peripheral Memory Power-Down Clear Register 0 (PCSPWRDWNCLR0) [offset = 70h]

31	0
PCS[31-0]PWRDNCLR	
R/WP-1	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-99. Peripheral Memory Power-Down Clear Register 0 (PCSPWRDWNCLR0) Field Descriptions

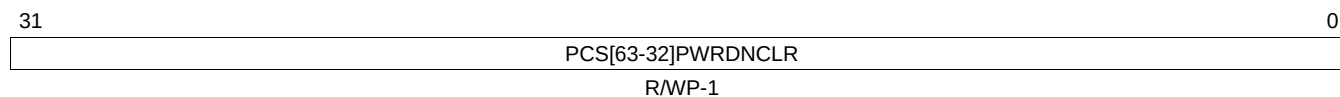
Bit	Field	Value	Description
31-0	PCS[31-0]PWRDNCLR	0	Peripheral memory clock power-down clear. <i>Read:</i> The peripheral memory clock[31-0] is active. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The peripheral memory clock[31-0] is inactive. <i>Write:</i> The corresponding bit in the PCSPWRDWNSET0 and PCSPWRDWNCLR0 registers is cleared to 0.

2.5.3.16 Peripheral Memory Power-Down Clear Register 1 (PCSPWRDWNCLR1)

This register is shown in [Figure 2-84](#) and described in [Table 2-100](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-84. Peripheral Memory Power-Down Clear Register 1 (PCSPWRDWNCLR1) [offset = 74h]



LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-100. Peripheral Memory Power-Down Set Register 1 (PCSPWRDWNCLR1) Field Descriptions

Bit	Field	Value	Description
31-0	PCS[63-32]PWRDNCLR	0	Peripheral memory clock power-down clear. <i>Read:</i> The peripheral memory clock[63-32] is active. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The peripheral memory clock[63-32] is inactive. <i>Write:</i> The corresponding bit in the PCSPWRDWNSET1 and PCSPWRDWNCLR1 registers is cleared to 0.

2.5.3.18 Peripheral Power-Down Set Register 1 (PSPWRDWNSET1)

There is one bit for each quadrant for PS8 to PS15. The protection scheme is described in [Section 2.5.3.17](#). This register is shown in [Figure 2-86](#) and described in [Table 2-102](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-86. Peripheral Power-Down Set Register 1 (PSPWRDWNSET1) [offset = 84h]

31	0
PS[15-8]QUAD[3-0]PWRDWNSET	
R/WP-1	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-102. Peripheral Power-Down Set Register 1 (PSPWRDWNSET1) Field Descriptions

Bit	Field	Value	Description
31-0	PS[15-8]QUAD[3-0] PWRDWNSET	0	Peripheral select quadrant clock power-down set. <i>Read:</i> The clock to the peripheral select quadrant is active. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The clock to the peripheral select quadrant is inactive. <i>Write:</i> The corresponding bit in PSPWRDWNSET1 and PSPWRDWNCLR1 registers is set to 1.

2.5.3.19 Peripheral Power-Down Set Register 2 (PSPWRDWNSET2)

There is one bit for each quadrant for PS16 to PS23. The protection scheme is described in [Section 2.5.3.17](#). This register is shown in [Figure 2-87](#) and described in [Table 2-103](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-87. Peripheral Power-Down Set Register 2 (PSPWRDWNSET2) [offset = 88h]

31	0
PS[23-16]QUAD[3-0]PWRDWNSET	
R/WP-1	

LEGEND: R/W = Read/Write: WP = Write in privileged mode only: -n = value after reset

Table 2-103. Peripheral Power-Down Set Register 2 (PSPWRDWNSET2) Field Descriptions

Bit	Field	Value	Description
31-0	PS[23-16]QUAD[3-0] PWRDWNSET	0	Peripheral select quadrant clock power-down set. <i>Read:</i> The clock to the peripheral select quadrant is active. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The clock to the peripheral select quadrant is inactive. <i>Write:</i> The corresponding bit in PSPWRDWNSET2 and PSPWRDWNCLR2 registers is set to 1.

2.5.3.20 Peripheral Power-Down Set Register 3 (PSPWRDWNSET3)

There is one bit for each quadrant for PS24 to PS31. The protection scheme is described in [Section 2.5.3.17](#). This register is shown in [Figure 2-88](#) and described in [Table 2-104](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-88. Peripheral Power-Down Set Register 3 (PSPWRDWNSET3) [offset = 8Ch]

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-104. Peripheral Power-Down Set Register 3 (PSPWRDWNSET3) Field Descriptions

Bit	Field	Value	Description
31-0	PS[31-24]QUAD[3-0] PWRDWNSET	0	Peripheral select quadrant clock power-down set. <i>Read:</i> The clock to the peripheral select quadrant is active. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The clock to the peripheral select quadrant is inactive. <i>Write:</i> The corresponding bit in PSPWRDWNSET3 and PSPWRDWNCLR3 registers is set to 1.

2.5.3.21 Peripheral Power-Down Clear Register 0 (PSPWRDWNCLR0)

There is one bit for each quadrant for PS0 to PS7. The protection scheme is described in [Section 2.5.3.17](#). This register is shown in [Figure 2-89](#) and described in [Table 2-105](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-89. Peripheral Power-Down Clear Register 0 (PSPWRDWNCLR0) [offset = A0h]

31 PS[7-0]QUAD[3-0]PWRDWNCLR 0
R/WP-1

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-105. Peripheral Power-Down Clear Register 0 (PSPWRDWNCLR0) Field Descriptions

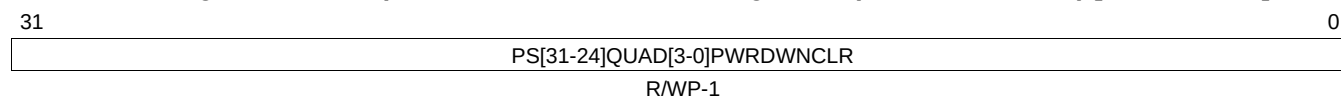
Bit	Field	Value	Description
31-0	PS[7-0]QUAD[3-0] PWRDWNCLR	0	Peripheral select quadrant clock power-down clear. <i>Read:</i> The clock to the peripheral select quadrant is active. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The clock to the peripheral select quadrant is inactive. <i>Write:</i> The corresponding bit in PSPWRDWNSET0 and PSPWRDWNCLR0 registers is cleared to 0.

2.5.3.24 Peripheral Power-Down Clear Register 3 (PSPWRDWNCLR3)

There is one bit for each quadrant for PS24 to PS31. The protection scheme is described in [Section 2.5.3.17](#). This register is shown in [Figure 2-92](#) and described in [Table 2-108](#).

NOTE: Only those bits that have a slave at the corresponding bit position are implemented. Writes to nonimplemented bits have no effect and reads are 0.

Figure 2-92. Peripheral Power-Down Clear Register 3 (PSPWRDWNCLR) [offset = ACh]



LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 2-108. Peripheral Power-Down Clear Register 3 (PSPWRDWNCLR3) Field Descriptions

Bit	Field	Value	Description
31-0	PS[31-24]QUAD[3-0] PWRDWNCLR	0	Peripheral select quadrant clock power-down clear. <i>Read:</i> The clock to the peripheral select quadrant is active. <i>Write:</i> The bit is unchanged.
		1	<i>Read:</i> The clock to the peripheral select quadrant is inactive. <i>Write:</i> The corresponding bit in PSPWRDWNSET3 and PSPWRDWNCLR3 registers is cleared to 0.